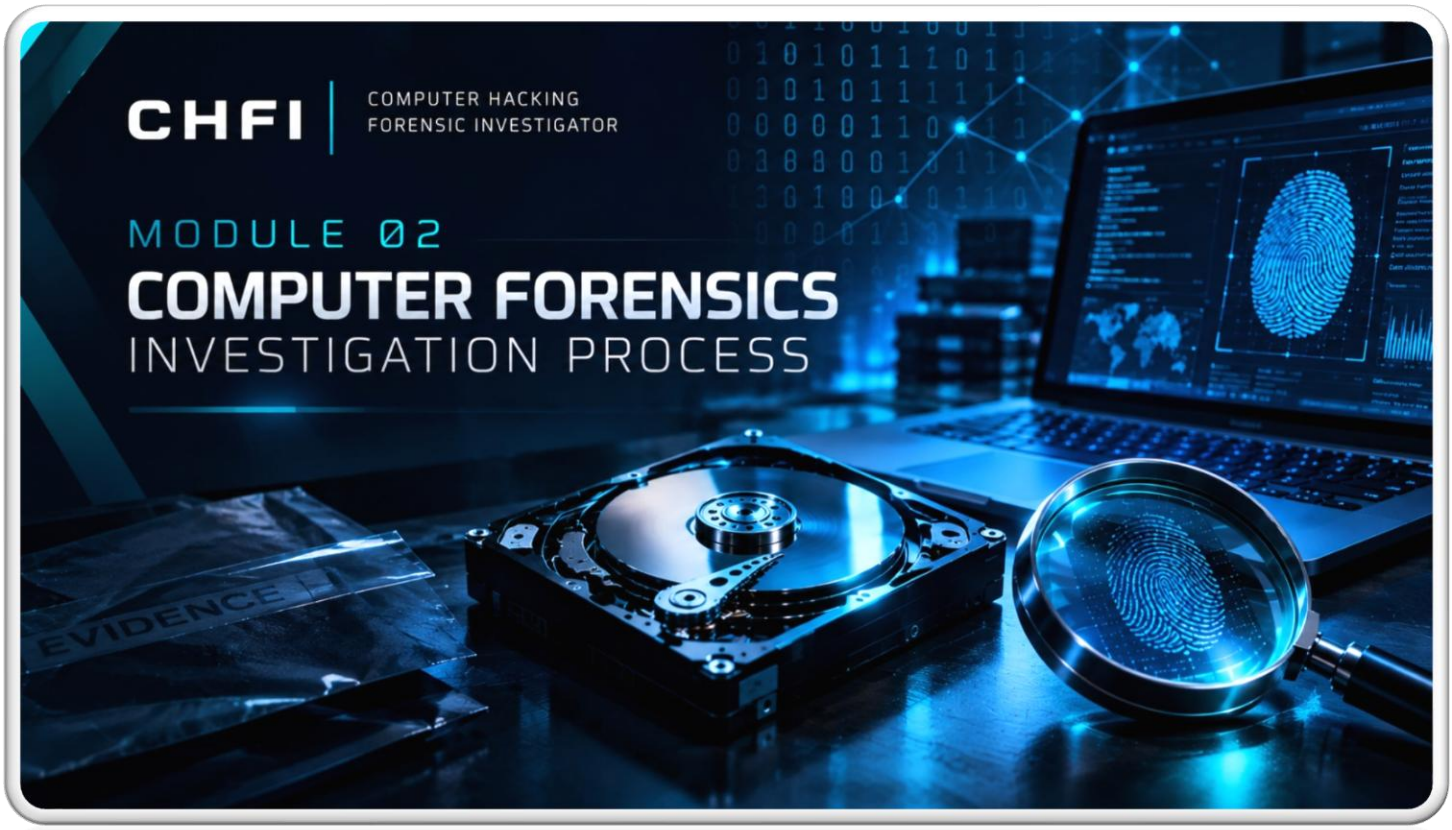


Abdul Wahid

EC-Council (Certified Hacking forensic investigator)

Computer Forensics

INVESTIGATION OVERVIEW

This investigation was conducted as part of CHFI Module 2. The subject evidence was a 32 GB Samsung USB drive that had been formatted prior to submission. Six progressive laboratory exercises were carried out to recover, examine, verify, and preserve the digital evidence found on the device. All procedures followed professional forensic standards with an unbroken chain of custody maintained throughout.

What Was Done Lab by Lab

Lab 1 — Files were recovered from the formatted USB drive using EaseUS Data Recovery Wizard. Approximately 2 GB of files were successfully retrieved and confirmed as intact.

Lab 2 — Hash values were calculated for key evidence files using MD5, SHA-1, SHA-256, and SHA-512. A file named Infected.pdf was submitted to VirusTotal and confirmed as a trojan by 44 of 62 antivirus vendors.

Lab 3 — File integrity verification was demonstrated by showing that modifying a file produces a completely different hash value, confirming hash comparison as a reliable tamper-detection method.

Lab 4 — A full forensic examination was conducted using five specialist tools. File signatures, metadata, archive contents, system artefacts, and keyword searches were all analyzed. Five critical findings were identified during this lab.

Lab 5 — Evidence handling procedures were applied. The evidence source was reviewed under read-only conditions using FTK Imager and MD5 and SHA-256 baseline hashes were established for use in Lab 6.

Lab 6 — A professional E01 forensic image was created using FTK Imager v4.7.3.81. MD5 and SHA-256 hashes were verified against the Lab 5 baseline and confirmed as an exact match, establishing bit-for-bit accuracy.

Key Findings

#	Finding	Discovered In	Priority
1	Infected.pdf confirmed as trojan.pdfka — 44 of 62 vendor detections	Lab 2	Critical
2	Email address alishba.shafique@thearzens.tech found in multiple documents	Lab 4	High
3	10 files flagged as suspicious by Autopsy v4.23.1	Lab 4	High
4	File timestamps spanning 2023 to 2026 found within recovered archive	Lab 4	Medium
5	Metadata anomalies — unnamed authors, identical timestamps, Python-generated PDFs	Lab 4	Medium

---

Evidence and Integrity Status

Evidence Item	Description	Status
EV-001	Original 32 GB Samsung USB drive	Preserved — unmodified
EV-002	Recovered files (~2 GB)	Verified — all files intact
EV-003	Workspace copy for Lab 4	Examined under read-only access
EV-004	Lab 5 evidence source	MD5 and SHA-256 baseline established
EV-006	E01 forensic image	Hashes matched — bit-for-bit accurate

Chain of custody remained unbroken across all six labs. The original media was never modified. All analysis was performed on working copies or the forensic image. The E01 image is verified and ready for long-term archival.

Tools Used

Tool	Version	Purpose	Labs
EaseUS Data Recovery	Free	File recovery	1
HashCalc	Latest	Hash calculation	2, 5, 6
VirusTotal	Online	Malware detection	2
HxD Hex Editor	Latest	File signature verification	4
exiftool	13.59	Metadata extraction	4
7-Zip	Latest	Archive inspection	4
Autopsy	4.23.1	Forensic case analysis	4
Registry Editor	Built-in	System artefact review	4
FTK Imager	4.7.3.81	Evidence handling and disk imaging	5, 6

DIGITAL FORENSICS FRAMEWORK

1. What Is Digital Forensics?

Digital forensics is the scientific discipline of identifying, preserving, recovering, examining, and presenting digital evidence in a manner that maintains its integrity and remains legally defensible. It serves as the bridge between technical investigation and legal proceedings — ensuring that evidence recovered from digital devices can withstand scrutiny in a court of law, an academic review, or a formal investigative body.

Unlike traditional forensics, digital forensics deal with evidence that is invisible to the naked eye, easily altered, and highly volatile. A single incorrect action — opening a file, connecting an unprotected drive, or saving data to the wrong location — can permanently compromise the integrity of an investigation. This is why every procedure, every tool, and every decision in digital forensics must be deliberate, documented, and defensible.

In this module, digital forensics was applied across six structured laboratory exercises. Each lab targets a distinct phase of a real-world investigation, beginning with the recovery of data from a formatted storage device and concluding with the creation of a professional, court-admissible forensic image verified through cryptographic hashing.

2. Scope of This Module

This module represents a complete, end-to-end digital forensic investigation conducted on a single piece of evidence — a 32 GB Samsung USB drive submitted under Case Number CHFI_Module2_Lab1-6_Case001-002. The investigation was carried out by Examiner Abdul Wahid at the Air University Forensic Lab between 12 and 13 September 2026, under course CY334 Digital Forensics.

Six-Lab Investigation Overview

Lab	Title	Primary Focus
Lab 1	Data Recovery from Formatted USB	Evidence collection and recovery
Lab 2	Hash Calculation and Integrity Verification	Integrity verification and malware detection
Lab 3	File Integrity Verification	Before-and-after comparison analysis
Lab 4	Advanced Forensic Analysis and Metadata Extraction	Deep examination and investigative analysis
Lab 5	Evidence Handling Procedures	Baseline establishment and professional standards
Lab 6	Disk Imaging and Professional Preservation	Archival and court-admissible imaging

Each lab builds directly upon the findings and evidence of the previous one, forming a continuous and unbroken investigative chain from first contact with the device through to final preservation.

3. The Seven-Phase Digital Forensics Investigation Model



Every professional digital forensic investigation follows a standardised seven-phase model. This model ensures that evidence is handled correctly at every stage, that nothing is overlooked, and that the investigation can be defended in a legal or formal setting. All six labs in this module map directly onto one or more of these phases.

Phase 1 — Identification

The investigator locates and recognises potential sources of digital evidence. The physical device is documented, its condition is recorded, and initial chain of custody entries are created. No data is collected at this stage — the focus is entirely on recognition and documentation.

Applied in this investigation: The 32 GB Samsung USB drive was identified, inspected, and logged as Evidence Item EV-001. Its physical state was recorded and the chain of custody was formally opened before any technical procedures began.

Phase 2 — Preservation

The original evidence is protected from any form of modification, accidental or deliberate. Working copies are created so that all analysis can be performed without ever touching the source media. Forensic images may also be created at this stage to provide a verified, bit-for-bit duplicate.

Applied in this investigation: The original USB drive was preserved in its unmodified state throughout all six labs. Working copies were created at Labs 1, 4, and 5. A professional E01 forensic image was produced in Lab 6 as the final preservation artefact.

Phase 3 — Collection

Evidence is gathered systematically using approved forensic tools and documented procedures. Every action taken during collection is recorded to ensure the process is reproducible and legally defensible.

Applied in this investigation: Approximately 2 GB of files were recovered from the formatted USB drive in Lab 1 using EaseUS Data Recovery Wizard. All recovered files were saved to a designated recovery folder and verified for accessibility before any further analysis was conducted.

Phase 4 — Examination

The evidence collected is inspected in depth. File signatures are verified against known headers, metadata is extracted, archive contents are reviewed, and system artefacts such as registry entries and event logs are examined. The goal is to surface as much raw information as possible from the evidence.

Applied in this investigation: Lab 4 applied five specialist tools — HxD Hex Editor, exiftool, 7-Zip, Windows Registry Editor, and Autopsy v4.23.1 — to conduct a

comprehensive examination of the recovered files. File signatures were confirmed valid, metadata anomalies were identified, and archive contents were inspected for timeline irregularities.

Phase 5 — Analysis

The findings from examination are interpreted and their investigative significance is assessed. The examiner identifies patterns, anomalies, and leads, connecting individual findings to the broader context of the investigation.

Applied in this investigation: Lab 4 analysis produced five critical findings — a confirmed trojan (Infected.pdf, detected by 44 of 62 VirusTotal vendors), an email address linked to a person of interest, 10 files flagged by Autopsy as suspicious, a timeline anomaly spanning 2023 to 2026, and metadata irregularities suggesting automated document creation and possible anonymisation.

Phase 6 — Reporting

All procedures, findings, and conclusions are compiled into a professional report suitable for submission to an academic institution, a formal investigative body, or a court of law. The report must be clear, accurate, and complete enough for an independent examiner to reproduce every step.

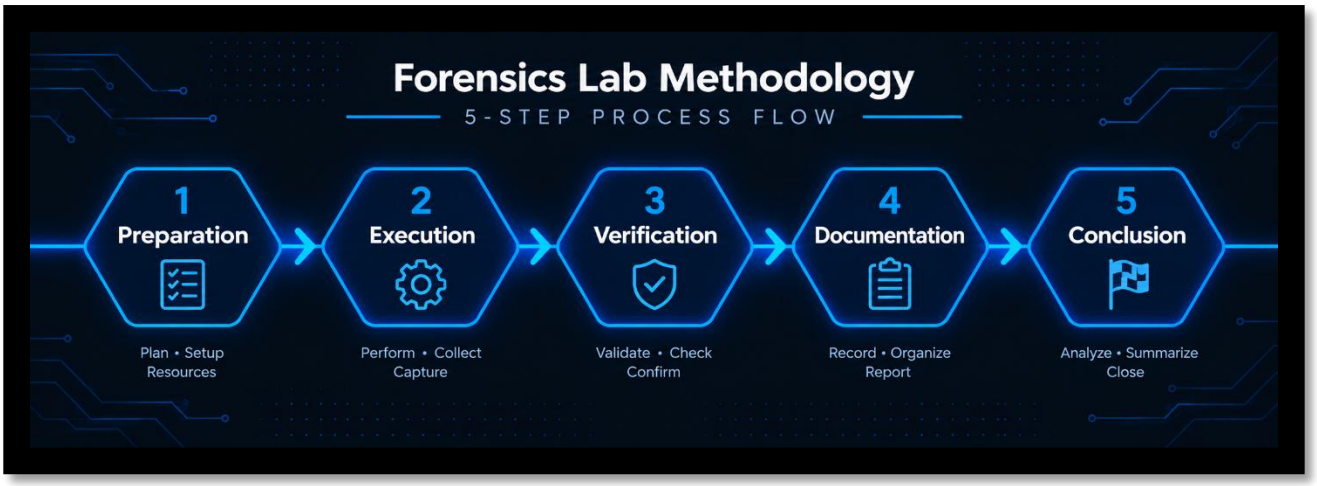
Applied in this investigation: Phase 7 professional reports were generated within Lab 4 and Lab 6. This full module report serves as the comprehensive documentation of all six labs, covering procedures, findings, evidence handling, and recommendations.

Phase 7 — Archival and Preservation

Evidence is stored in a durable, verifiable format designed for long-term retention and potential re-examination. Cryptographic hash values are recorded alongside the archived evidence to allow future verification that nothing has changed since the investigation concluded.

Applied in this investigation: Lab 6 produced a professional E01 forensic image using FTK Imager v4.7.3.81. The image was verified through both MD5 and SHA-256 hash comparison against the Lab 5 baseline. Hashes matched exactly, confirming bit-for-bit accuracy and establishing the image as ready for long-term archival and court submission.

4. Investigation Methodology



While the seven-phase model governs the overall investigation, each individual lab in this module follows a consistent five-step internal methodology. This structure ensures that every lab is approached with the same level of rigour and produces documentation of equivalent professional standard.

Step 1 — Preparation

Before any technical work begins, the lab objectives are reviewed, the required folder structure is created, all tools are verified and version-checked, and the case details are documented. Evidence items relevant to the lab are identified and their current status is confirmed in the chain of custody log.

Step 2 — Execution

Procedures are carried out in a deliberate, sequential order. Each action is accompanied by a screenshot and a written observation. No steps are skipped, no shortcuts are taken, and any unexpected outcomes are recorded rather than ignored.

Step 3 — Verification

At the conclusion of each lab's technical work, cryptographic hash values are calculated and compared where applicable. This step confirms that evidence has not been altered during the lab and provides a documented integrity check that can be referenced in subsequent labs or legal proceedings.

Step 4 — Documentation

All findings are recorded in a structured format. Chain of custody entries are updated to reflect every action taken during the lab. Tool versions, file paths, hash values, timestamps, and observations are all captured and retained as part of the permanent case record.

Step 5 — Conclusion

The lab is formally closed with a summary of results, an assessment of investigative significance, and a note of any leads identified for follow-up. The status of all evidence items is confirmed, and readiness for the next lab is verified before proceeding.

5. Core Forensic Principles

The following principles were applied without exception across every phase of this investigation. They are not procedural preferences — they are the non-negotiable standards that determine whether digital evidence is admissible and defensible.

Original evidence is never modified. Every piece of analysis in this investigation was performed on a working copy or forensic image. The original USB drive, logged as EV-001, was never connected to an unprotected system, never written to, and never opened directly for analysis. It remains in its original state from the moment it was received.

All analysis is performed on verified working copies. Before any examination was conducted, working copies were created from verified sources and their integrity was confirmed through hashing. This ensures that any finding from the analysis is traceable back to a verified copy of the original evidence.

Cryptographic hashes are calculated and compared at every critical stage. MD5, SHA-1, SHA-256, and SHA-512 hashes were calculated during Labs 2, 5, and 6. Hash comparison was used to confirm file integrity, detect modification, verify baseline evidence, and validate the final forensic image. A matching hash is mathematical proof that no change occurred.

Chain of custody is maintained continuously. Every action taken on every piece of evidence was logged with a date, time, action description, and the name of the handler. There are no gaps, no undocumented transfers, and no unrecorded accesses across the entire six-lab investigation.

Read-only access is enforced wherever possible. FTK Imager was configured in read-only mode during Labs 5 and 6. Evidence folders were set to read-only permissions

before examination. No tool used for analysis was permitted write access to an evidence item.

All tools, versions, and procedures are fully recorded. Nine professional forensic tools were used across the six labs. Every tool is documented with its version number and the specific labs in which it was applied, ensuring the investigation is fully reproducible.

6. Legal Admissibility Standards

For digital evidence to be accepted in a court of law, it must satisfy five fundamental requirements. The following table confirms the status of this investigation against each requirement:

Requirement	Our Status	How It Was Met
Unbroken Chain of Custody	Confirmed	Logged continuously across all 6 labs with no gaps
Proper Collection Procedures	Confirmed	Industry-standard tools and CHFI-certified procedures used throughout
Integrity Verification	Confirmed	Hash verification completed at Labs 2, 5, and 6 — all values matched
Professional Documentation	Confirmed	Phase-by-phase reports generated; this module report compiled in full
Qualified Examiner	Confirmed	CHFI Module 2 trained examiner; proficiency demonstrated across 9 tools

This investigation meets all five requirements. The evidence produced is suitable for academic submission and, where applicable, for presentation in formal legal proceedings.

LAB 1 — DATA RECOVERY FROM FORMATTED USB DRIVE

Lab Overview

Field	Detail
Lab Number	1
Lab Title	Data Recovery from Formatted USB Drive
Case Reference	CHFI_Module2_Lab1-6_Case001
Examiner	Abdul Wahid
Date Conducted	12 September 2026
Investigation Phase	Collection and Preservation

Purpose

When a storage device is formatted, the operating system removes the file system index but does not physically erase the underlying data. The files remain on the disk until new data overwrites them. This makes formatting an ineffective method of evidence destruction and gives forensic investigators the ability to recover data that a suspect believed was permanently deleted.

This laboratory demonstrates that process. A 32 GB Samsung USB drive was formatted to simulate evidence destruction. Professional data recovery software was then used to retrieve the deleted files, verify their integrity, and establish them as the evidence base for all subsequent laboratory exercises.

Objectives

- Recover deleted files from a formatted USB drive using a professional forensic tool
- Verify that all recovered files are intact and accessible
- Save recovered files to a designated location without writing to the original media

- Create and log the recovered file set as Evidence Item EV-002
- Establish the chain of custody for all evidence items entering the investigation

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Tools and Evidence

Tool	Version	Purpose
EaseUS Data Recovery Wizard	Free	File recovery from formatted storage

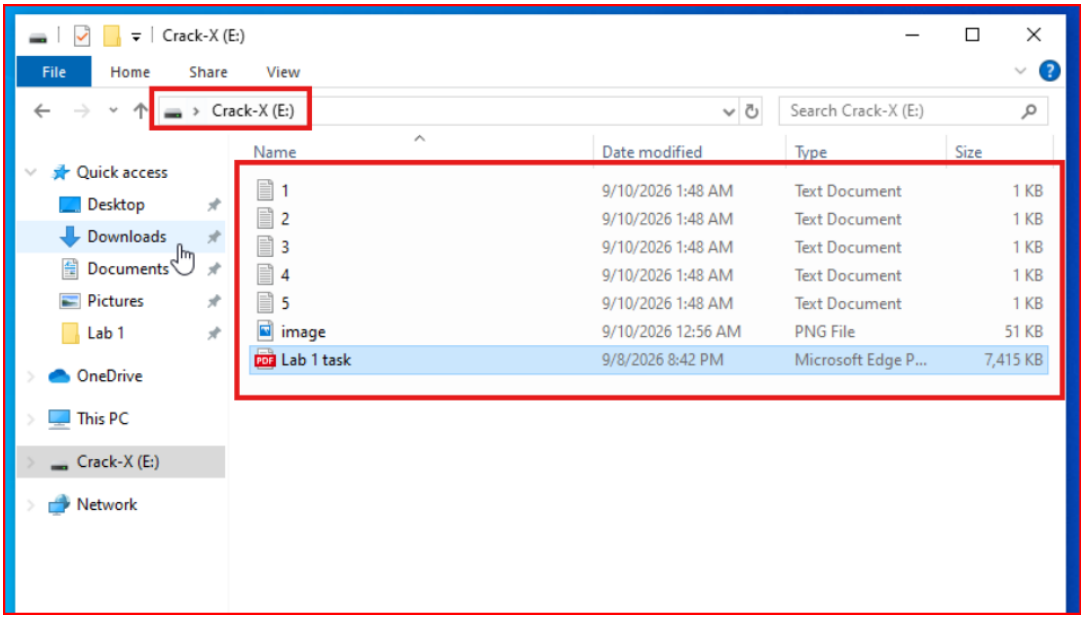
Evidence Item	Description	Status at Start of Lab
EV-001	Original 32 GB Samsung USB drive	Received — unmodified

Real World Relevance

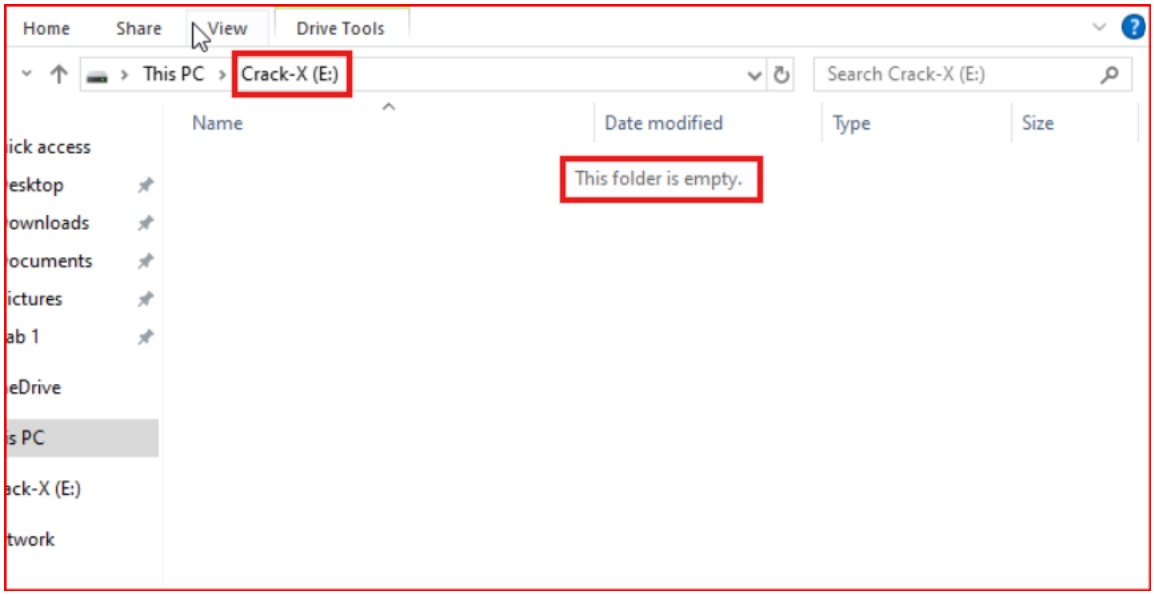
In live criminal investigations, suspects routinely format drives, delete partitions, or perform factory resets in an attempt to destroy incriminating data. Digital forensic examiners are trained to counter this using specialised recovery tools that read raw disk sectors rather than relying on the file system index. The techniques applied in this laboratory reflect the same methodology used by law enforcement and professional DFIR teams worldwide.

Lab Setup

1. Take a USB drive (preferably 4GB–16GB).
2. Copy files into the USB (photos, PDFs, Word documents, etc.).

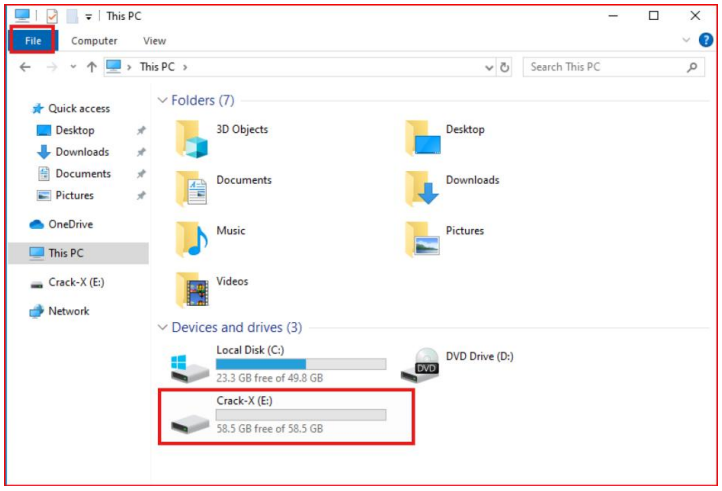


3. Format the USB (use Quick Format).



Phase 1: Preparation

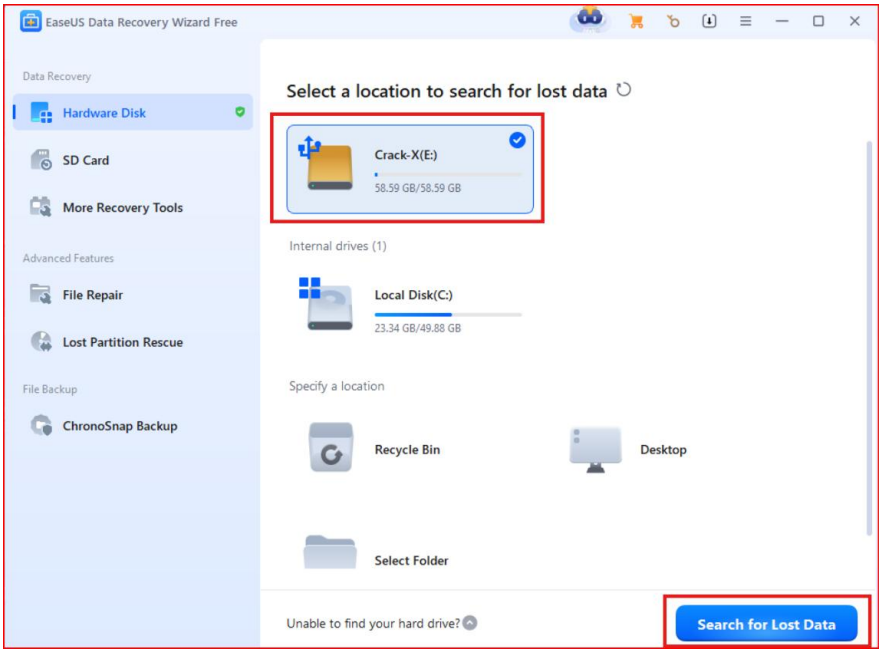
- Insert the USB drive.
- Open File Explorer and confirm that the USB is empty after formatting.



Phase 2: Launch Recovery Tool

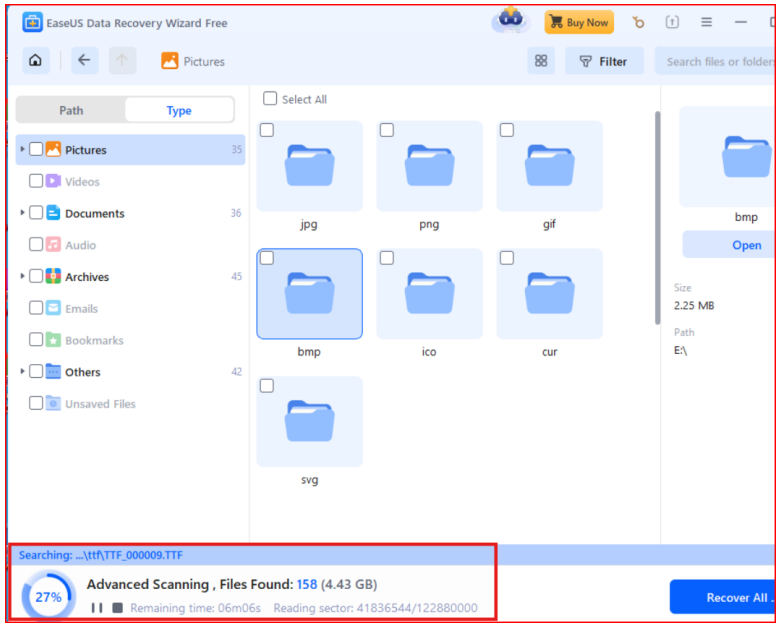
If using EaseUS Data Recovery Wizard:

1. Open EaseUS Data Recovery Wizard.
2. Select your USB drive.
3. Click Search for Lost Data or Scan.



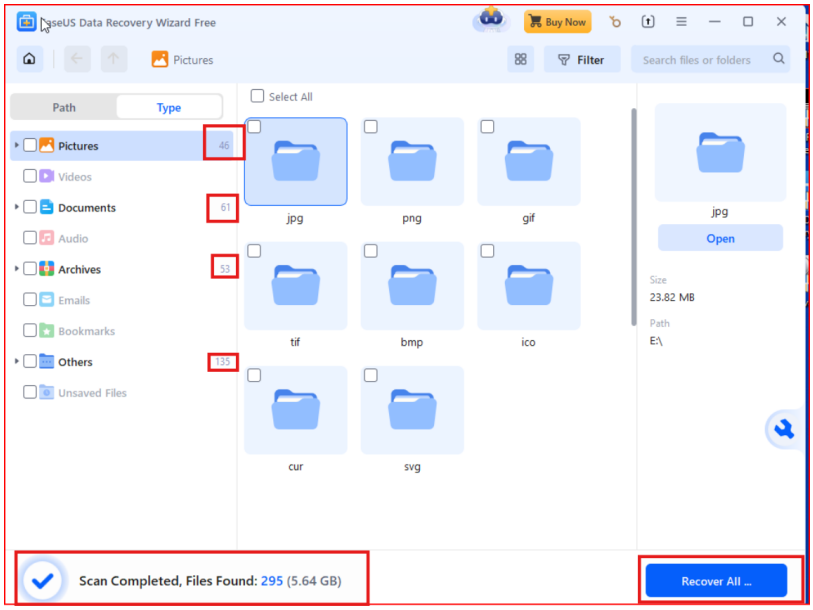
Phase 3: Scanning

- Let the scan complete (it may take some time).

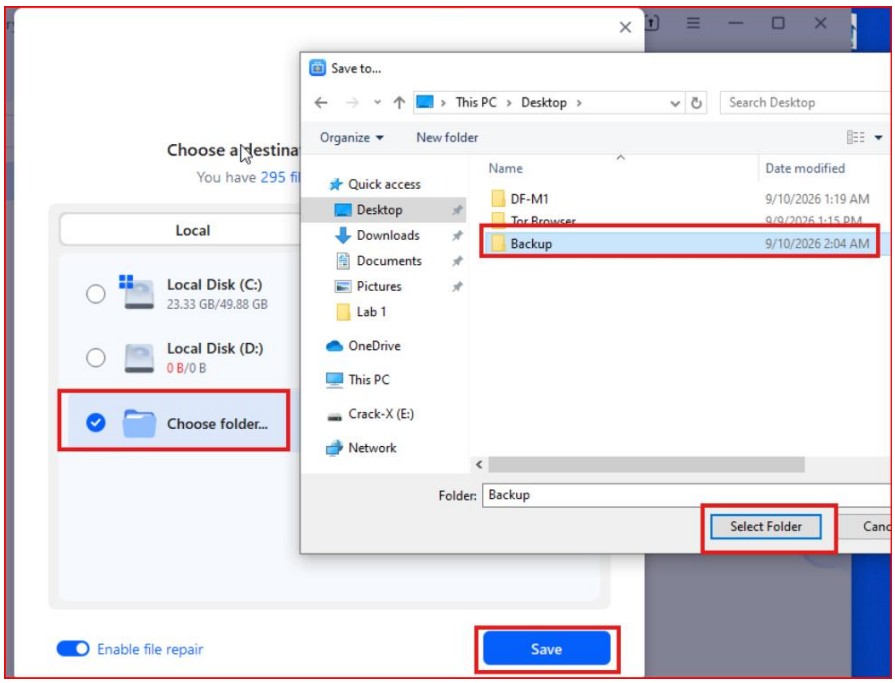


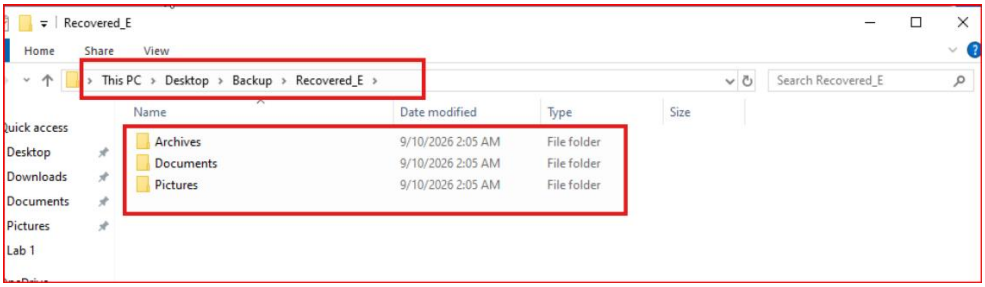
Phase 4: Recover the Files

1. After the scan finishes, you will see the list of recoverable files.
2. Select the files you want to recover (or Select All).
3. Click the **Recover** button.



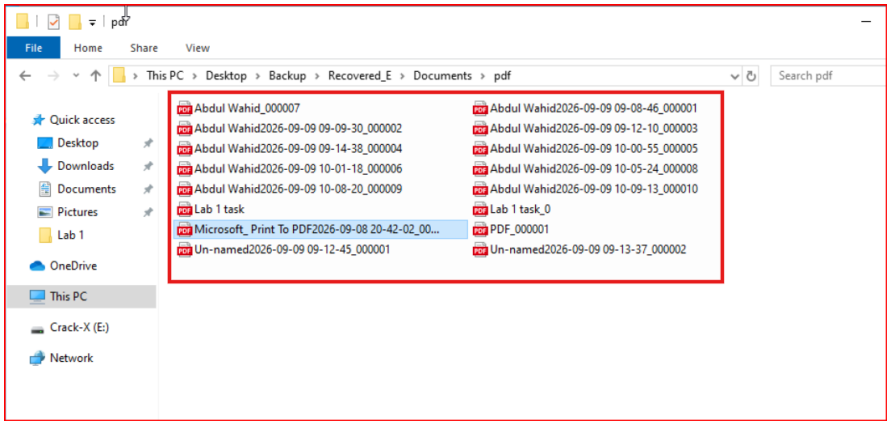
4. Save the recovered files to a different location (Do not save them back to the same USB).





Phase 5: Verification

- Open the folder where you saved the recovered files.
- Check that the files open correctly.



Note: As I don't have premium access and tried to recover almost more than 5gb data I only recovered about 2GB data here so the files and folder we created didn't showed here to check the specific files try recovering specific files you delete recently instead of recovering all the files you once placed in your USB

Lab 1 Conclusion

The objective of this laboratory was fully achieved. Approximately 2 GB of files were successfully recovered from the formatted USB drive using EaseUS Data Recovery Wizard. All recovered files were confirmed as intact and accessible, with no corruption detected across any file type examined.

The recovery demonstrated a critical forensic principle — formatting a storage device does not destroy the underlying data. Every file retrieved in this lab existed because the format operation only removed the file system index, leaving the raw data on disk untouched and available for professional recovery.

Evidence Status at Close of Lab 1

Evidence Item	Description	Status
EV-001	Original 32 GB Samsung USB drive	Preserved — unmodified throughout
EV-002	Recovered files, approximately 2 GB	Verified — all files intact and accessible

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Key Finding

The USB drive yielded approximately 2 GB of recoverable files spanning multiple categories including documents, images, archives, and system files. These files form the complete evidence base for Labs 2 through 6. No data was written to the original media at any point during this laboratory.

LAB 2 & 3 — HASH CALCULATION AND INTEGRITY VERIFICATION

Lab Overview

Field	Detail
Lab Numbers	2 and 3 — Combined
Lab Title	Hash Calculation and Integrity Verification
Case Reference	CHFI_Module2_Lab1-6_Case001
Examiner	Abdul Wahid
Date Conducted	12 September 2026
Investigation Phase	Examination and Verification

Purpose

Recovering files from a device proves that data existed on it. Proving that those files have not been altered since recovery is an equally critical and entirely separate requirement. Without integrity verification, no recovered file can be confidently presented as evidence — there is no way to demonstrate that what is being examined is exactly what was originally collected from the device.

Cryptographic hashing addresses this directly. A hash function processes a file and produces a unique fixed-length value based on its exact byte-level contents. If the file changes in any way — a single character, a single bit — the resulting hash value changes completely and immediately. This property makes hash comparison the most reliable and legally accepted method of proving digital evidence integrity worldwide.

Lab 2 introduces the hashing process using HashCalc, establishes integrity baselines across four algorithms, and performs malware detection on a suspicious recovered file using VirusTotal. Lab 3 extends that work using a dedicated MD5 tool to conduct a structured and formal before and after integrity comparison — demonstrating that any modification to a file is immediately and unambiguously detectable. Together these two laboratories form a complete and professionally documented integrity verification workflow.

Objectives

- Calculate MD5, SHA-1, SHA-256, and SHA-512 hash values for key evidence files using HashCalc
- Establish cryptographic integrity baselines for recovered evidence
- Submit Infected.pdf to VirusTotal and document the malware detection result
- Apply a dedicated MD5 tool in Lab 3 to perform structured before and after integrity comparison
- Demonstrate that an unmodified file produces an identical hash value across repeated calculations
- Demonstrate that a modified file produces a completely different hash value
- Record all hash values and verification results as part of the permanent case record

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Tools and Evidence

Tool	Version	Purpose	Lab
HashCalc	Latest	MD5, SHA-1, SHA-256, SHA-512 hash calculation	Lab 2
VirusTotal	Online	Malware detection via file and hash submission	Lab 2
MD5 Tool	Latest	Dedicated integrity verification and before/after comparison	Lab 3

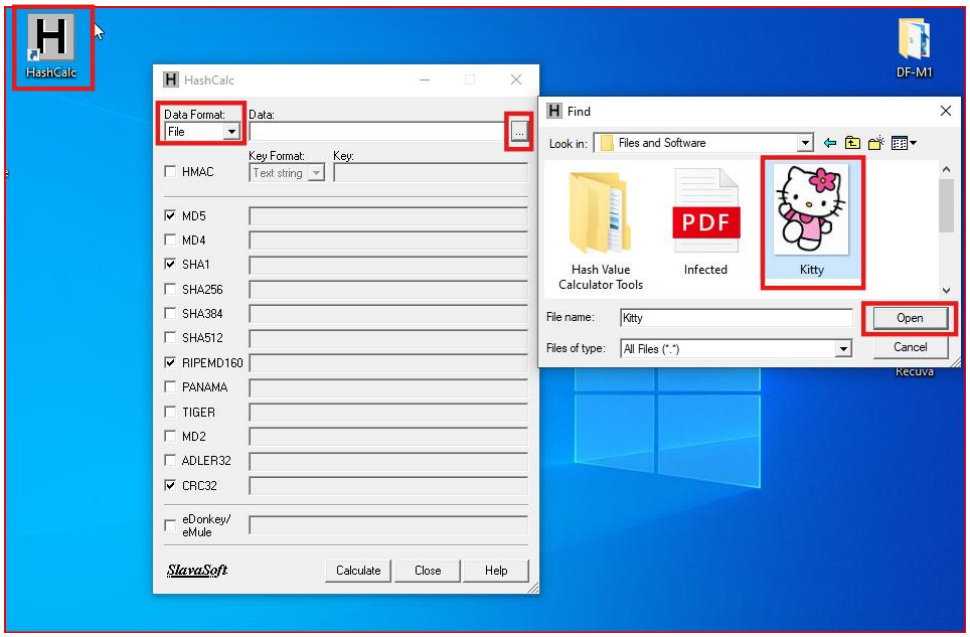
Evidence Item	Description	Status at Start of Labs
EV-002	Recovered files, approximately 2 GB	Verified — carried forward from Lab 1

Real World Relevance

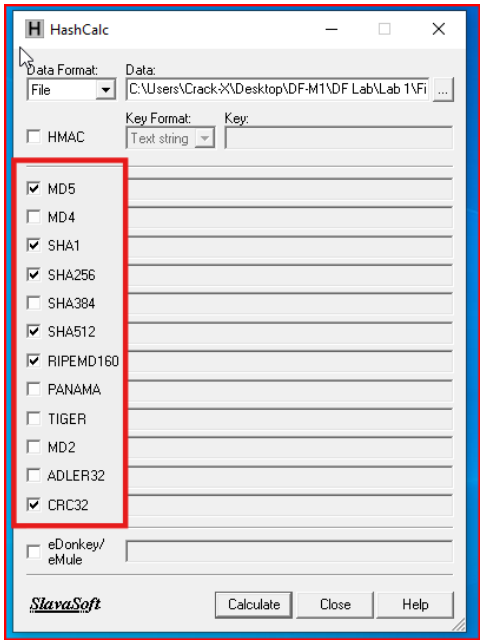
Every piece of digital evidence submitted in a legal proceeding must be accompanied by hash values recorded at the point of collection and verifiable at any later stage. Courts and international forensic standards recognise MD5 and SHA-256 as the minimum acceptable algorithms for evidence integrity verification. Using HashCalc to calculate multiple algorithm values simultaneously provides a comprehensive baseline, while a dedicated MD5 tool for Lab 3 mirrors the real-world practice of using purpose-specific verification tools at different stages of an investigation. Identifying malicious files early through VirusTotal additionally prevents accidental execution on the examination workstation and establishes criminal intent on the part of the device owner.

Phase 1: Hash a File (Kitty.jpg)

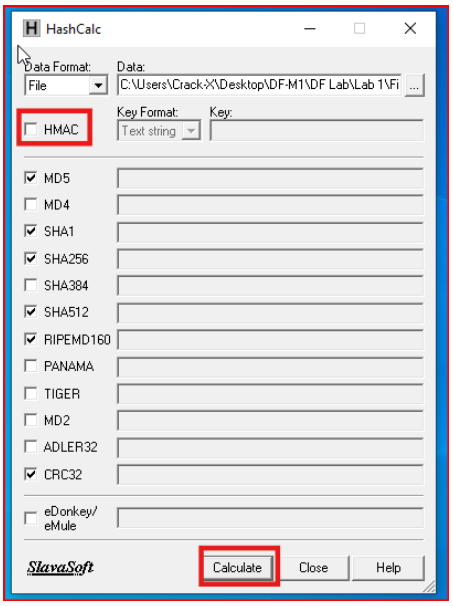
1. Open **HashCalc**.
2. Set **Data Format** to **File**.
3. Click the **...** (browse) button and select Kitty.jpg.



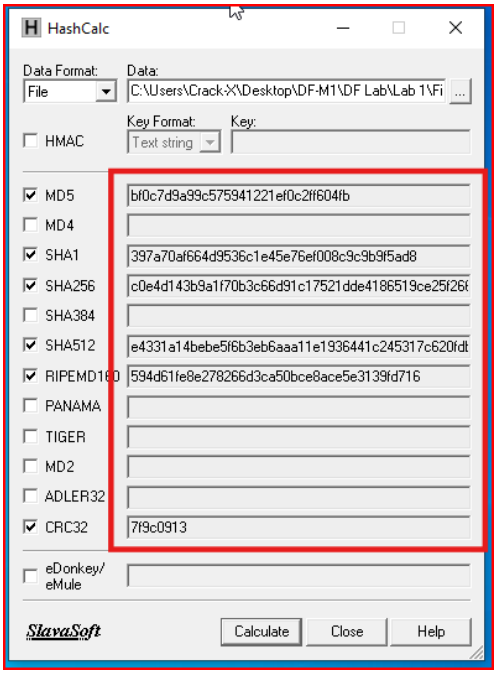
4. Check the boxes for **MD5, SHA1, SHA256, SHA512** (you can select more if you want).



5. Make sure **HMAC** is **unchecked**.
6. Click **Calculate**.

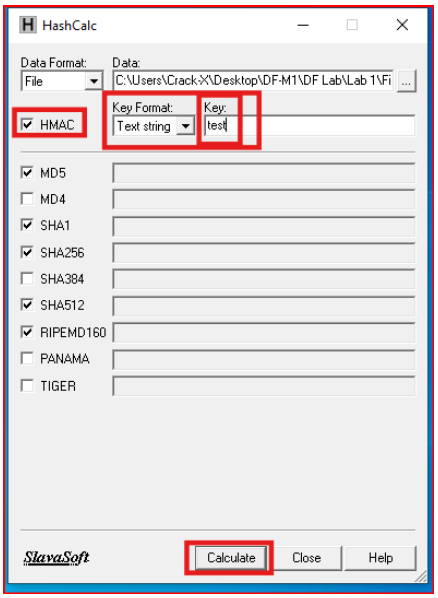


7. Generated hash values.

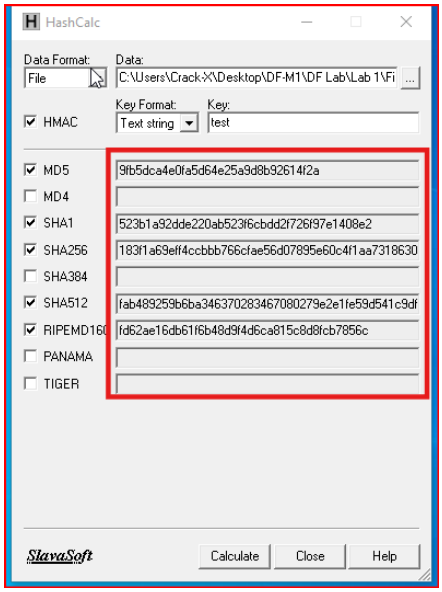


Phase 2: Calculate HMAC

1. In the same HashCalc window, check the **HMAC** box.
2. Set **Key Format** to **Text string**.
3. In the **Key** field, type: test
4. Keep the same algorithms selected.
5. Click **Calculate**.

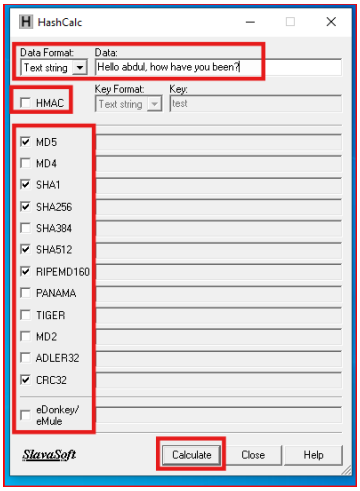


6. HMAC values.



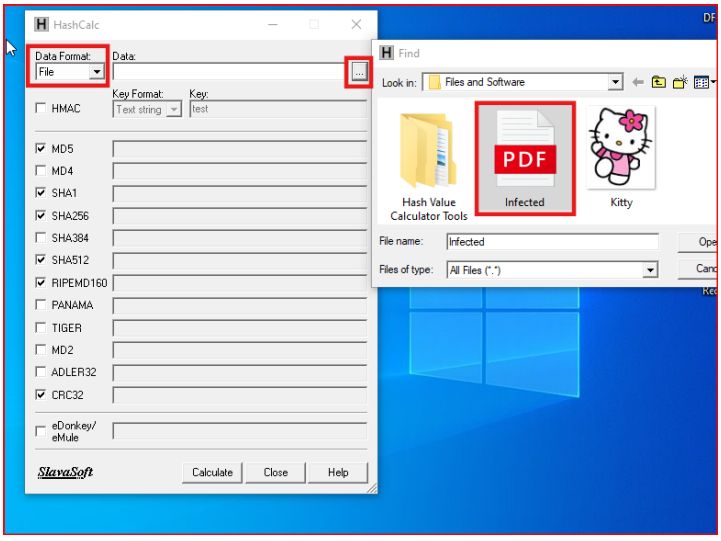
Phase 3: Hash a Text String

1. Change **Data Format** to **Text string**.
2. Uncheck the **HMAC** box.
3. In the Data field, type:
   Hello abdul, how have you been?
4. Select the algorithms (MD5, SHA1, SHA256, etc.).
5. Click **Calculate**.

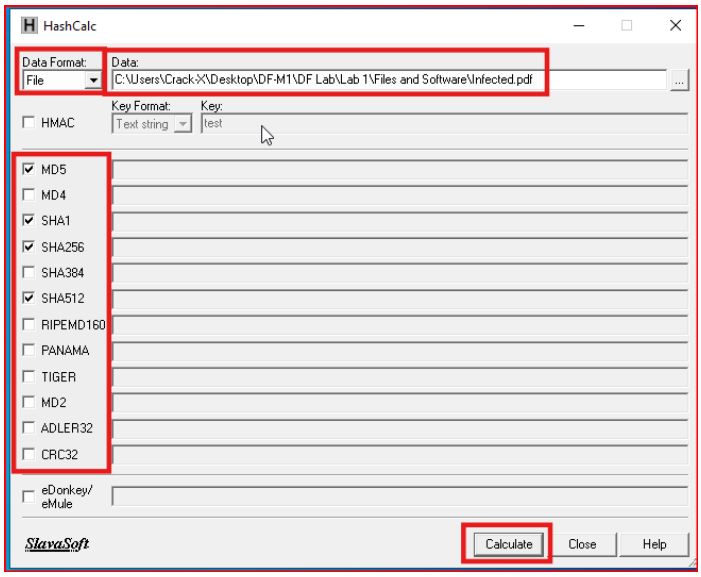


Phase 4: Hash Infected.pdf + VirusTotal Check

1. Set **Data Format** back to **File**.
2. Browse and select Infected.pdf.



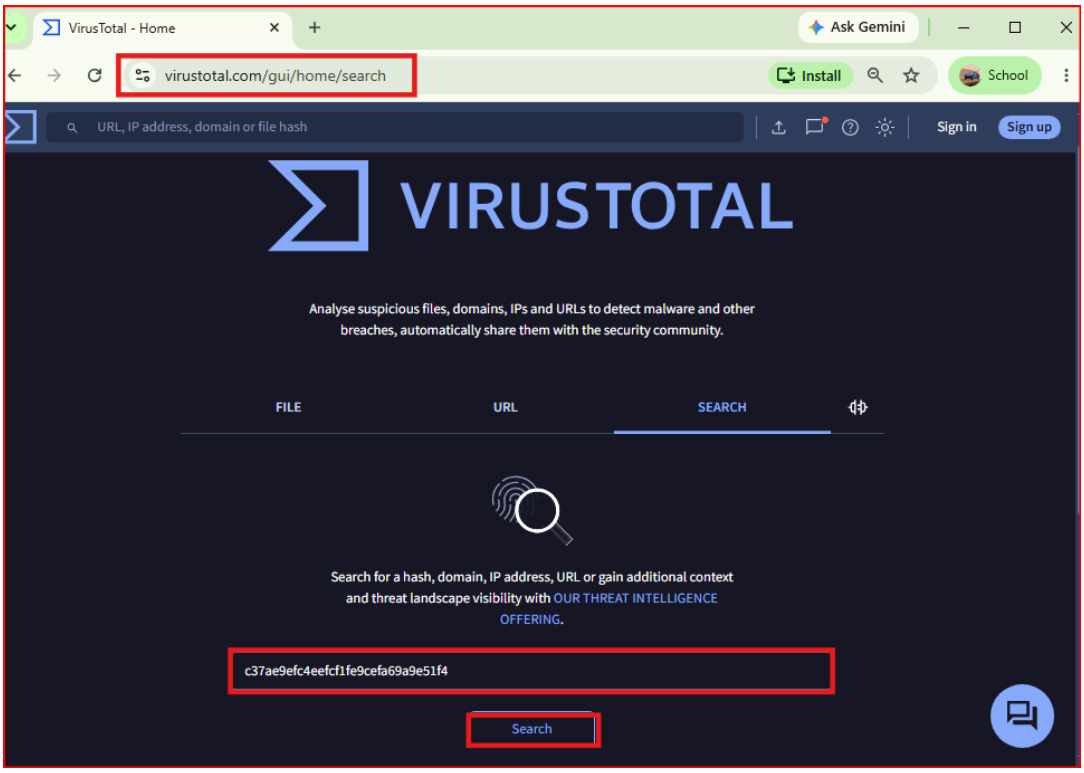
3. Select at least **MD5**.
4. Click **Calculate**.



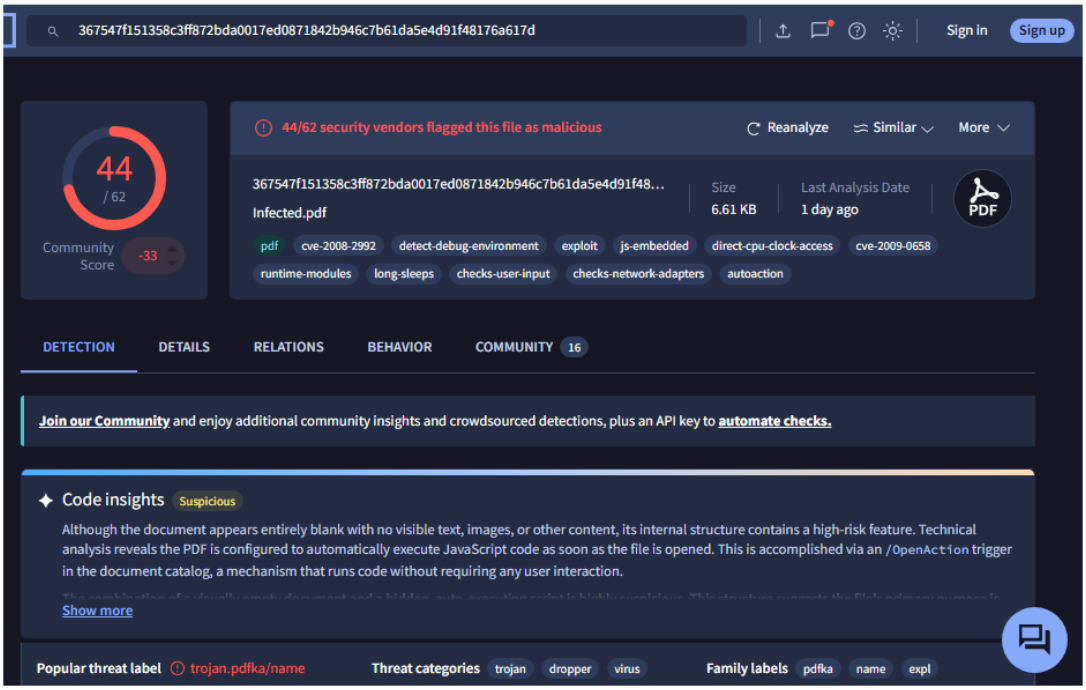
5. Copy the **MD5** hash value.



6. Open browser and go to: https://www.virustotal.com
7. Paste the MD5 hash in the search bar and press Enter.



8. Virustotal Output



Phase 5: Integrity Verification

Goal:

Prove the integrity of digital evidence using cryptographic hashing, exactly as done in real investigations.

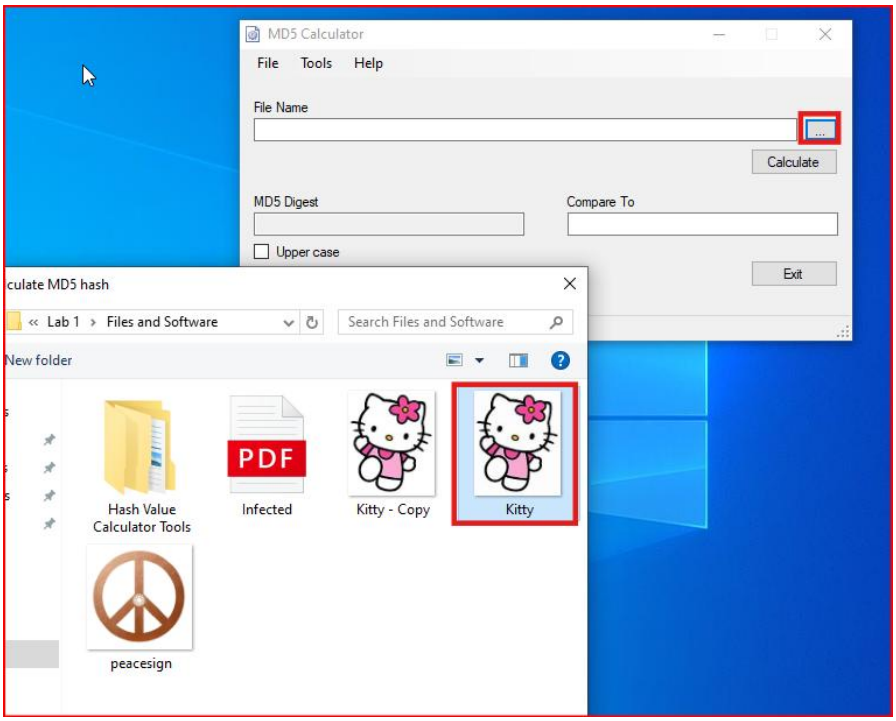
Before using MD5 Calculator, make sure .NET Framework is installed on your system.

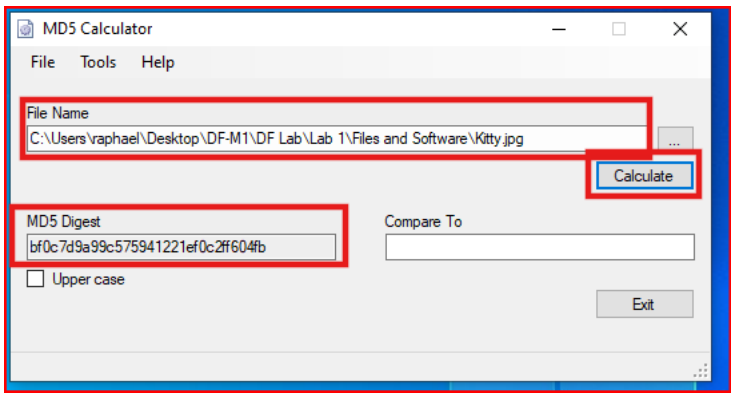
Without .NET Framework, MD5 Calculator will not run properly.

Steps:

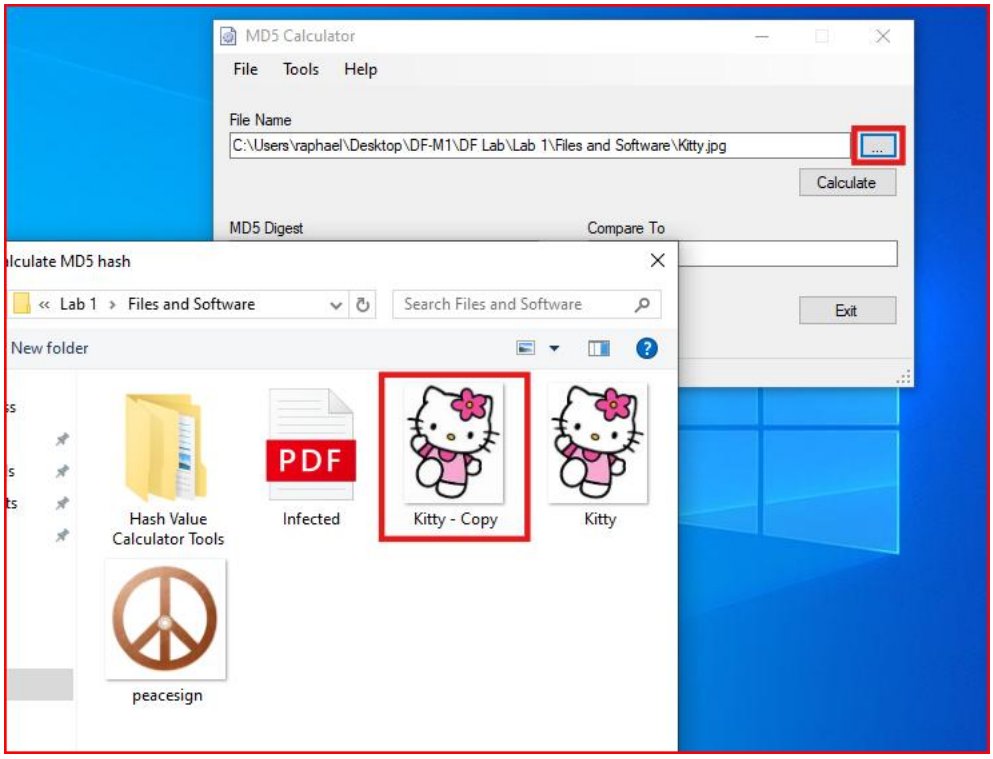
1. Open **MD5 Calculator**.

2. Calculate the MD5 hash of the original Kitty.jpg and record the value.

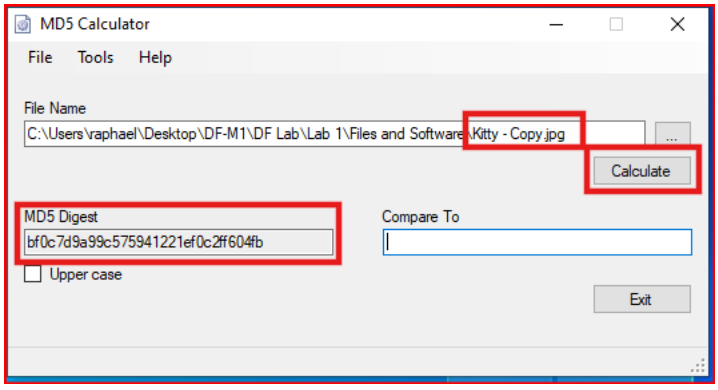




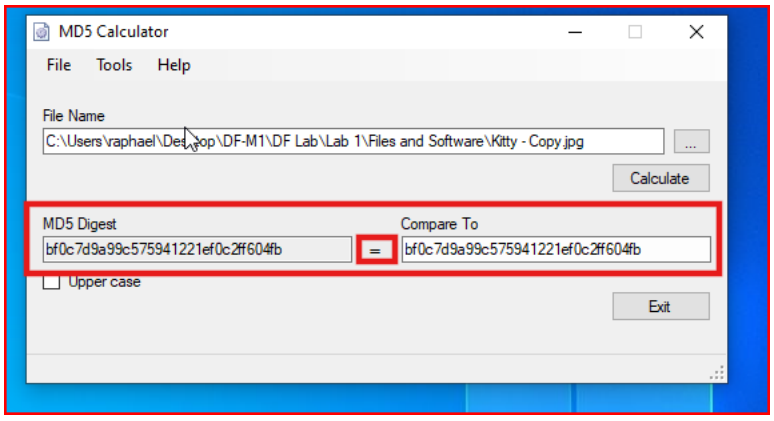
3. Create an exact copy of the file and name it Kitty-Copy.jpg.



4. Calculate the MD5 hash of Kitty_Copy.jpg.

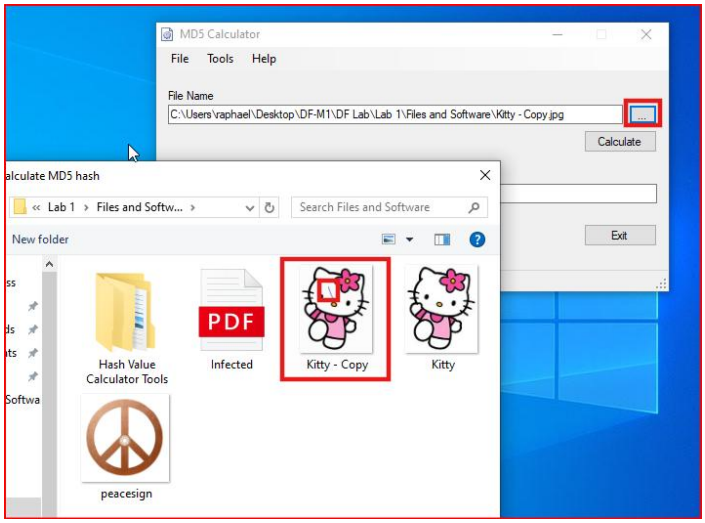


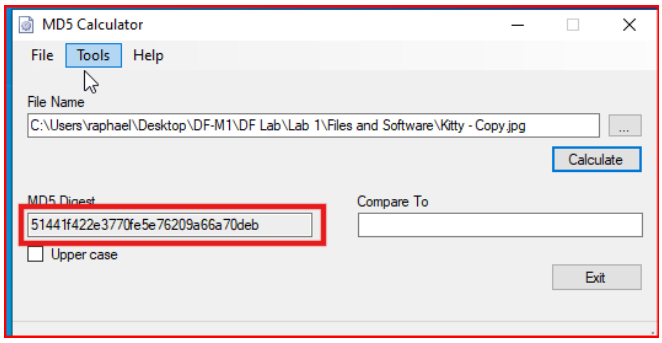
5. Compare both hashes side by side.
   Both hashes must be identical. This confirms that the file has not been altered and the hashing tool is working correctly.



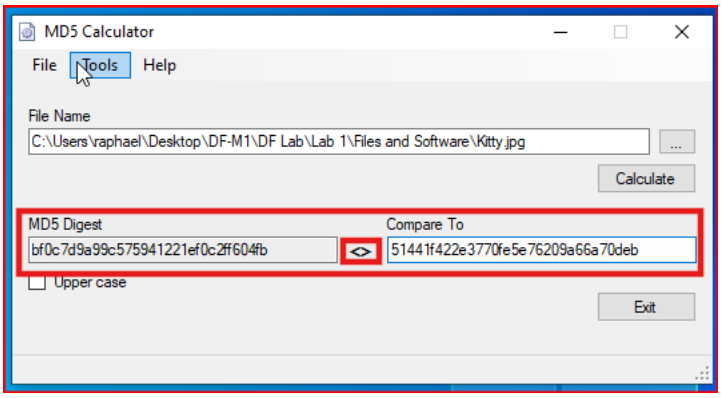
6. Now modify Kitty_Copy.jpg slightly (open in Paint → add a small mark → Save).

7. Calculate the MD5 hash of the modified file.





8. Compare the new hash with the original hash.
   The hash will be completely different. This demonstrates that even a minor change destroys the previous hash value and proves the file's integrity has been compromised.



Lab 2 Conclusion

Hash values were successfully calculated for all key evidence files using HashCalc across MD5, SHA-1, SHA-256, and SHA-512 algorithms. All baseline values were recorded and retained as part of the permanent case record.

The critical outcome of Lab 2 was the identification of malware within the recovered evidence. Infected.pdf was submitted to VirusTotal and identified as trojan.pdfka by 44 of 62 antivirus vendors — a detection rate of 71 percent. This eliminates any reasonable possibility of a false positive and confirms that malicious content was deliberately present on the subject device. This finding is carried forward as the highest priority investigative item in this investigation.

Hash Calculation Summary — Lab 2

File	Algorithms Applied	Result
Kitty.jpg — Original	MD5, SHA-1, SHA-256, SHA-512	Baseline recorded
Kitty-Copy.jpg — Unmodified	MD5, SHA-1, SHA-256, SHA-512	Matched baseline exactly
Kitty-Copy.jpg — Modified	MD5, SHA-1, SHA-256, SHA-512	Completely different — modification confirmed
Infected.pdf	MD5, SHA-256	Recorded — submitted to VirusTotal

Malware Detection Result — Lab 2

Field	Detail
File Name	Infected.pdf
Threat Classification	trojan.pdfka
Tool Used	VirusTotal
Detection Rate	44 of 62 antivirus vendors — 71 percent
Severity	Critical
Action Taken	File isolated and flagged for priority investigation

Lab 3 Conclusion

The dedicated MD5 tool was used to perform a structured before and after integrity comparison across key evidence files. Unmodified files returned hash values identical to the Lab 2 baselines, confirming that the recovered evidence has remained intact and untampered with since collection. The controlled modification test produced a completely different MD5 value, confirming that any change to a file — however minor — is immediately and unambiguously detectable.

Integrity Verification Results — Lab 3

File	Tool Used	Status	Result
Kitty.jpg — Original	MD5 Tool	Unmodified	Matched Lab 2 baseline exactly
Kitty-Copy.jpg — Unmodified	MD5 Tool	Unmodified	Matched Lab 2 baseline exactly
Kitty-Copy.jpg — Modified	MD5 Tool	Modified	Completely different — change immediately detected

Evidence Status at Close of Labs 2 and 3

Evidence Item	Description	Status
EV-001	Original 32 GB Samsung USB drive	Preserved — unmodified
EV-002	Recovered files, approximately 2 GB	Integrity verified through hash comparison

Chain of custody remained unbroken across both laboratories. All hash values, malware detection results, and integrity verification outcomes have been recorded and retained as part of the permanent case record.

LAB 4 — ADVANCED METADATA AND FILE ANALYSIS

Lab Overview

Field	Detail
Lab Number	4
Lab Title	Advanced Metadata and File Analysis
Case Reference	CHFI_Module2_Lab1-6_Case001
Examiner	Abdul Wahid
Date Conducted	13 September 2026
Investigation Phase	Examination and Analysis

Purpose

File recovery and integrity verification confirm that evidence exists and is intact. They do not tell you what that evidence means. This laboratory moves the investigation forward by conducting a deep forensic examination of the recovered files — treating each file not just as data but as a potential source of investigative intelligence.

As a junior digital forensic examiner, the task in this laboratory is to look beyond the surface of each file. A file's name and extension tell you very little. Its internal signature, embedded metadata, archive contents, system artefacts, and behaviour within a forensic platform tell you significantly more. Every anomaly identified, every lead uncovered, and every inconsistency documented in this lab has the potential to advance the investigation in a meaningful direction.

All work in this laboratory was performed exclusively on working copies. The original recovered files were never modified, accessed directly, or moved from their designated evidence location.

Objectives

- Verify the authenticity of recovered files by examining their internal file signatures using HxD Hex Editor
- Extract and interpret metadata from recovered files using exiftool to identify creation details, authorship, timestamps, and anomalies
- Inspect recovered archive files using 7-Zip to examine their contents and identify any timeline irregularities
- Review system artefacts including registry entries and event logs using Windows Registry Editor
- Conduct a comprehensive forensic case analysis using Autopsy v4.23.1 to identify suspicious items, keyword matches, and investigative leads
- Document all findings professionally and flag items of investigative significance

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Folder Structure

Lab-4\

├── Evidence-Files\ ← Copy of all recovered files from Lab 1

├── Analysis-Workspace\ ← Working copies used for examination

├── Tool-Outputs\ ← All tool results saved here

├── Screenshots\ ← Supporting visual documentation

└── Reports\ ← Professional findings report

Tools and Evidence

Tool	Version	Purpose
HxD Hex Editor	Latest	File signature verification
exiftool	13.59	Metadata extraction and analysis
7-Zip	Latest	Archive inspection and contents review
Windows Registry Editor	Built-in	System artefact examination
Autopsy	4.23.1	Comprehensive forensic case analysis

Evidence Item	Description	Status at Start of Lab
EV-002	Recovered files, approximately 2 GB	Verified — integrity confirmed in Lab 2
EV-003	Workspace copy for Lab 4 analysis	Created at start of this lab

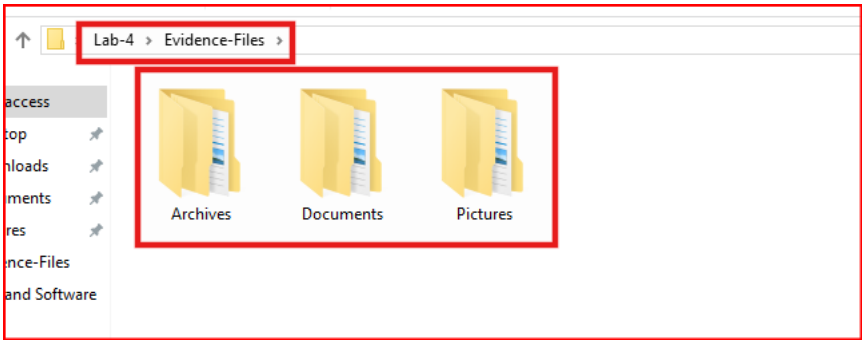
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Real World Relevance

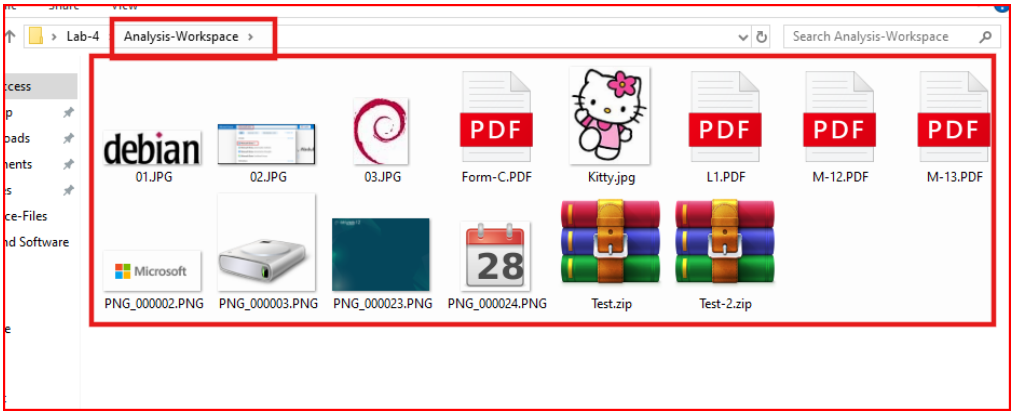
In a live investigation, metadata and file analysis frequently produces the most actionable leads. A document's author field can name a suspect. A file's creation timestamp can place a person at a device at a specific time. A mismatched file signature can reveal that a file has been deliberately disguised. Archive contents can expose a timeline of activity spanning months or years. Registry entries can show which devices were connected and when. Every tool applied in this laboratory reflects techniques used daily by professional DFIR examiners and law enforcement forensic units.

PHASE 1: Case Preparation & Evidence Handling

1. Create the folder structure above.
2. Copy (do not move) selected files from Lab 1 into Evidence-Files.

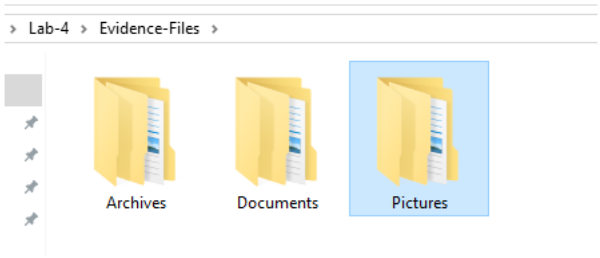


3. Make working copies in Analysis-Workspace.



4. Create a simple Evidence Log:

Item ID	File/Folder Name	Source	Date Copied	Examiner
EV-001	Pictures	Lab 1 Recovered	9-12-2026	Abdul Wahid
EV-002	Documents	Lab 1 Recovered	9-12-2026	Abdul Wahid
EV-003	Archives	Lab 1 Recovered	9-12-2026	Abdul Wahid

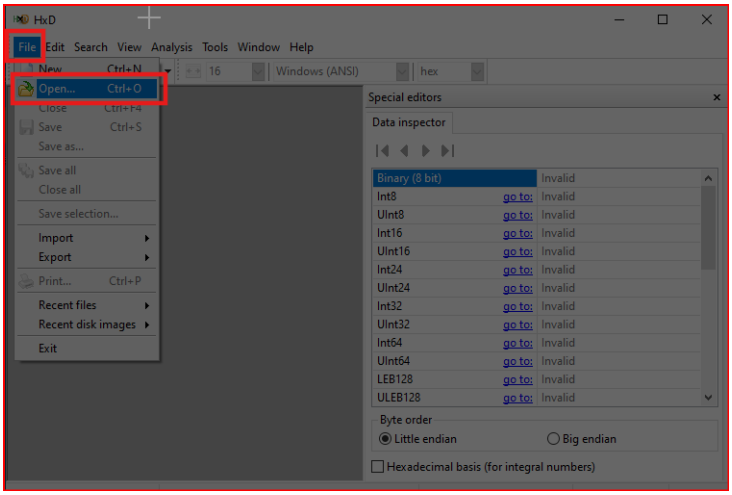


PHASE 2: Deep File Signature Analysis (HxD)

Do not stop at the first few bytes.

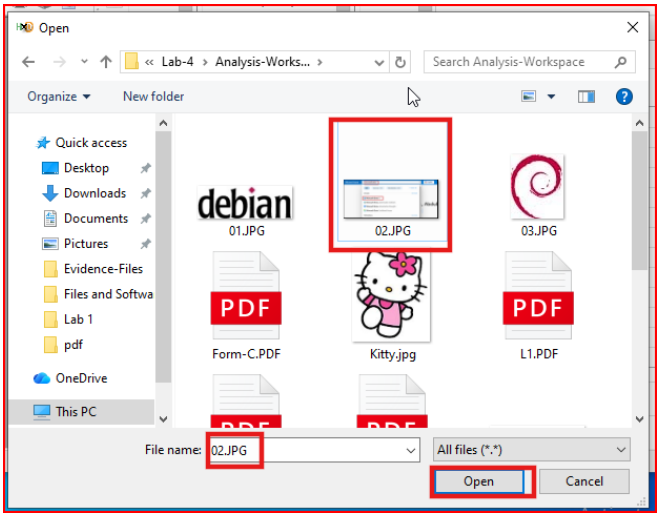
For each major file type (JPG, PNG, PDF, ZIP, DOCX):

1. **Open the file in HxD.**



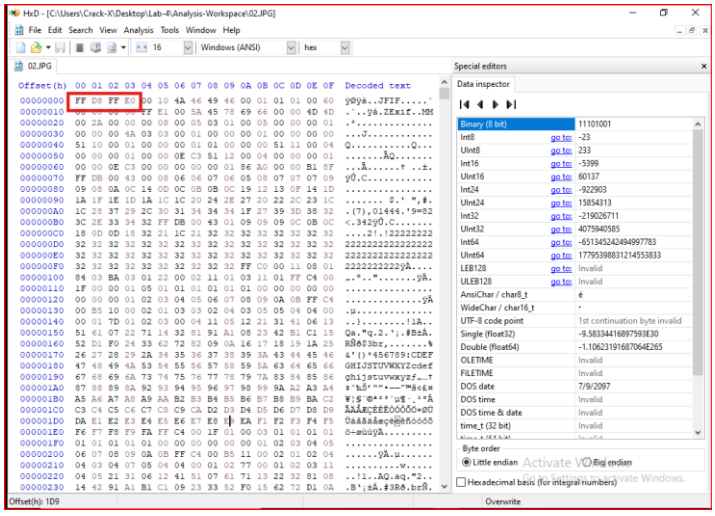
2. **Images (Jpg)**

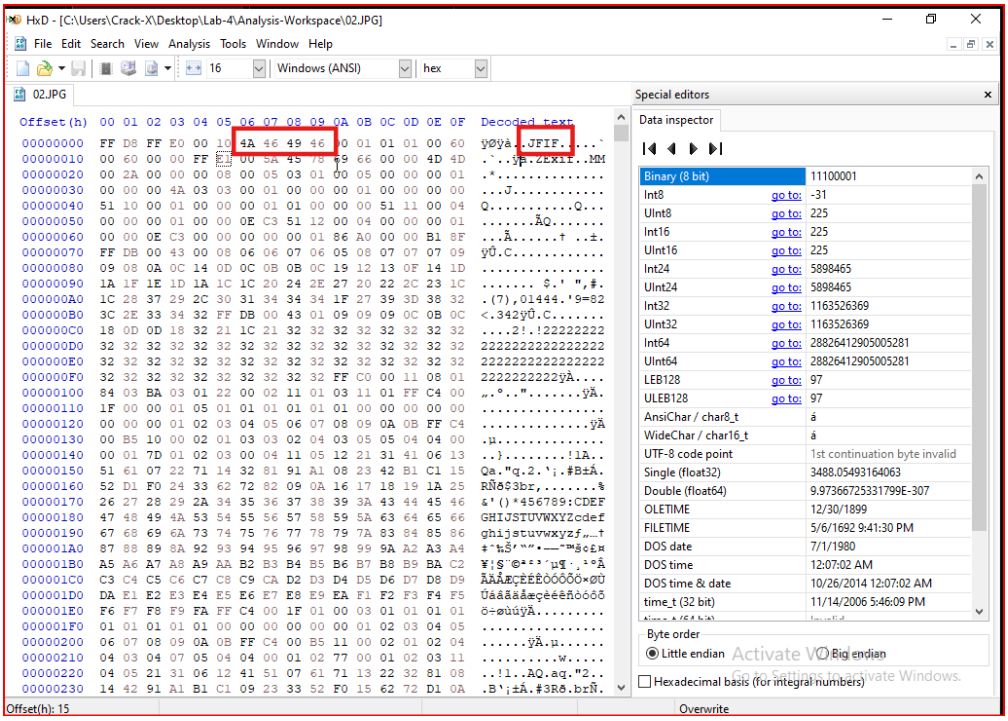
Deep Analysis of JPG in HxD



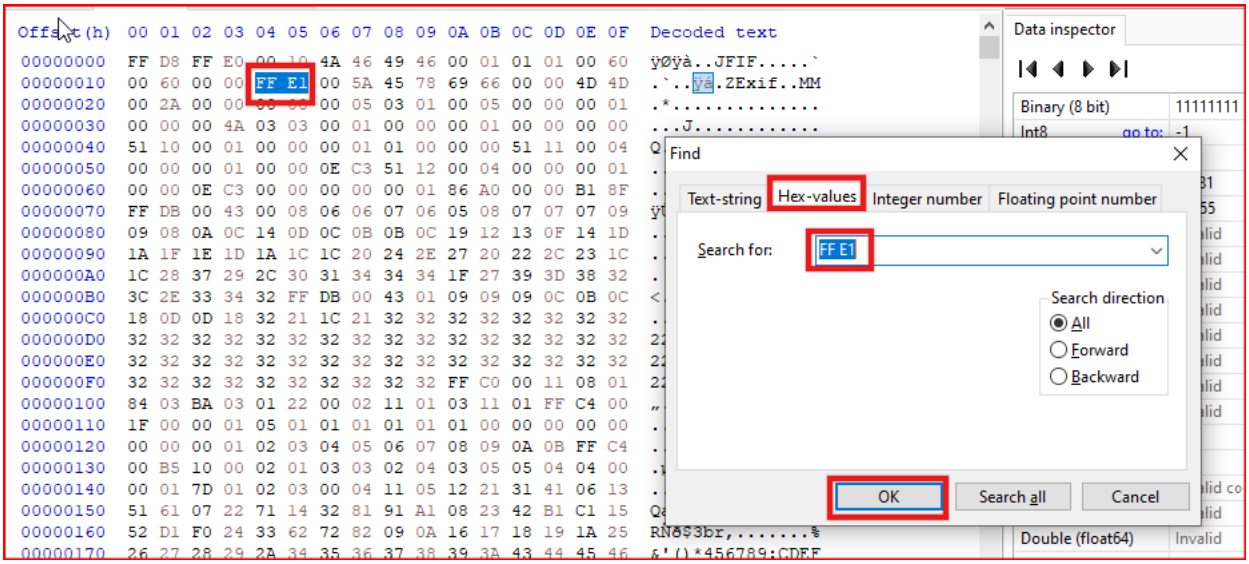
A. Signature & Header

- Must start with: FF D8 FF
- Common next bytes: FF E0 (JFIF) or FF E1 (EXIF)



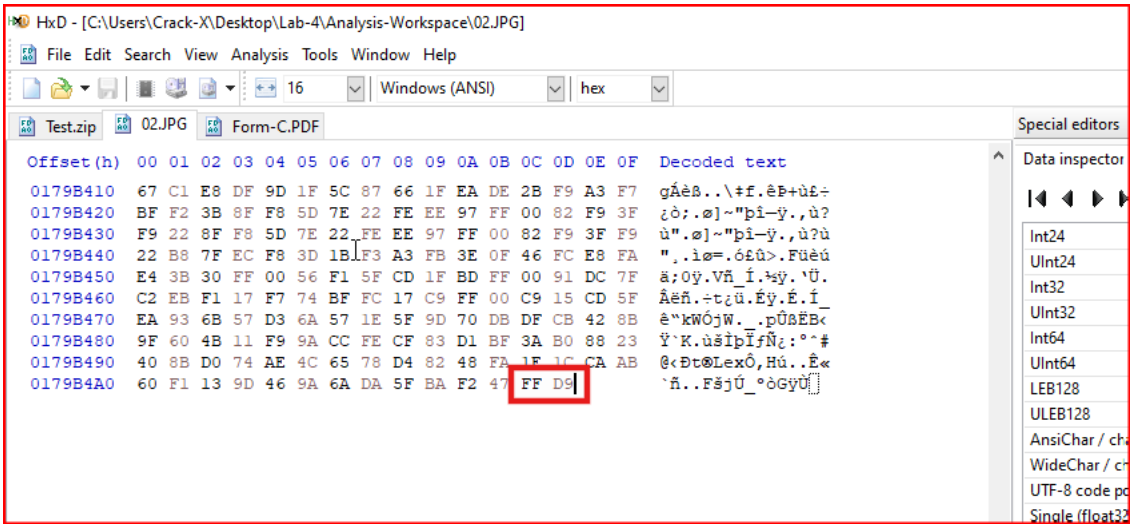


Search for FF E1 → this often contains camera, date, GPS data.

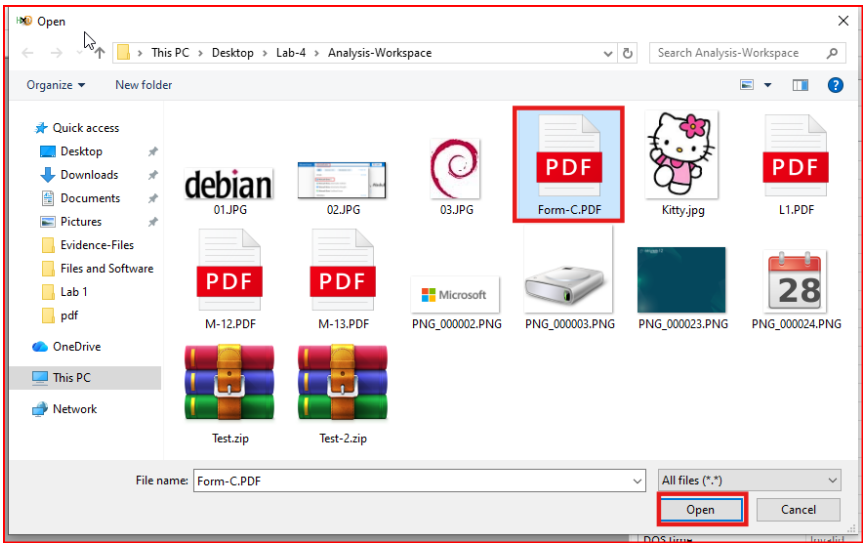


Look near the end of the file for FF D9 (End of Image).

Check if there is any data **after** FF D9 (this can be extra/hidden data).



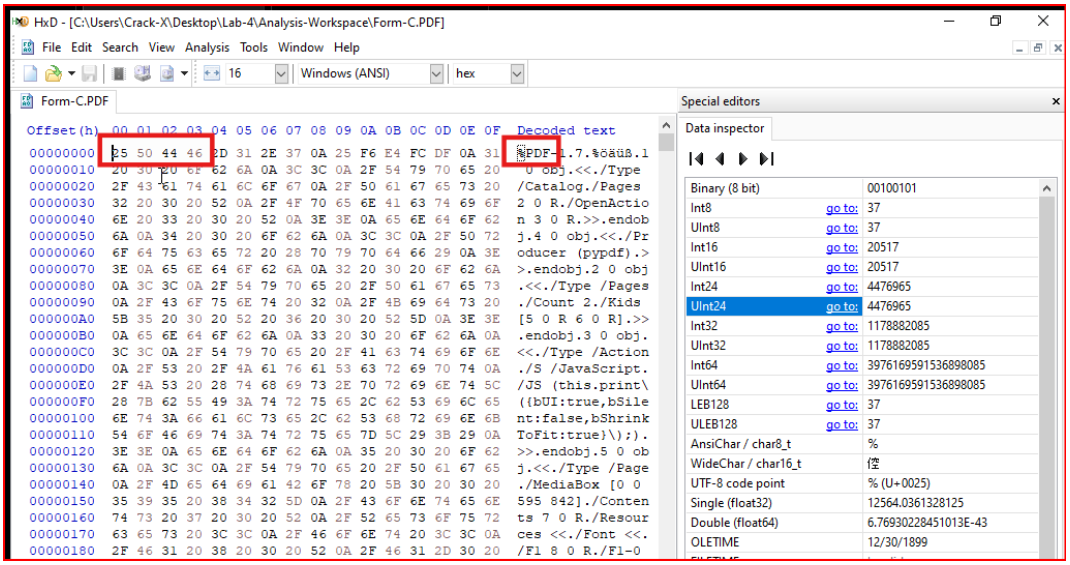
3. **PDF file (Form-C)**

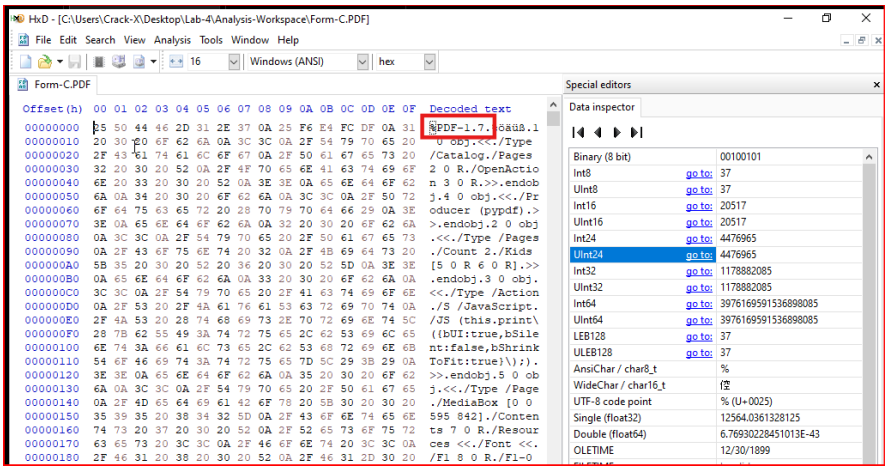


Deep Analysis of PDF in HxD

A. Signature

- **Must start with: 25 50 44 46 → %PDF**
- **Followed by version (e.g. %PDF-1.7)**



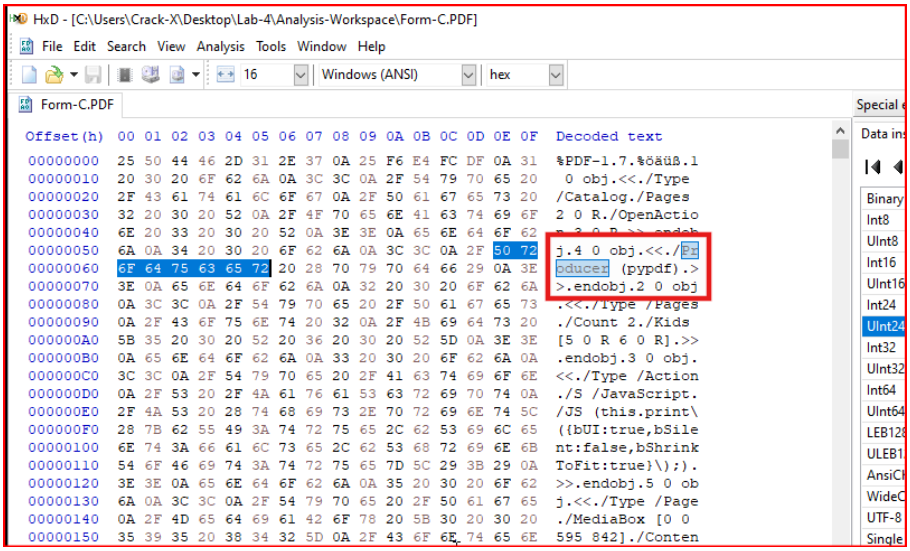


B. Important things to examine

Item	What to look for	Meaning
Version	%PDF-1.x	PDF version
Producer	/Producer	Software that created the PDF
Creator	/Creator	Original application
Title	/Title	Document title
Author	/Author	Author name
Creation date	/CreationDate	When it was created
Mod date	/ModDate	Last modification date
EOF marker	%%EOF	End of file

Let's check if we found anything here in this Form-C

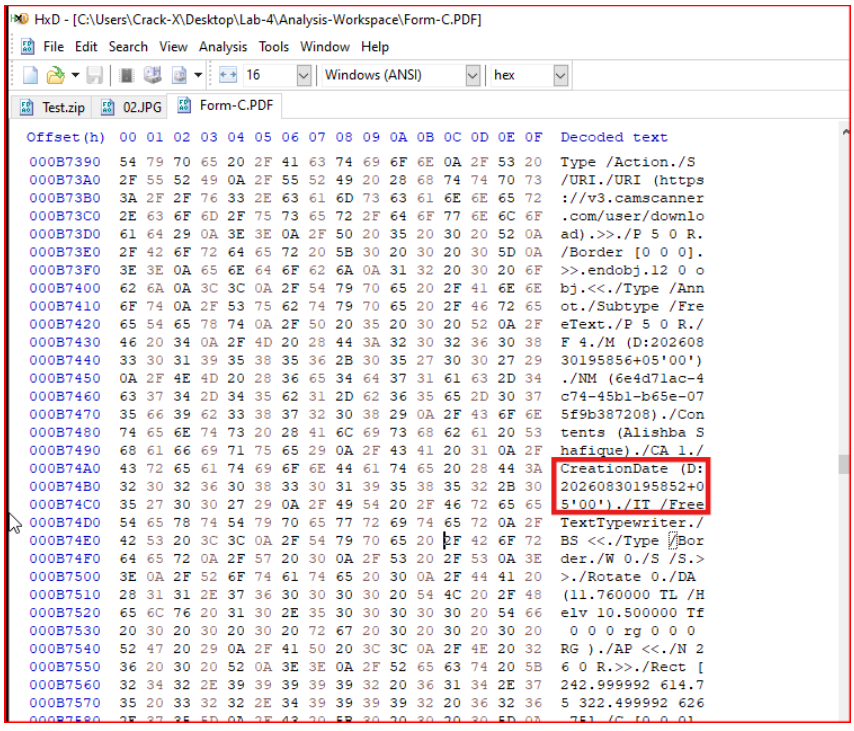
Producer=pypdf



NOTE:

pypdf is a Python library used to create/modify PDFs. This tells you the tool that generated or processed this PDF.

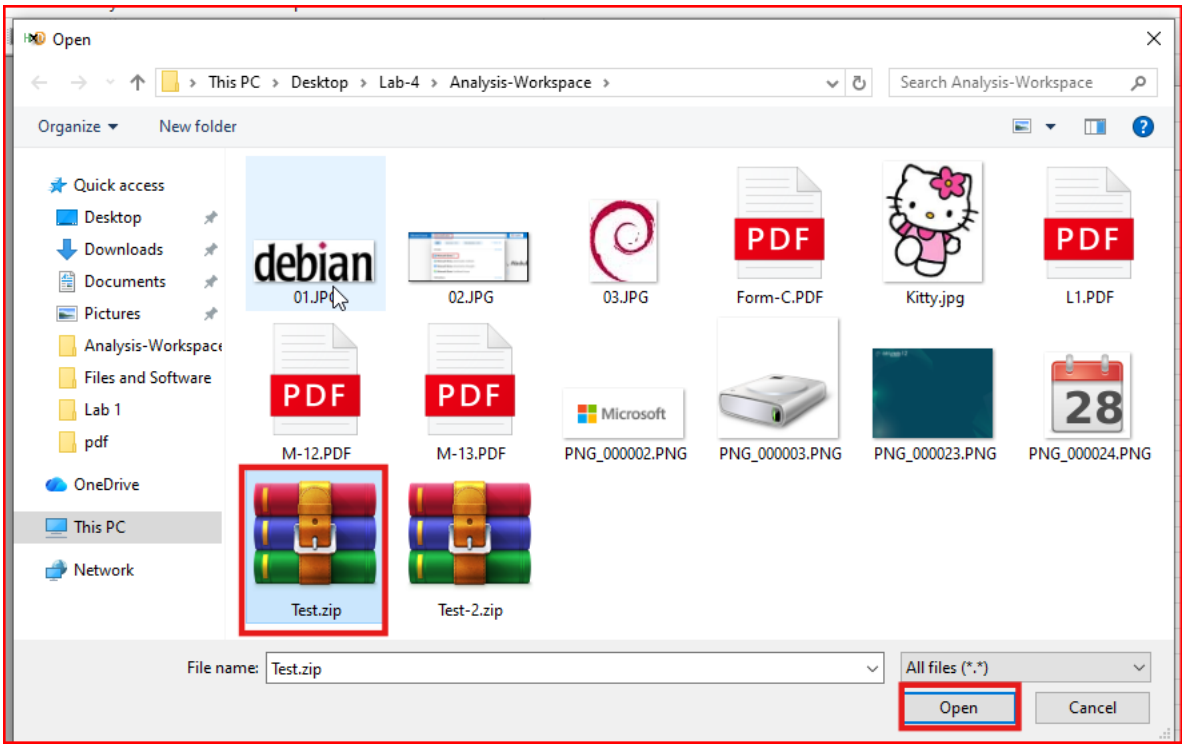
CreationDate



%%EOF = END of File

```
0016A760  5B 3C 38 41 36 45 30 45 38 38 34 30 45 34 37 46  [<8A6E0E8840E47F
0016A770  32 41 37 38 42 39 42 30 37 33 33 32 46 33 41 45  2A78B9B07332F3AE
0016A780  42 46 3E 20 3C 38 41 36 45 30 45 38 38 34 30 45  BF> <8A6E0E8840E
0016A790  34 37 46 32 41 37 38 42 39 42 30 37 33 33 32 46  47F2A78B9B07332F
0016A7A0  33 41 45 42 46 3E 5D 0A 3E 3E 0A 73 74 61 72 74  3AEBF>].>>.start
0016A7B0  78 72 65 66 0A 31 34 38 32 31 34 30 0A 25 25 45  xref.1482140.%%E
0016A7C0  4F 46 0A 00                                      OF..
```

4. Zip File (Test)



1. Signature

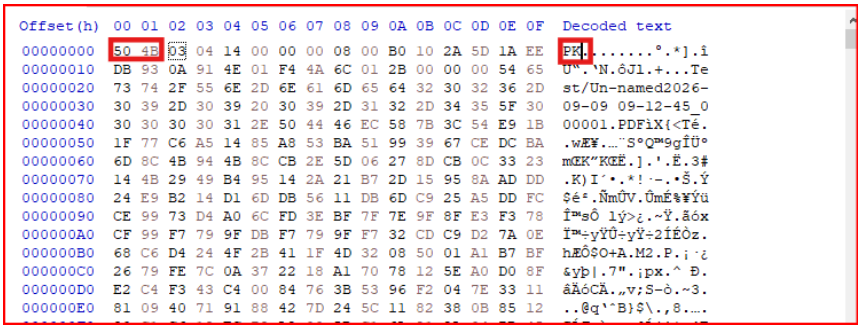
Look at the first 4 bytes:

text

50 4B 03 04

- 50 4B = “PK” (Phil Katz – creator of ZIP)
- 03 04 = Local File Header

Result: Valid ZIP file

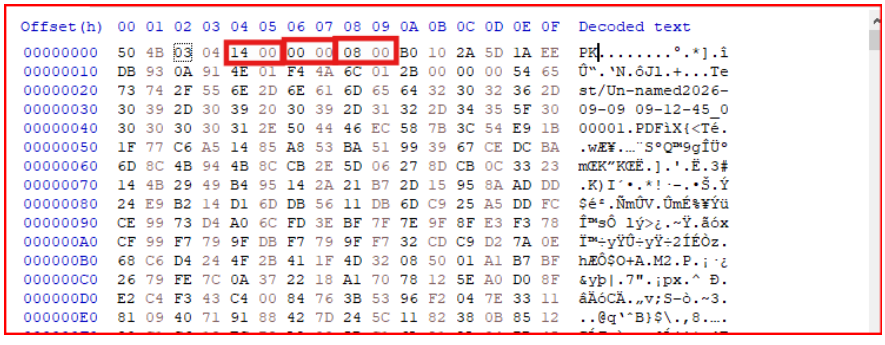


Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	50	4B	03	04	14	00	00	00	08	00	B0	10	2A	5D	1A	EE	PK.....°.*].î
00000010	DB	93	0A	91	4E	01	F4	4A	6C	01	2B	00	00	00	54	65	Û“.‘N.ôJl.+...Te
00000020	73	74	2F	55	6E	2D	6E	61	6D	65	64	32	30	32	36	2D	st/Un-named2026-
00000030	30	39	2D	30	39	20	30	39	2D	31	32	2D	34	35	5F	30	09-09 09-12-45_0
00000040	30	30	30	30	31	2E	50	44	46	EC	58	7B	3C	54	E9	1B	00001.PDFìX{<Té.
00000050	1F	77	C6	A5	14	85	A8	53	BA	51	99	39	67	CE	DC	BA	.wÆ¥.…¨S°Q™9gÎÜ°
00000060	6D	8C	4B	94	4B	8C	CB	2E	5D	06	27	8D	CB	0C	33	23	mŒK”KŒË.].'.Ë.3#
00000070	14	4B	29	49	B4	95	14	2A	21	B7	2D	15	95	8A	AD	DD	.K)I´•.*!·-.•Š.Ý
00000080	24	E9	B2	14	D1	6D	DB	56	11	DB	6D	C9	25	A5	DD	FC	$é².ÑmÛV.ÛmÉ%¥Ýü
00000090	CE	99	73	D4	A0	6C	FD	3E	BF	7F	7E	9F	8F	E3	F3	78	Î™sÔ lý>¿.~Ÿ.ãóx
000000A0	CF	99	F7	79	9F	DB	F7	79	9F	F7	32	CD	C9	D2	7A	0E	Ï™÷yŸÛ÷yŸ÷2ÍÉÒz.
000000B0	68	C6	D4	24	4F	2B	41	1F	4D	32	08	50	01	A1	B7	BF	hÆÔ$O+A.M2.P.¡·¿
000000C0	26	79	FE	7C	0A	37	22	18	A1	70	12	5E	A0	D0	8F		&yþ|.7".¡px.^ Ð.
000000D0	E2	C4	F3	43	C4	00	84	76	3B	53	96	F2	04	7E	33	11	âÄóCÄ.„v;S–ò.~3.
000000E0	81	09	40	71	91	88	42	7D	24	5C	11	82	38	0B	85	12	..@q‘ˆB}$\.‚8.….

What else to examine in the header

Right after the signature, the ZIP header contains:

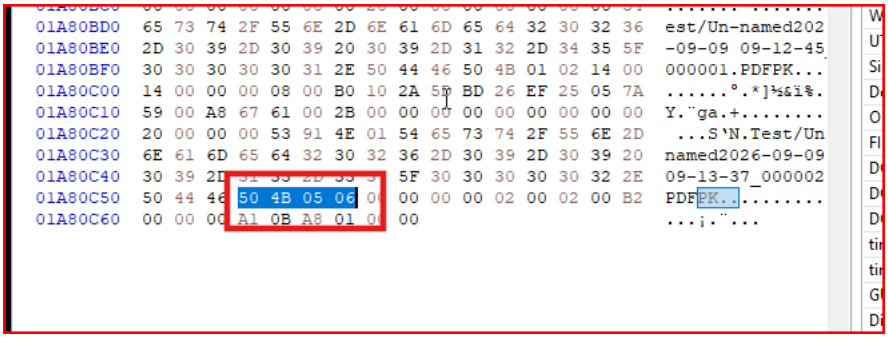
Offset	Meaning	What you can see
04-05	Version needed	Usually 14 00
06-07	General purpose flag	Encryption, etc.
08-09	Compression method	00 00 = Stored, 08 00 = Deflate

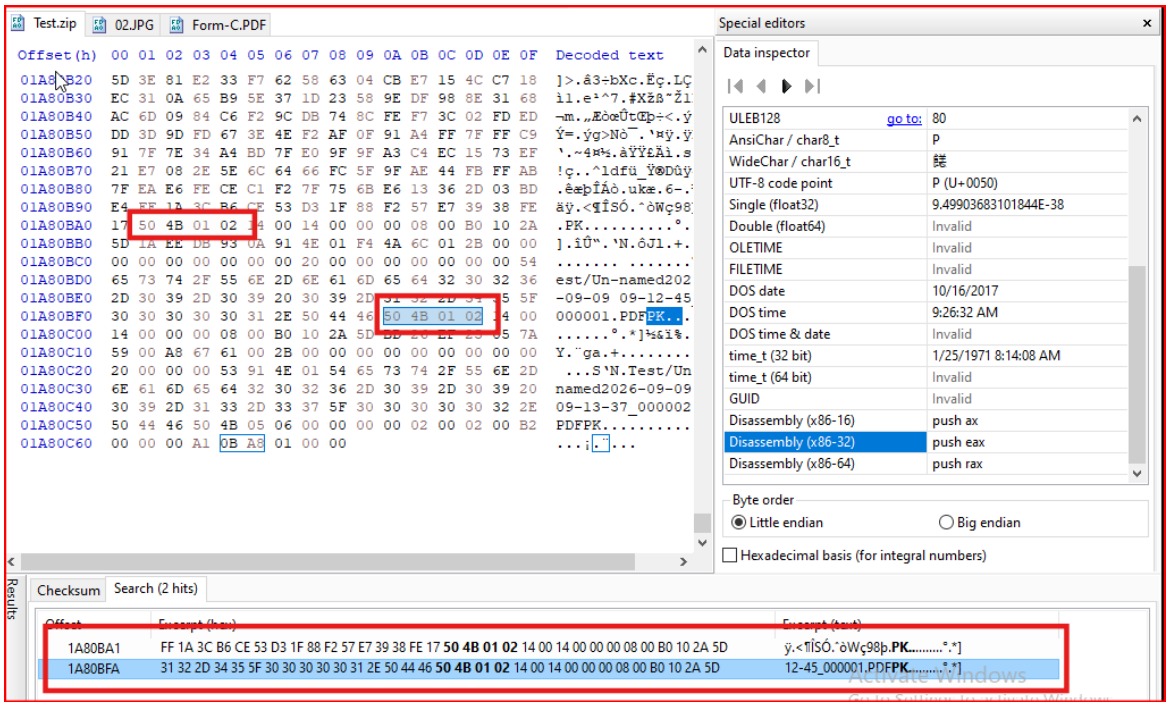


Offset(h)	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F	Decoded text
00000000	50	4B	03	04	14	00	00	00	08	00	B0	10	2A	5D	1A	EE	PK.....°.*].î
00000010	DB	93	0A	91	4E	01	F4	4A	6C	01	2B	00	00	00	54	65	Û“.‘N.ôJl.+...Te
00000020	73	74	2F	55	6E	2D	6E	61	6D	65	64	32	30	32	36	2D	st/Un-named2026-
00000030	30	39	2D	30	39	20	30	39	2D	31	32	2D	34	35	5F	30	09-09 09-12-45_0
00000040	30	30	30	30	31	2E	50	44	46	EC	58	7B	3C	54	E9	1B	00001.PDFìX{<Té.
00000050	1F	77	C6	A5	14	85	A8	53	BA	51	99	39	67	CE	DC	BA	.wÆ¥.…¨S°Q™9gÎÜ°
00000060	6D	8C	4B	94	4B	8C	CB	2E	5D	06	27	8D	CB	0C	33	23	mŒK”KŒË.].'.Ë.3#
00000070	14	4B	29	49	B4	95	14	2A	21	B7	2D	15	95	8A	AD	DD	.K)I´•.*!·-.•Š.Ý
00000080	24	E9	B2	14	D1	6D	DB	56	11	DB	6D	C9	25	A5	DD	FC	$é².ÑmÛV.ÛmÉ%¥Ýü
00000090	CE	99	73	D4	A0	6C	FD	3E	BF	7F	7E	9F	8F	E3	F3	78	Î™sÔ lý>¿.~Ÿ.ãóx
000000A0	CF	99	F7	79	9F	DB	F7	79	9F	F7	32	CD	C9	D2	7A	0E	Ï™÷yŸÛ÷yŸ÷2ÍÉÒz.
000000B0	68	C6	D4	24	4F	2B	41	1F	4D	32	08	50	01	A1	B7	BF	hÆÔ$O+A.M2.P.¡·¿
000000C0	26	79	FE	7C	0A	37	22	18	A1	70	12	5E	A0	D0	8F		&yþ|.7".¡px.^ Ð.
000000D0	E2	C4	F3	43	C4	00	84	76	3B	53	96	F2	04	7E	33	11	âÄóCÄ.„v;S–ò.~3.
000000E0	81	09	40	71	91	88	42	7D	24	5C	11	82	38	0B	85	12	..@q‘ˆB}$\.‚8.….

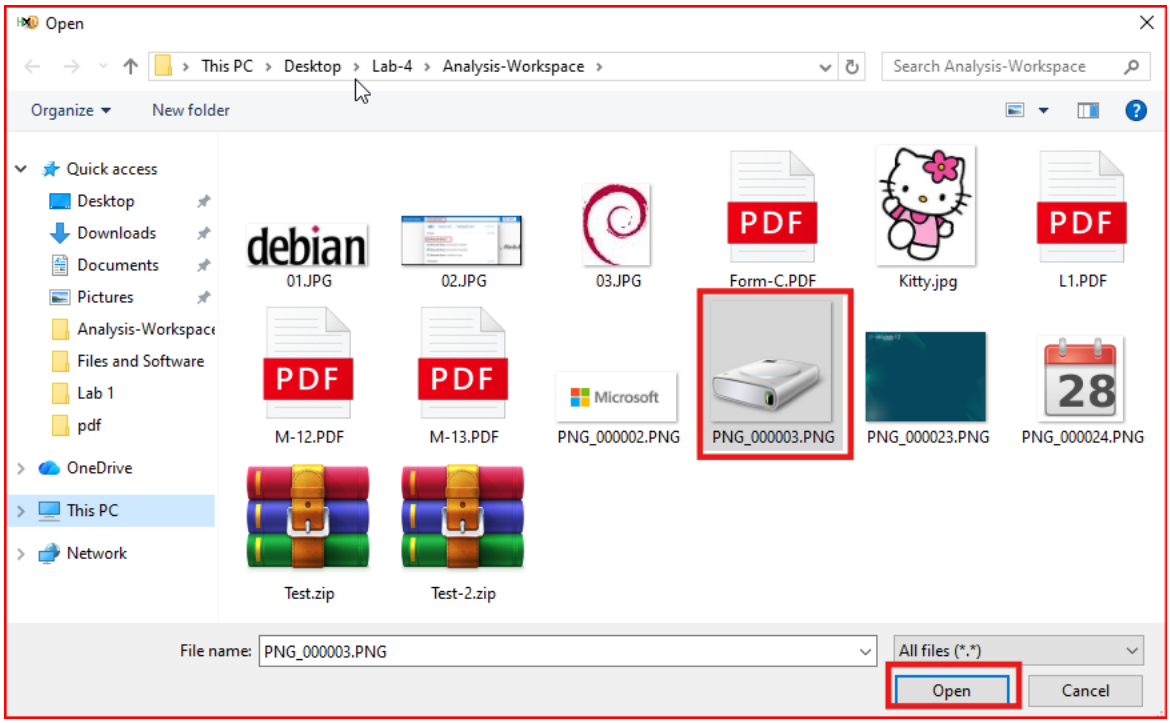
Look near the end of the file for:

- 50 4B 01 02 → Central Directory Header
- 50 4B 05 06 → End of Central Directory





5. PNG (PNG_000003.png)



Deep Analysis of PNG in HxD

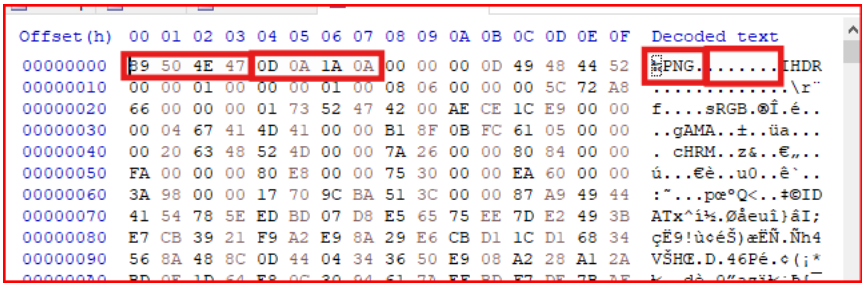
1. Signature

Look at the first 8 bytes:

89 50 4E 47 0D 0A 1A 0A

- 89 50 4E 47 = ".PNG"
- 0D 0A 1A 0A = standard PNG line endings

Result: Valid PNG



2. Important Chunks to Examine

PNG files are made of chunks. Look for these (search in Hex or ASCII):

Chunk	Hex	Meaning
IHDR	49 48 44 52	Image Header (width, height, bit depth, color type)
IDAT	49 44 41 54	Image Data
IEND	49 45 4E 44	End of PNG file
tEXt	74 45 58 74	Textual metadata (Author, Title, etc.)
iTXt	69 54 58 74	International text metadata
zTXt	7A 54 58 74	Compressed text metadata
pHYs	70 48 59 73	Physical pixel dimensions
tIME	74 49 4D 45	Last modification time

Image Header

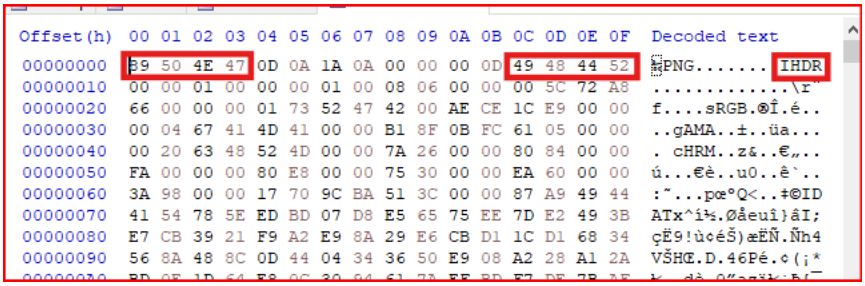
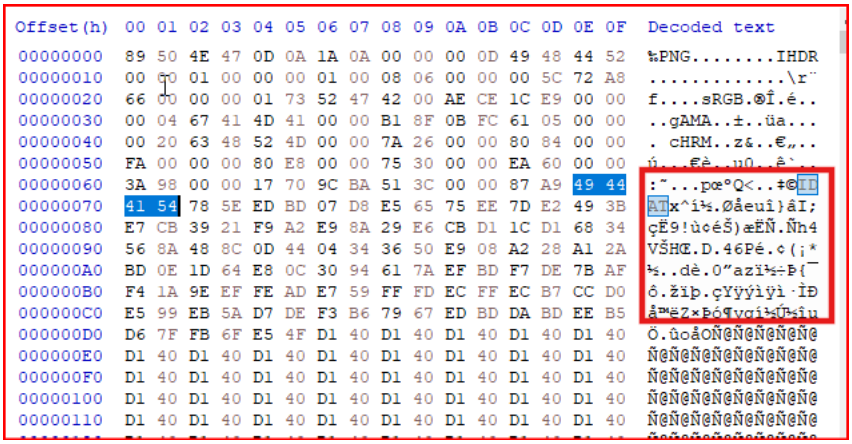
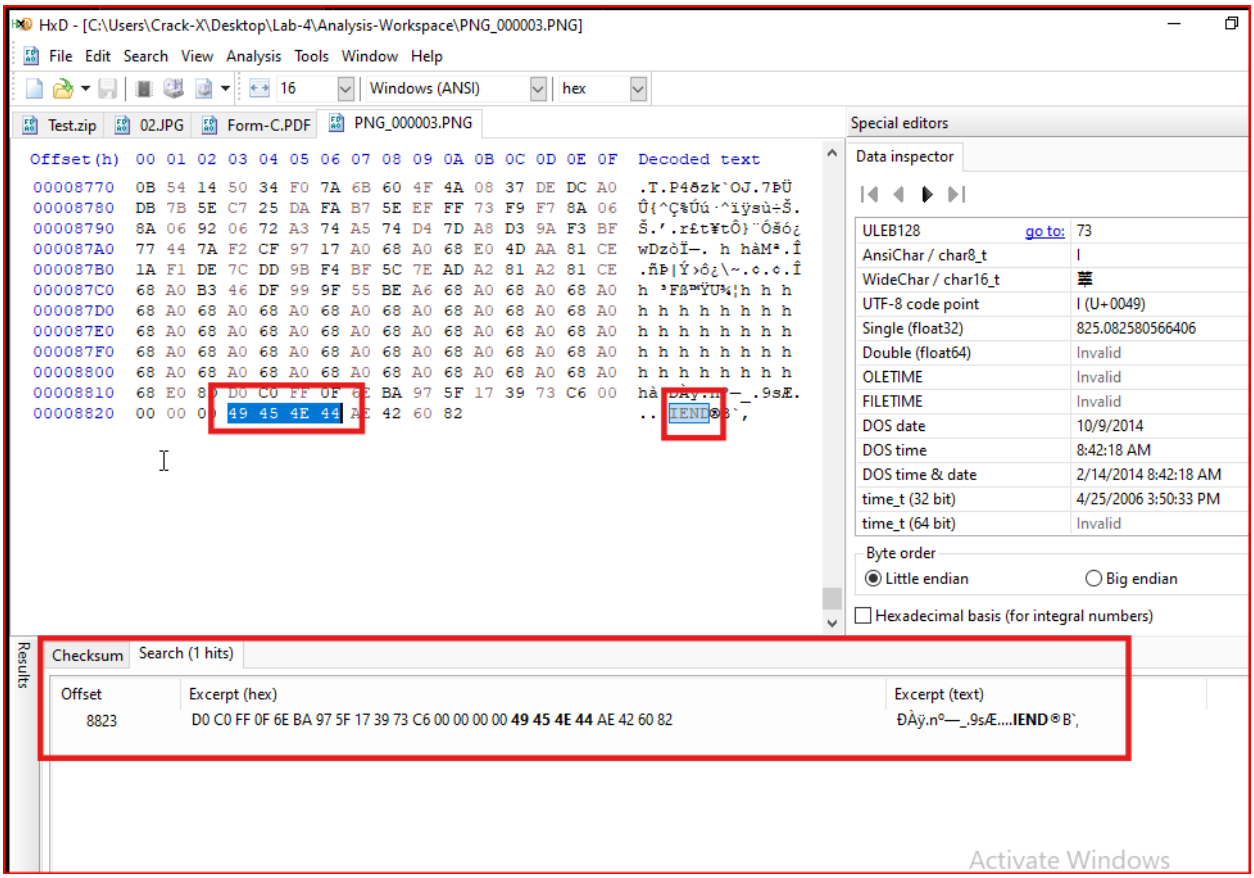


Image Data



End of File



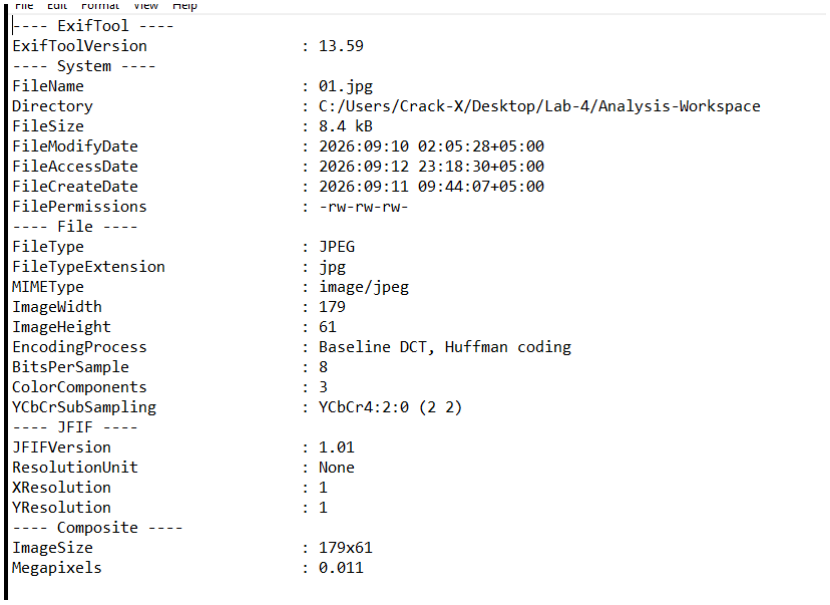
Output: Tool_Outputs\Signature_Analysis.txt

PHASE 3: Deep Metadata Extraction (exiftool)

Full metadata + structured output

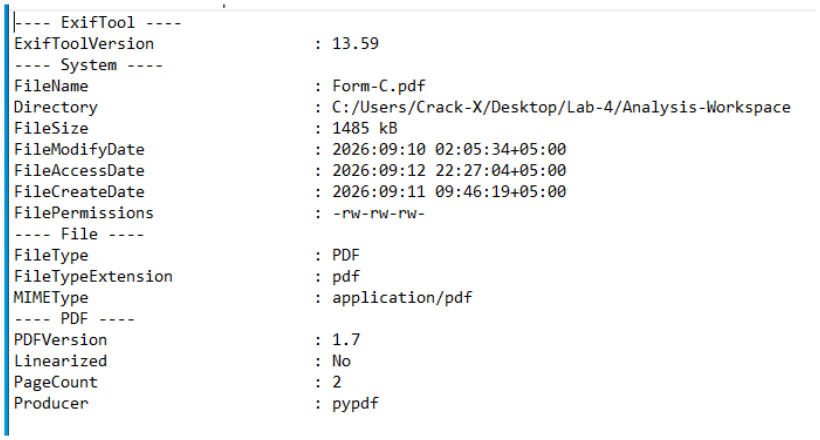
C:\ExifTool\exiftool.exe -a -u -g1 -s "C:\Users\Abdul\Pictures\photo.jpg" > "C:\Users\Abdul\Pictures\output.txt"

JPG



```
File  Edit  Format  View  Help
---- ExifTool ----
ExifToolVersion                 : 13.59
---- System ----
FileName                        : 01.jpg
Directory                       : C:/Users/Crack-X/Desktop/Lab-4/Analysis-Workspace
FileSize                        : 8.4 kB
FileModifyDate                  : 2026:09:10 02:05:28+05:00
FileAccessDate                  : 2026:09:12 23:18:30+05:00
FileCreateDate                  : 2026:09:11 09:44:07+05:00
FilePermissions                 : -rw-rw-rw-
---- File ----
FileType                        : JPEG
FileTypeExtension               : jpg
MIMEType                        : image/jpeg
ImageWidth                      : 179
ImageHeight                     : 61
EncodingProcess                 : Baseline DCT, Huffman coding
BitsPerSample                   : 8
ColorComponents                 : 3
YCbCrSubSampling                : YCbCr4:2:0 (2 2)
---- JFIF ----
JFIFVersion                     : 1.01
ResolutionUnit                  : None
XResolution                     : 1
YResolution                     : 1
---- Composite ----
ImageSize                       : 179x61
Megapixels                      : 0.011
```

PDF



```
---- ExifTool ----
ExifToolVersion                 : 13.59
---- System ----
FileName                        : Form-C.pdf
Directory                       : C:/Users/Crack-X/Desktop/Lab-4/Analysis-Workspace
FileSize                        : 1485 kB
FileModifyDate                  : 2026:09:10 02:05:34+05:00
FileAccessDate                  : 2026:09:12 22:27:04+05:00
FileCreateDate                  : 2026:09:11 09:46:19+05:00
FilePermissions                 : -rw-rw-rw-
---- File ----
FileType                        : PDF
FileTypeExtension               : pdf
MIMEType                        : application/pdf
---- PDF ----
PDFVersion                      : 1.7
Linearized                      : No
PageCount                       : 2
Producer                        : pypdf
```

Zip File

File Edit Format View Help

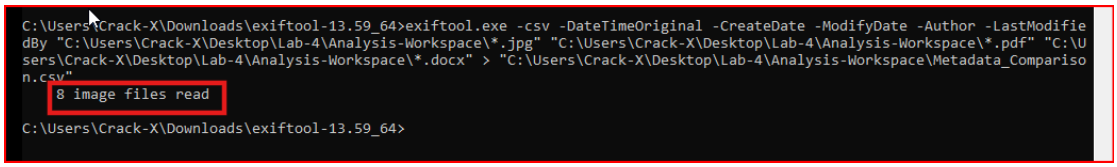
```
---- ExifTool ----
ExifToolVersion                 : 13.59
---- System ----
FileName                        : Test.zip
Directory                       : C:/Users/Crack-X/Desktop/Lab-4/Analysis-Workspace
FileSize                        : 28 MB
FileModifyDate                  : 2026:09:11 09:47:51+05:00
FileAccessDate                  : 2026:09:12 23:02:37+05:00
FileCreateDate                  : 2026:09:11 09:48:42+05:00
FilePermissions                 : -rw-rw-rw-
---- File ----
FileType                        : ZIP
FileTypeExtension               : zip
MIMEType                        : application/zip
---- ZIP ----
ZipRequiredVersion              : 20
ZipBitFlag                      : 0
ZipCompression                  : Deflated
ZipModifyDate                   : 2026:09:10 02:05:32
ZipCRC                          : 0x93dbee1a
ZipCompressedSize               : 21926154
ZipUncompressedSize             : 23874292
ZipFileName                     : Test/Un-named2026-09-09 09-12-45_000001.PDF
ZipRequiredVersion              : 20
ZipBitFlag                      : 0
ZipCompression                  : Deflated
ZipModifyDate                   : 2026:09:10 02:05:32
ZipCRC                          : 0x25ef26bd
ZipCompressedSize               : 5863941
ZipUncompressedSize             : 6383528
ZipFileName                     : Test/Un-named2026-09-09 09-13-37_000002.PDF
```

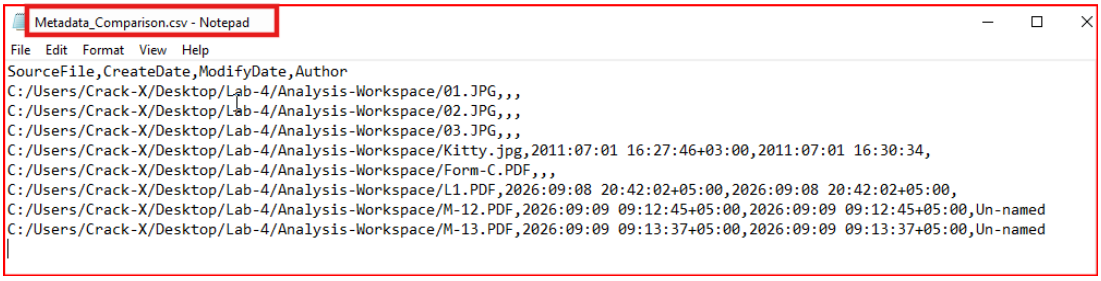
PNG

```
---- ExifTool ----
ExifToolVersion                 : 13.59
---- System ----
FileName                        : PNG_000003.PNG
Directory                       : C:/Users/Crack-X/Desktop/Lab-4/Analysis-Workspace
FileSize                        : 35 kB
FileModifyDate                  : 2026:09:10 02:05:16+05:00
FileAccessDate                  : 2026:09:12 23:02:36+05:00
FileCreateDate                  : 2026:09:11 09:44:51+05:00
FilePermissions                 : -rw-rw-rw-
---- File ----
FileType                        : PNG
FileTypeExtension               : png
MIMEType                        : image/png
---- PNG ----
ImageWidth                      : 256
ImageHeight                     : 256
BitDepth                        : 8
ColorType                       : RGB with Alpha
Compression                     : Deflate/Inflate
Filter                          : Adaptive
Interlace                       : Noninterlaced
SRGBRendering                   : Perceptual
Gamma                           : 2.2
WhitePointX                     : 0.3127
WhitePointY                     : 0.329
RedX                            : 0.64
RedY                            : 0.33
GreenX                          : 0.3
GreenY                          : 0.6
BlueX                           : 0.15
BlueY                           : 0.06
---- Composite ----
ImageSize                       : 256x256
Megapixels                      : 0.066
```

Activate Windo
Go to Settings to a

Compare multiple files





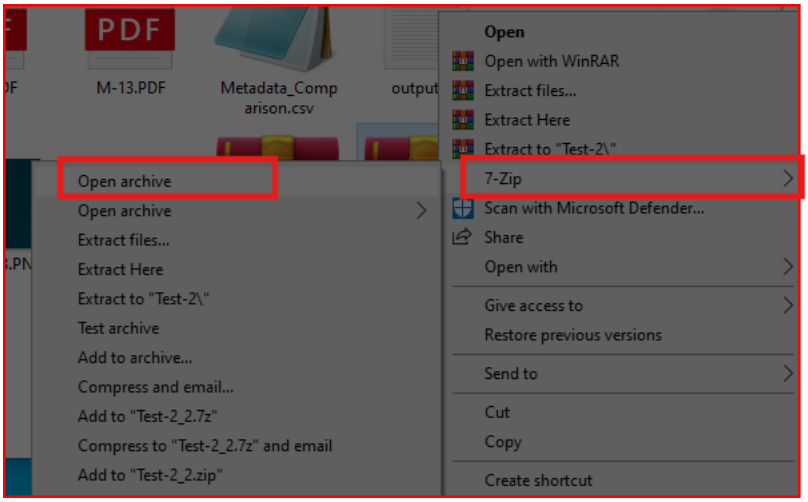
Most image files contained no extractable metadata. Only Kitty.jpg retained original creation and modification dates (2011).

Two PDF files (M-12.PDF and M-13.PDF) showed Author as “Un-named” and identical Create/Modify timestamps, which may indicate automated generation or minimal editing.

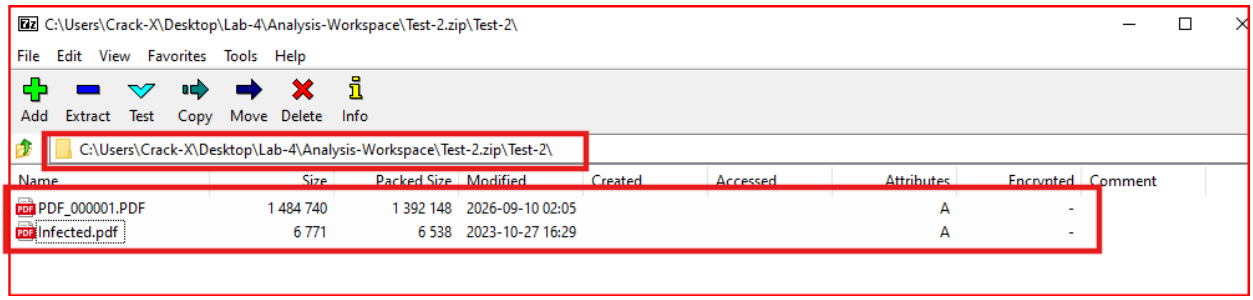
Save all outputs in 03_Tool_Outputs.

PHASE 4: Archive Deep Inspection (Test2.zip)

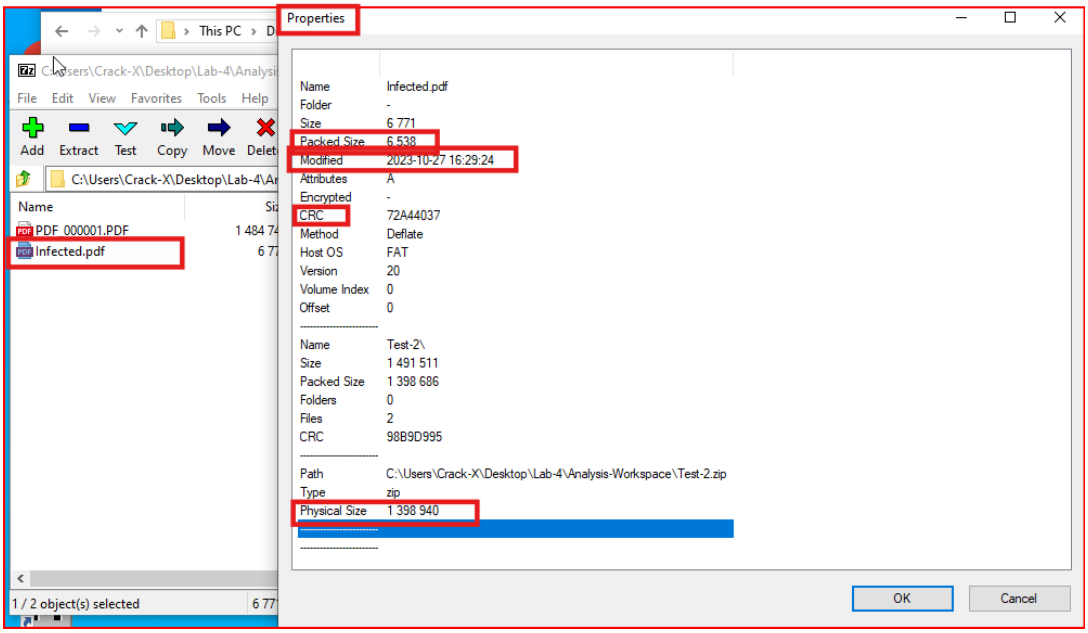
Open ZIP/RAR with 7-Zip.



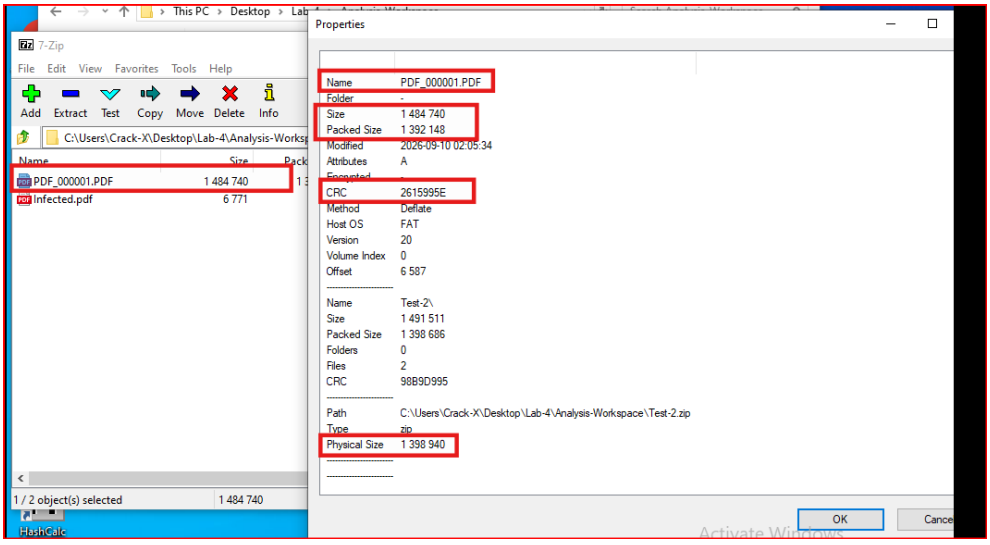
Details:



Checking Properties of the pdf



Checking Properties of another pdf



Archive Summary – Test-2.zip

Item	Details	Observation
Total Files	2	Small archive
Files Inside	PDF_000001.PDF + Infected.pdf	One large PDF + one small PDF
Compression Method	Deflate	Normal & common
Password Protected	No	Easy to extract
Nested Archive	No	Clean structure
Host OS	FAT	Typical for USB/removable media

Individual File Analysis

File Name	Original Size	Packed Size	Modified Date	Notes
PDF_000001.PDF	1,484,740	1,392,148	2026-09-10 02:05:34	Large file, recent date
Infected.pdf	6,771	6,538	2023-10-27 16:29:24	Small file, older date (2023)

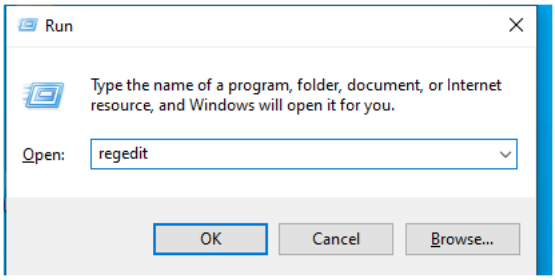
Phase 5: System Artifact Analysis

This phase focuses on **Windows Registry** and **Event Logs**.

Follow these steps carefully:

Part A: Registry Analysis

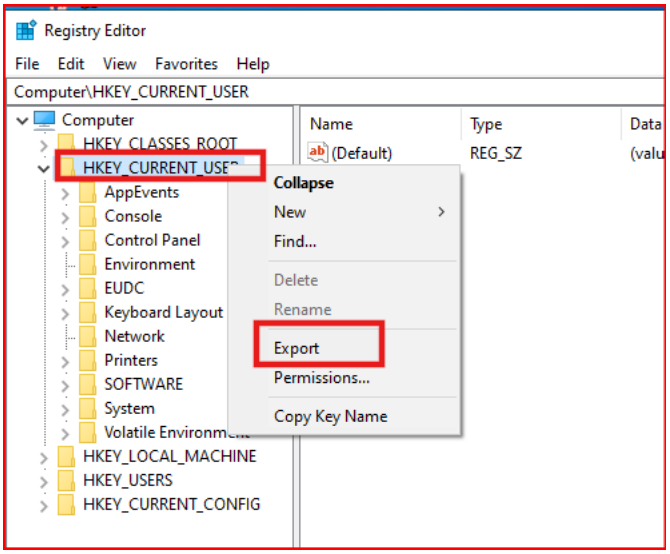
1. Press Windows + R → type regedit → press Enter
   (Registry Editor will open)



2. On the left side, go to this location:

HKEY_CURRENT_USER

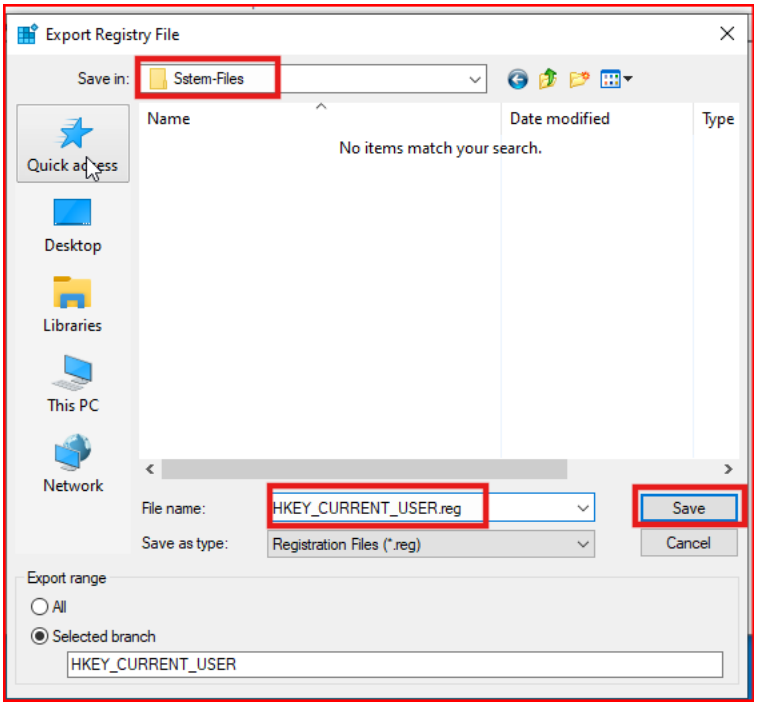
3. Right-click on **HKEY_CURRENT_USER** → click **Export**



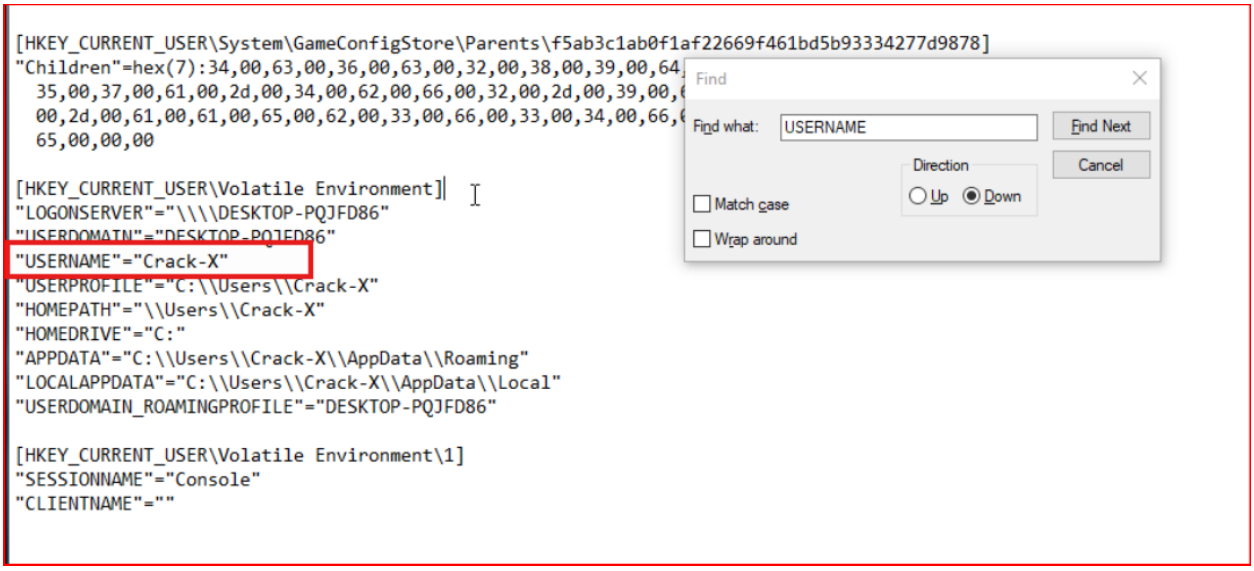
4. Save the file here: **(Create System-Files folder)**

C:\Users\Crack-X\Desktop\Lab-4\System_Files\

Name it: HKEY_CURRENT_USER.reg



5. After exporting, open the .reg file with Notepad and look for:
    - Registered owner / username



Recent Docs

```
[HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs]
"MRUListEx"=hex:34,00,00,00,51,00,00,00,0c,00,00,00,50,00,00,00,0b,00,00,00,3d,\
  00,00,00,48,00,00,00,4f,00,00,00,19,00,00,00,4e,00,00,00,43,00,00,00,0e,00,\
  00,00,0d,00,00,00,4d,00,00,00,4c,00,00,00,4b,00,00,00,4a,00,00,00,49,00,00,\
  00,44,00,00,00,42,00,00,00,47,00,00,00,46,00,00,00,45,00,00,00,3f,00,00,00,\
  41,00,00,00,22,00,00,00,40,00,00,00,3e,00,00,00,3b,00,00,00,1f,00,00,00,00,\
  00,00,00,3c,00,00,00,3a,00,00,00,39,00,00,00,38,00,00,00,37,00,00,00,36,00,\
  00,00,35,00,00,00,33,00,00,00,32,00,00,00,29,00,00,00,31,00,00,00,07,00,00,\
  00,06,00,00,00,2e,00,00,00,2d,00,00,00,30,00,00,00,2f,00,00,00,2c,00,00,00,\
  05,00,00,00,2b,00,00,00,2a,00,00,00,28,00,00,00,27,00,00,00,26,00,00,00,25,\
  00,00,00,24,00,00,00,23,00,00,00,21,00,00,00,20,00,00,00,1e,00,00,00,1b,00,\
```

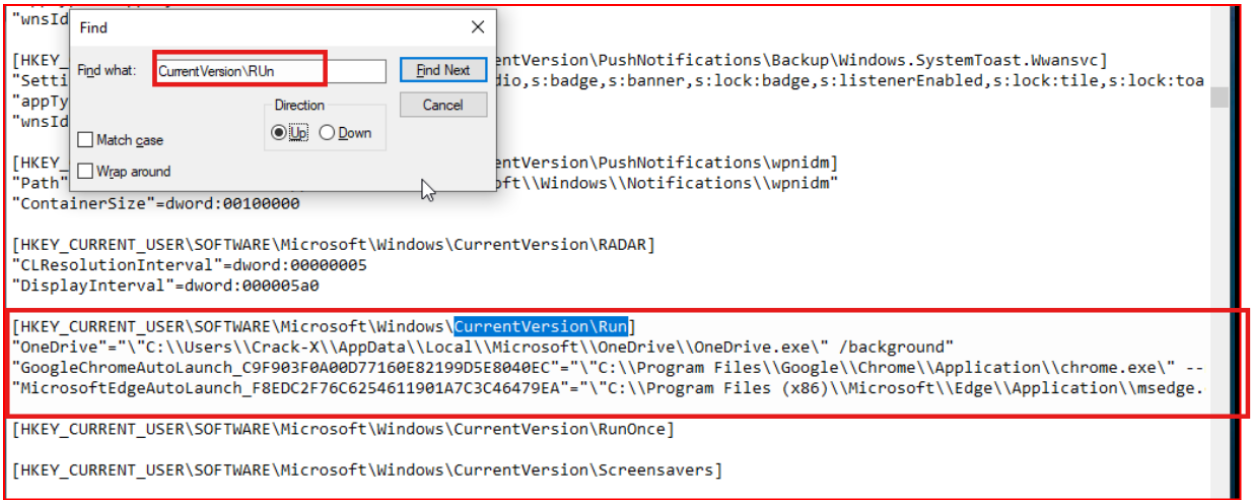
```
[HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.csv]
"0"=hex:4d,00,65,00,74,00,61,00,64,00,61,00,74,00,61,00,5f,00,43,00,6f,00,6d,\
  00,70,00,61,00,72,00,69,00,73,00,6f,00,6e,00,2e,00,63,00,73,00,76,00,00,00,\
  92,00,32,00,00,00,00,00,00,00,00,00,00,00,4d,65,74,61,64,61,74,61,5f,43,6f,\
  6d,70,61,72,69,73,6f,6e,2e,63,73,76,2e,6c,6e,6b,00,68,00,09,00,04,00,ef,be,\
  00,00,00,00,00,00,00,00,2e,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,\
  00,00,00,00,00,00,00,00,00,00,00,00,00,4d,00,65,00,74,00,61,00,64,00,61,00,\
  74,00,61,00,5f,00,43,00,6f,00,6d,00,70,00,61,00,72,00,69,00,73,00,6f,00,6e,\
  00,2e,00,63,00,73,00,76,00,2e,00,6c,00,6e,00,6b,00,00,00,2a,00,00,00
"MRUListEx"=hex:00,00,00,00,ff,ff,ff,ff

[HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.docx]
"0"=hex:43,00,59,00,33,00,33,00,34,00,5f,00,44,00,46,00,5f,00,4c,00,61,00,62,\
  00,30,00,31,00,2e,00,64,00,6f,00,63,00,78,00,00,00,78,00,32,00,00,00,00,00,\
  00,00,00,00,00,00,43,59,33,33,34,5f,44,46,5f,4c,61,62,30,31,2e,6c,6e,6b,00,\
  00,56,00,09,00,04,00,ef,be,00,00,00,00,00,00,00,00,2e,00,00,00,00,00,00,00,\
  00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,43,00,59,\
  00,33,00,33,00,34,00,5f,00,44,00,46,00,5f,00,4c,00,61,00,62,00,30,00,31,00,\
  2e,00,6c,00,6e,00,6b,00,00,00,22,00,00,00
"MRUListEx"=hex:00,00,00,00,ff,ff,ff,ff

[HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.jpg]
"0"=hex:4b,00,69,00,74,00,74,00,79,00,2e,00,6a,00,70,00,67,00,00,00,5c,00,32,\
  00,00,00,00,00,00,00,00,00,00,00,4b,69,74,74,79,2e,6c,6e,6b,00,44,00,09,00,\
  04,00,ef,be,00,00,00,00,00,00,00,00,2e,00,00,00,00,00,00,00,00,00,00,00,00,\
  00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,4b,00,69,00,74,00,74,00,\
  79,00,2e,00,6c,00,6e,00,6b,00,00,00,18,00,00,00
```

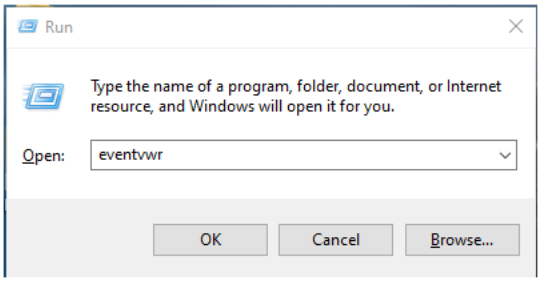
```
[HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.pdf]
"MRUListEx"=hex:02,00,00,00,0b,00,00,00,0d,00,00,00,0c,00,00,00,0a,00,00,00,07,\
  00,00,00,09,00,00,00,08,00,00,00,00,00,00,00,06,00,00,00,05,00,00,00,04,00,\
  00,00,03,00,00,00,01,00,00,00,ff,ff,ff,ff
"1"=hex:4d,00,6f,00,64,00,75,00,6c,00,65,00,2d,00,32,00,2d,00,43,00,6f,00,6d,\
  00,70,00,75,00,74,00,65,00,72,00,2d,00,46,00,6f,00,72,00,65,00,6e,00,73,00,\
  69,00,63,00,73,00,2d,00,49,00,6e,00,76,00,65,00,73,00,74,00,69,00,67,00,61,\
  00,74,00,69,00,6f,00,6e,00,2d,00,50,00,72,00,6f,00,63,00,65,00,73,00,73,00,\
  2e,00,70,00,64,00,66,00,00,00,e0,00,32,00,00,00,00,00,00,00,00,00,00,00,4d,\
```

Current Version Run



Part B: Event Log Analysis

1. Press Windows + R → type eventvwr → press Enter
   (Event Viewer will open)

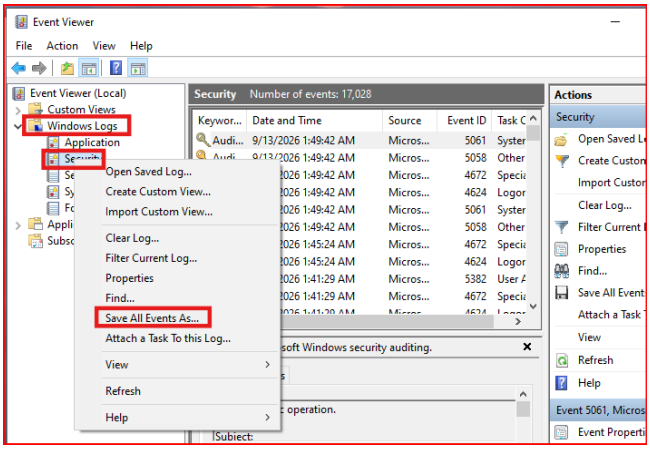


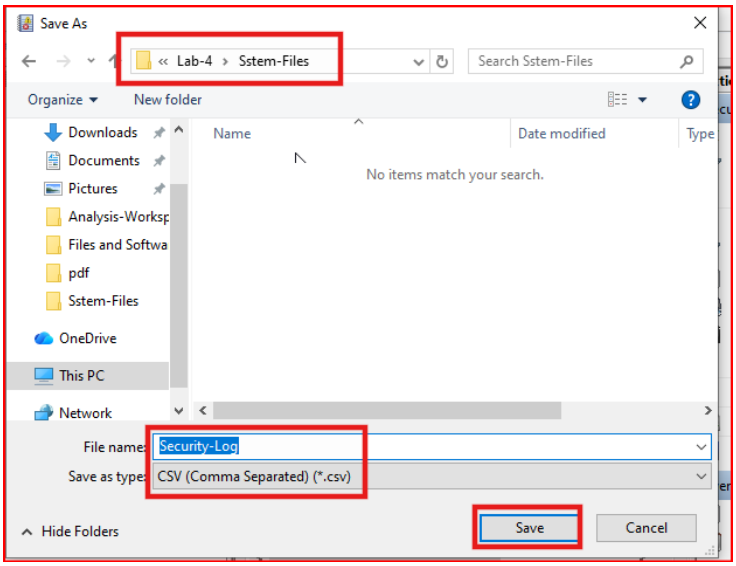
2. On the left side go to:

Windows Logs → Security

3. Right-click on **Security** → click **Save All Events As...**
4. Choose format: **CSV**
5. Save it here:

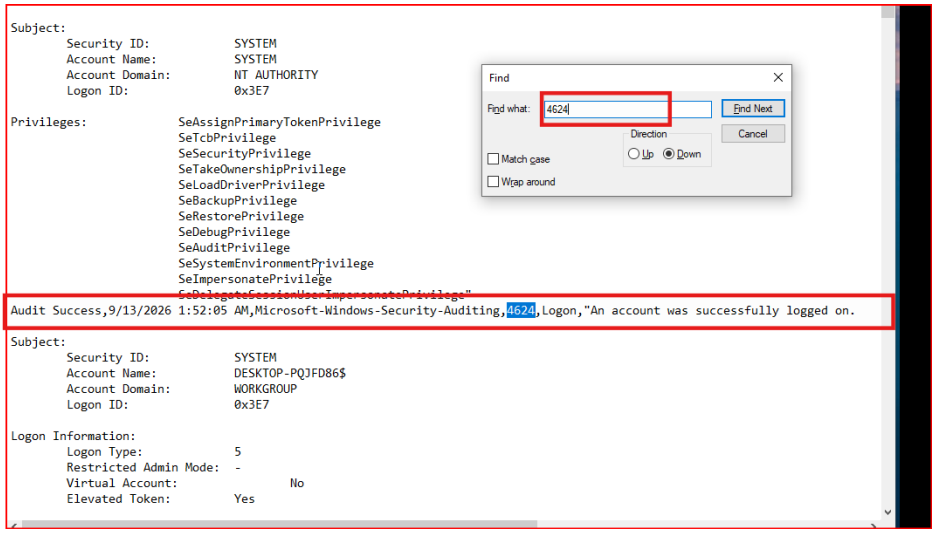
C:\Users\Crack-X\Desktop\Lab-4\System-Files\Security-Log.csv



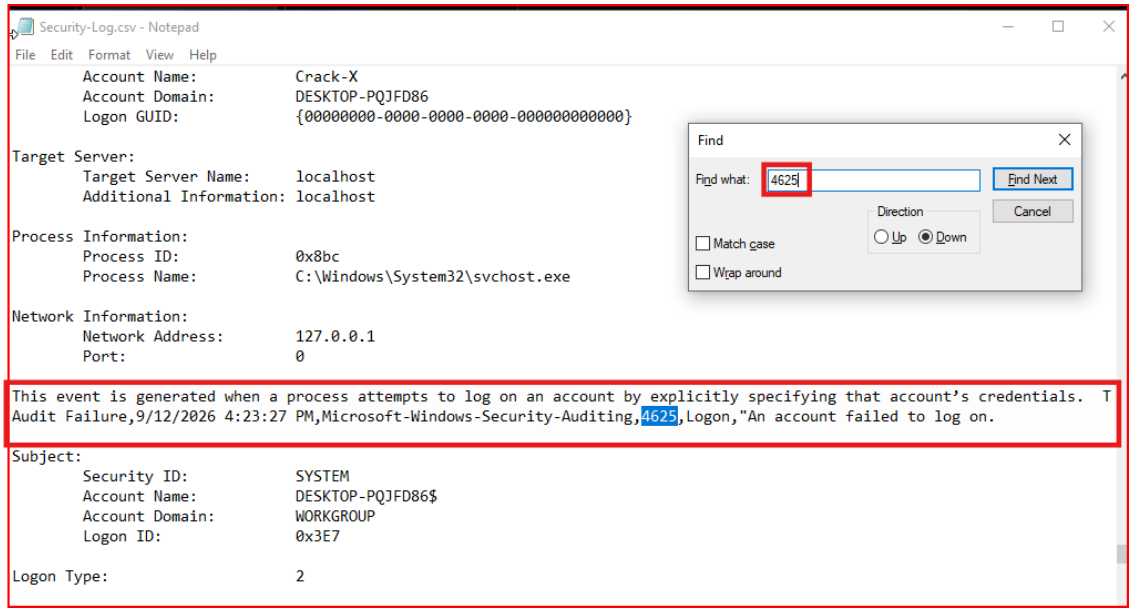


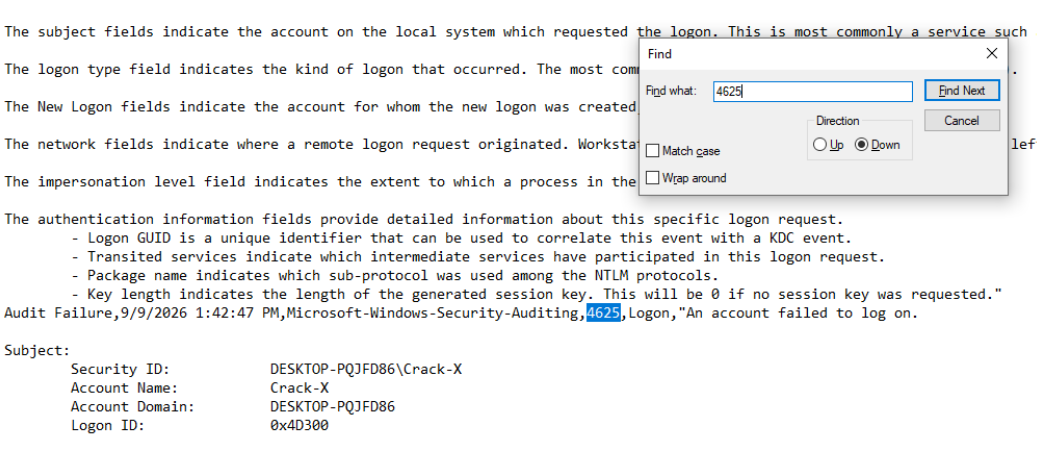
6. Open the CSV file in Excel or Notepad and look for:

   - Event ID **4624** → Successful logon



- Event ID **4625** → Failed logon






```
Account For Which Logon Failed:
        Security ID:            NULL SID
        Account Name:           Crack-X
        Account Domain:         DESKTOP-PQJFD86

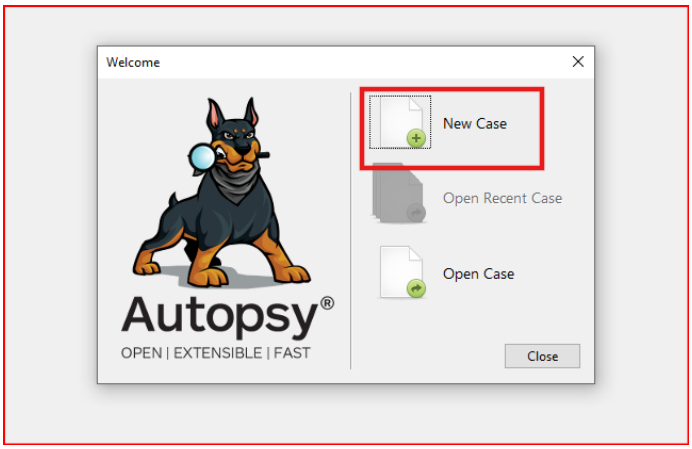
Failure Information:
        Failure Reason:         Unknown user name or bad password.
        Status:                 0xC000006D
        Sub Status:             0xC000006A
```

Phase 6: Autopsy

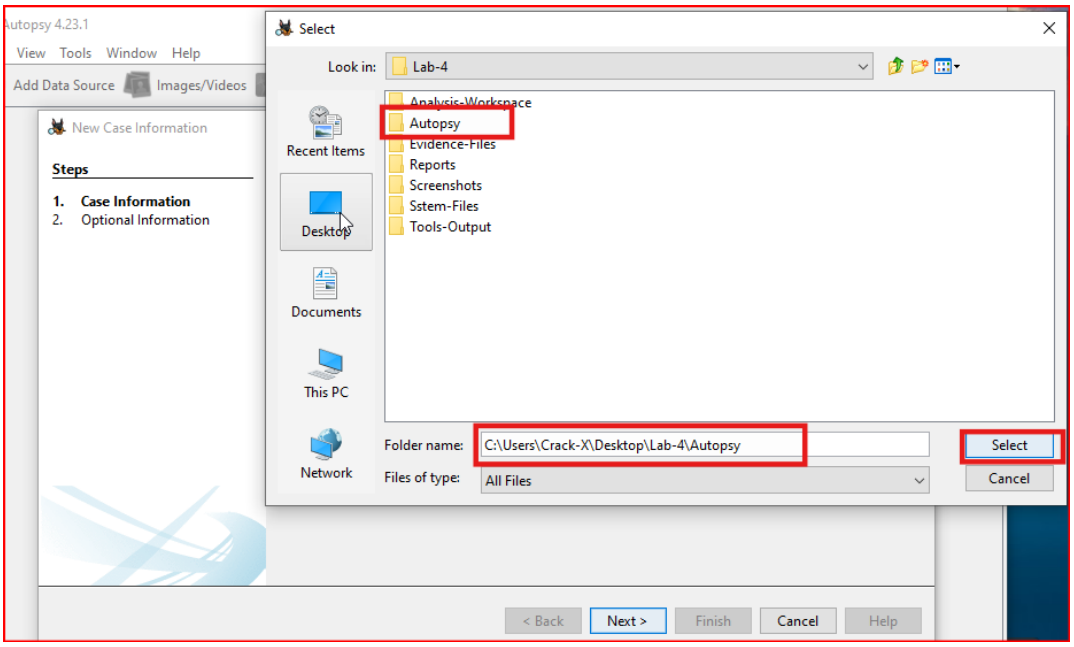
---

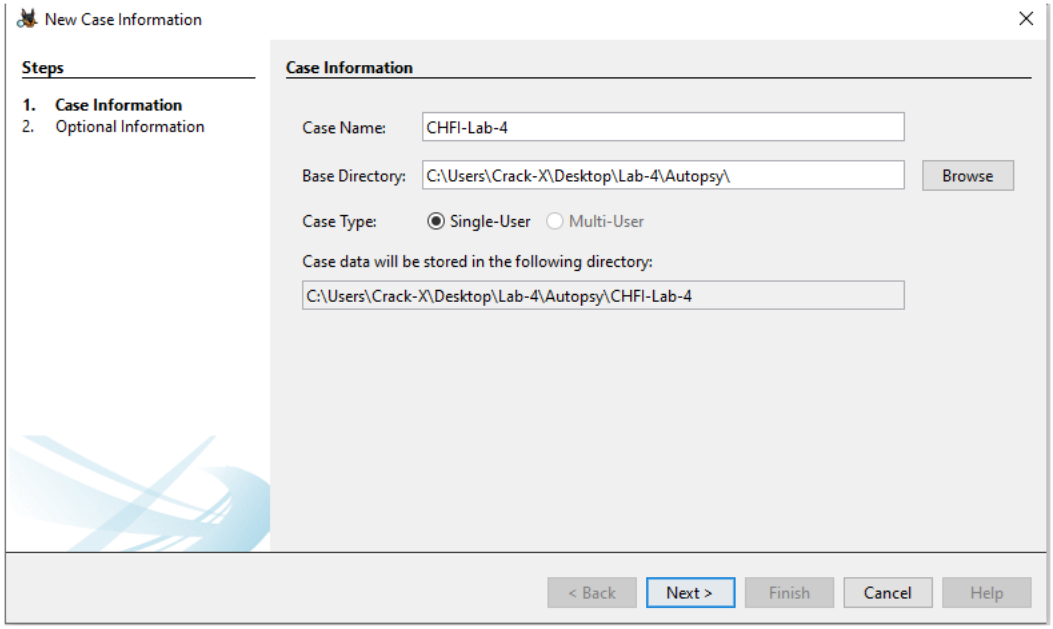
1. Create a Proper Case

- Open Autopsy.
- Click **"New Case"**.

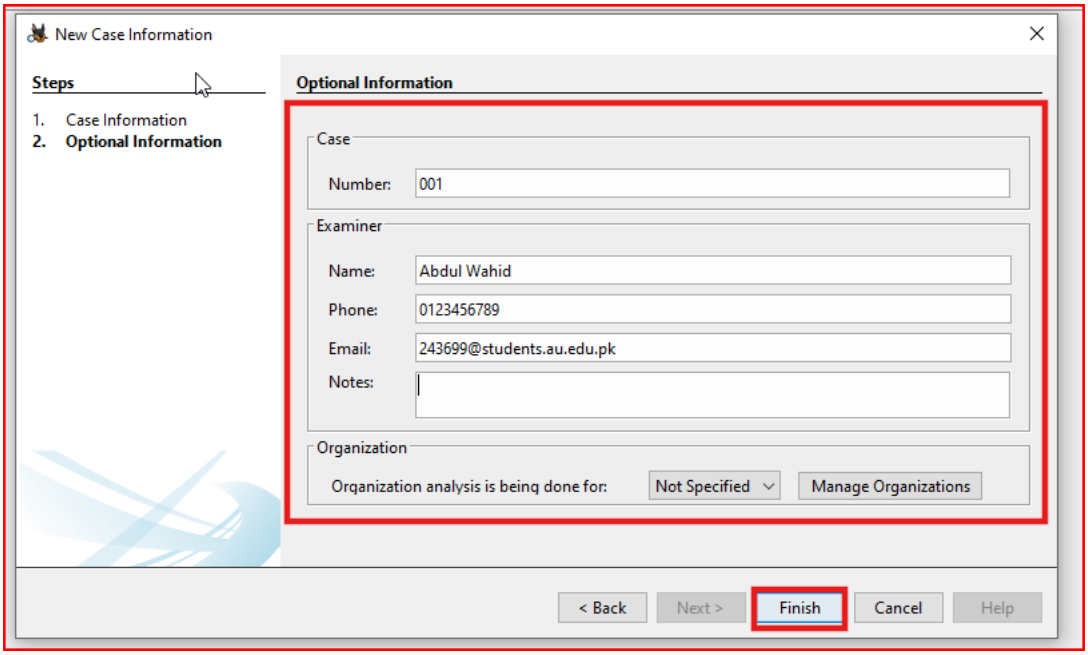


- Give the case a name (example: CHFI-Lab-4).
- Choose a folder to save the case (recommended: Lab-4\Autopsy).



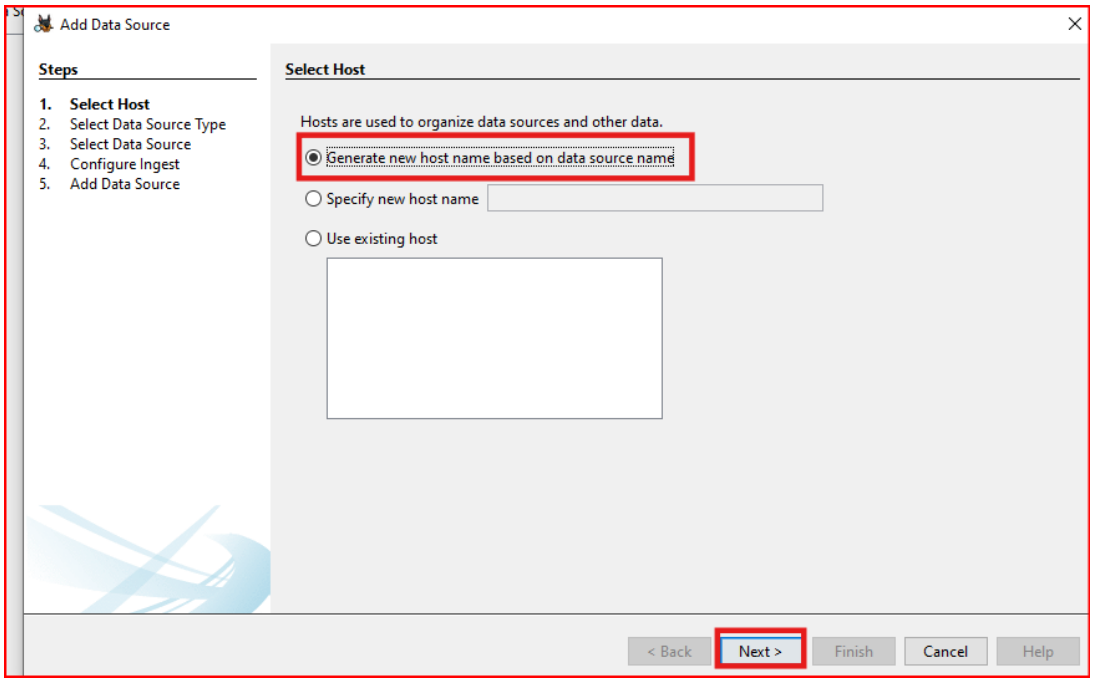


- Enter your name as the examiner **(Abdul Wahid).**
- Click **Next → Finish**.

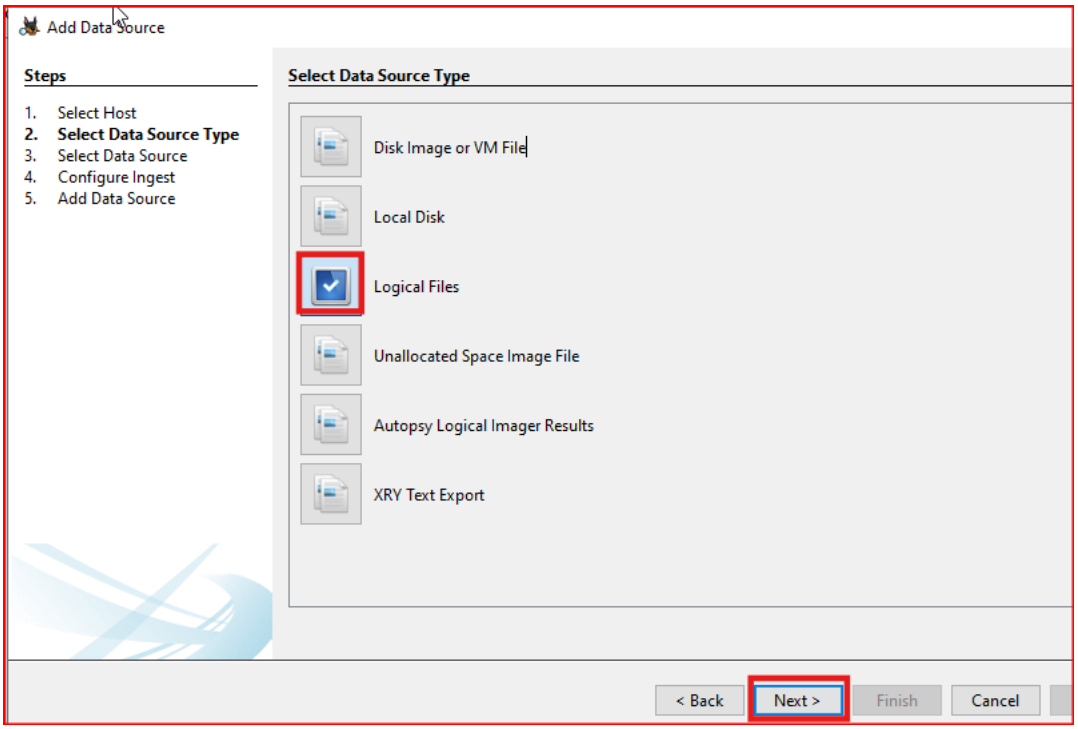


This creates an official case file (like opening a real investigation).

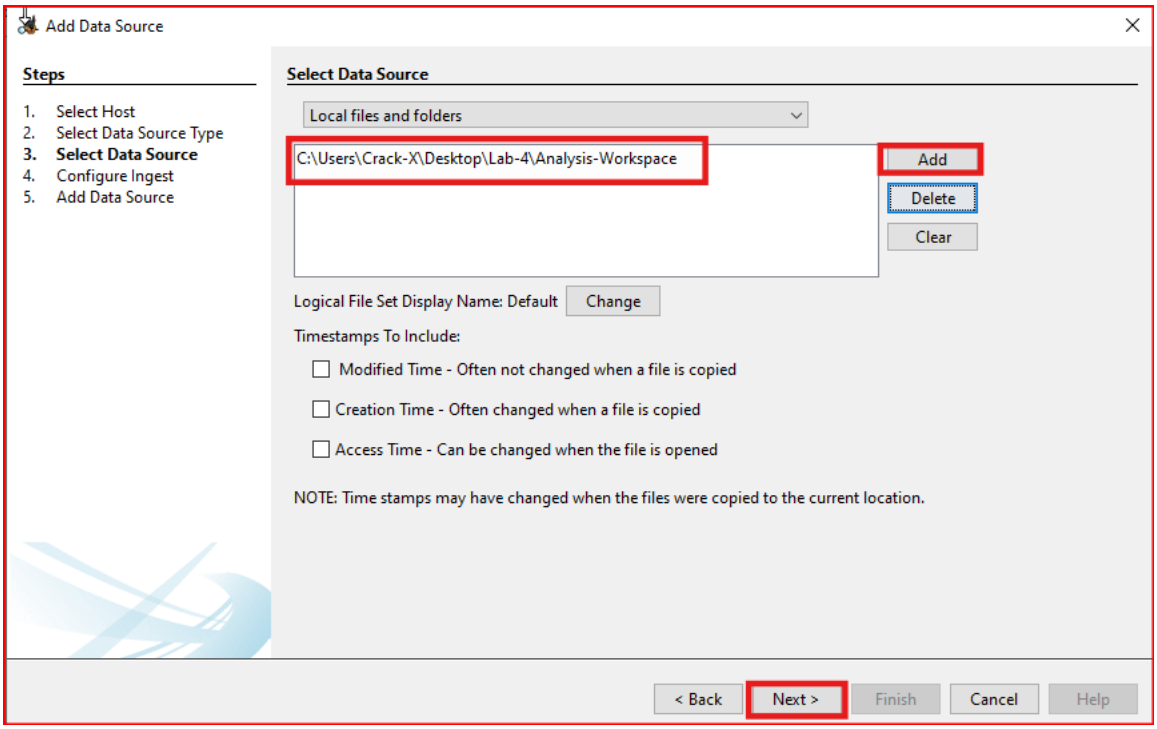
2. Add the Working Copies as Logical Files



- Select **"Logical Files"** (because we are adding normal folders/files, not a full disk image).



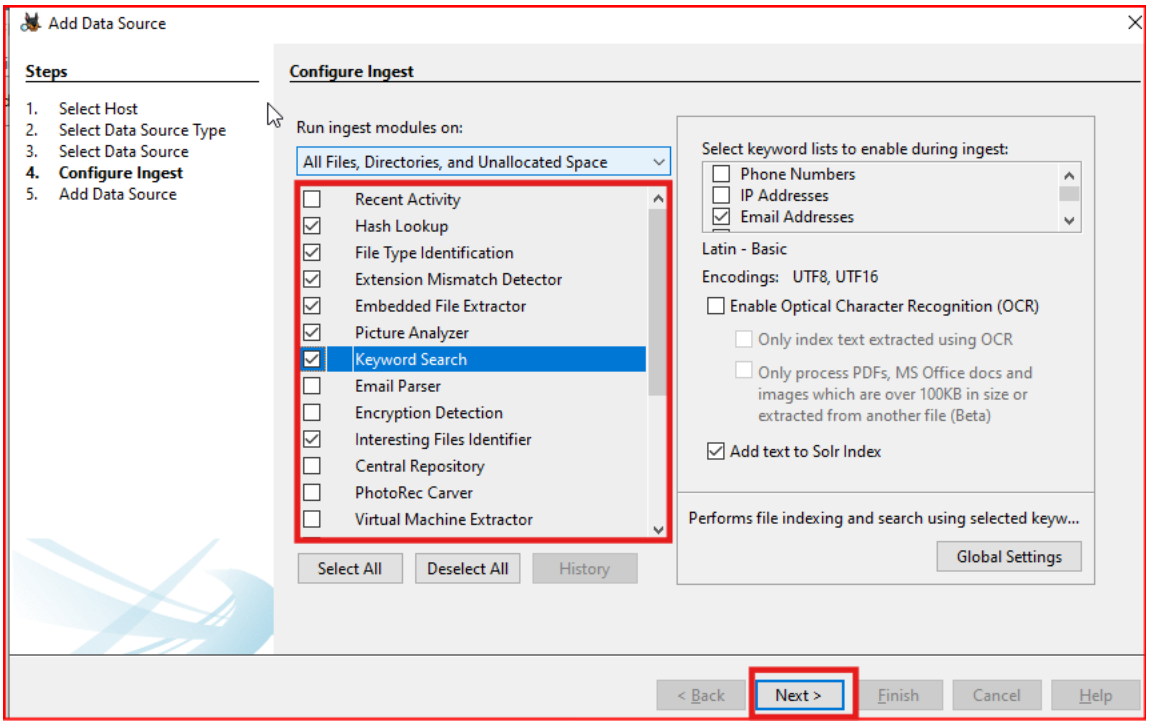
- Browse and select your Analysis-Workspace folder (the one that contains JPG, PDF, ZIP, etc.).
- Click **Next**.



3. Enable Important Modules

When Autopsy asks which modules to run, enable these:

Module	Why it is important
File Type Identification	Confirms real file type
Extension Mismatch Detector	Finds files with wrong extensions (hiding)
Hash Lookup	Calculates MD5/SHA hashes
Keyword Search	Searches for important words
Embedded File Extractor	Finds files hidden inside other files

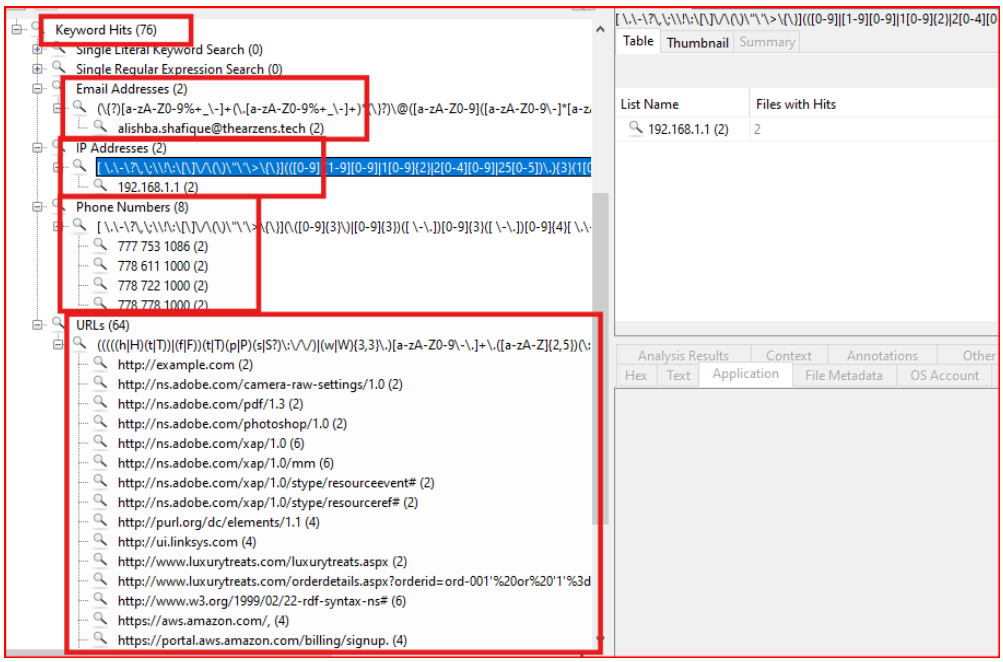


Then click Next → Finish. Autopsy will start analyzing (this is called Ingest).

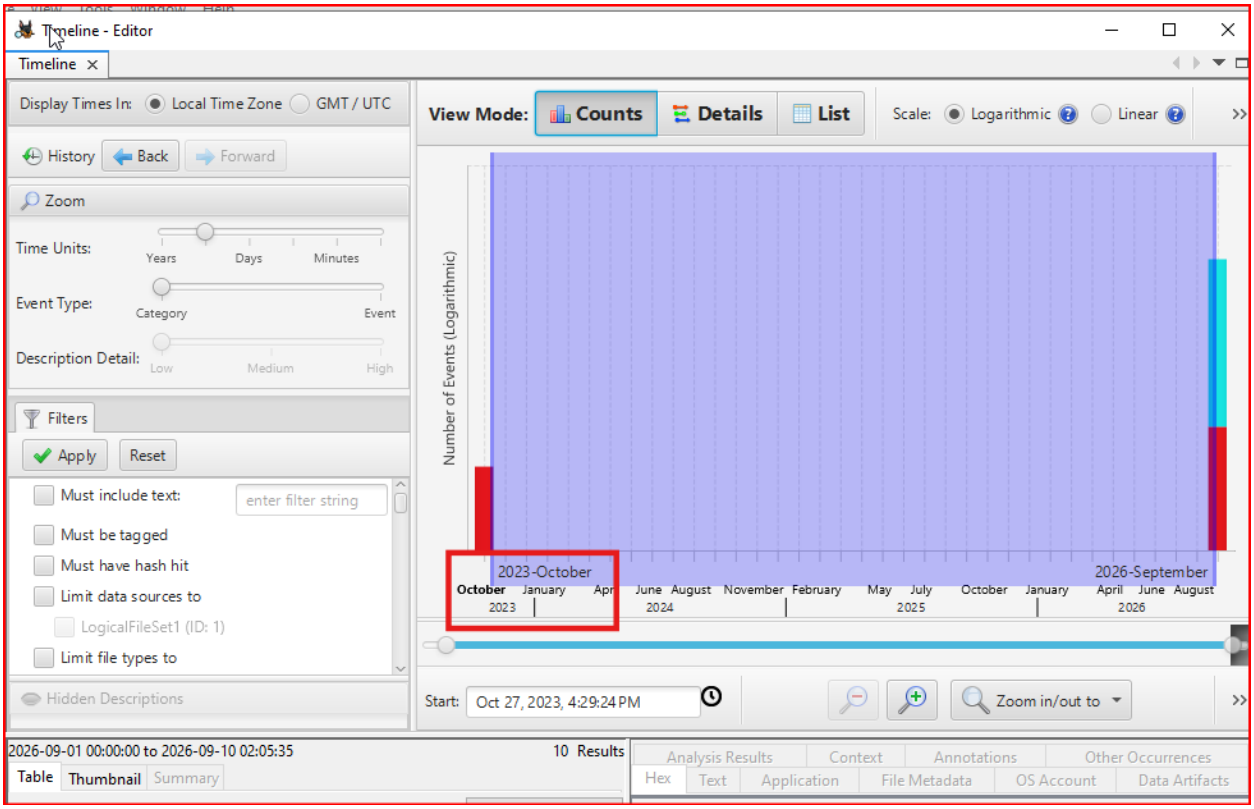
4. After Ingest (Analysis Results)

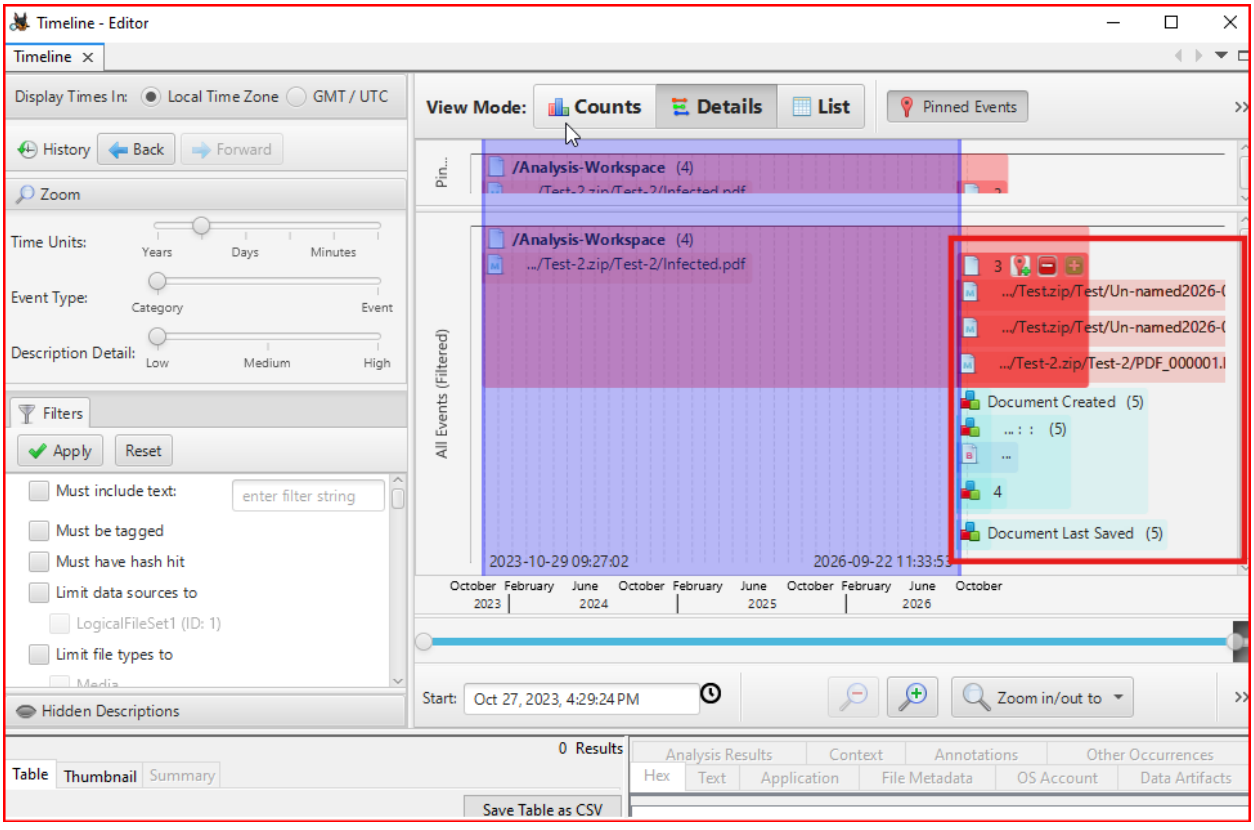
Once the analysis finishes, check these things:

- **Keyword Hits** → If you searched for words like "password", "confidential", "infected", etc.

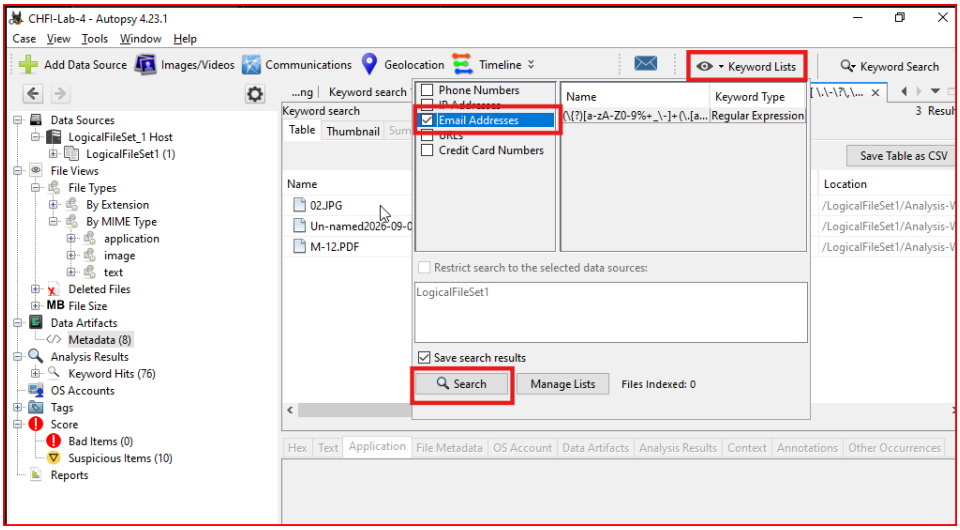


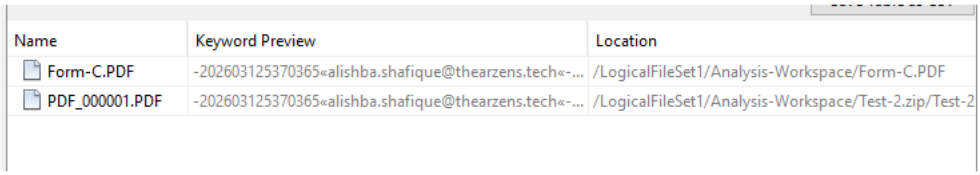
- **File Timeline** → Shows when files were created/modified (if available).



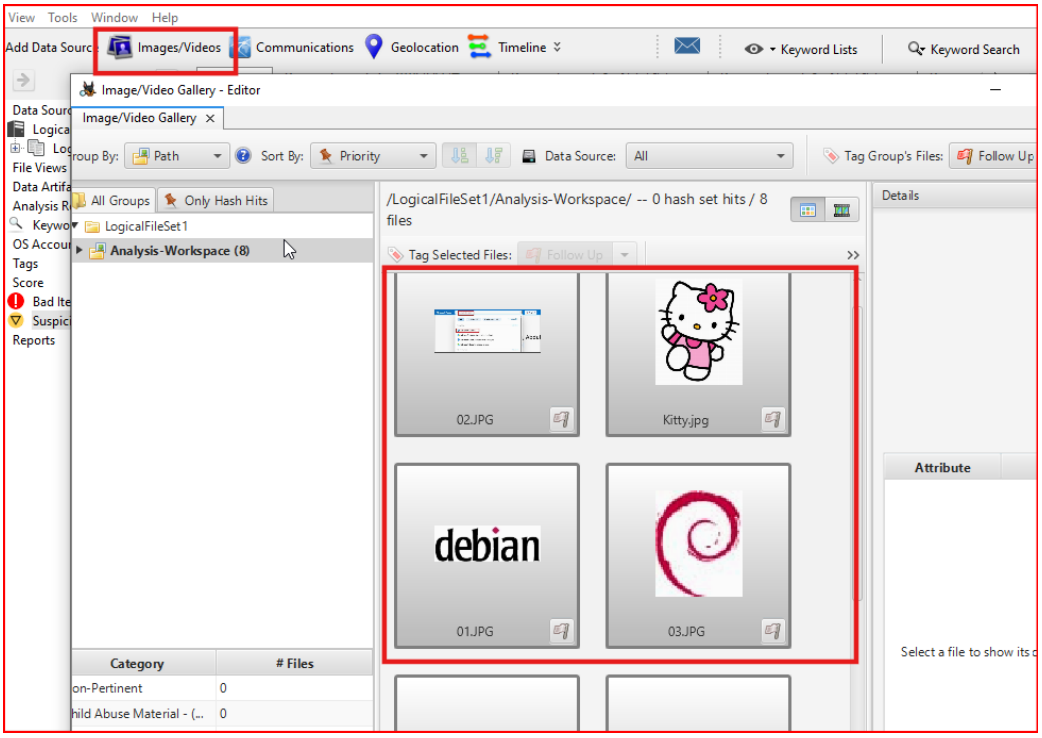


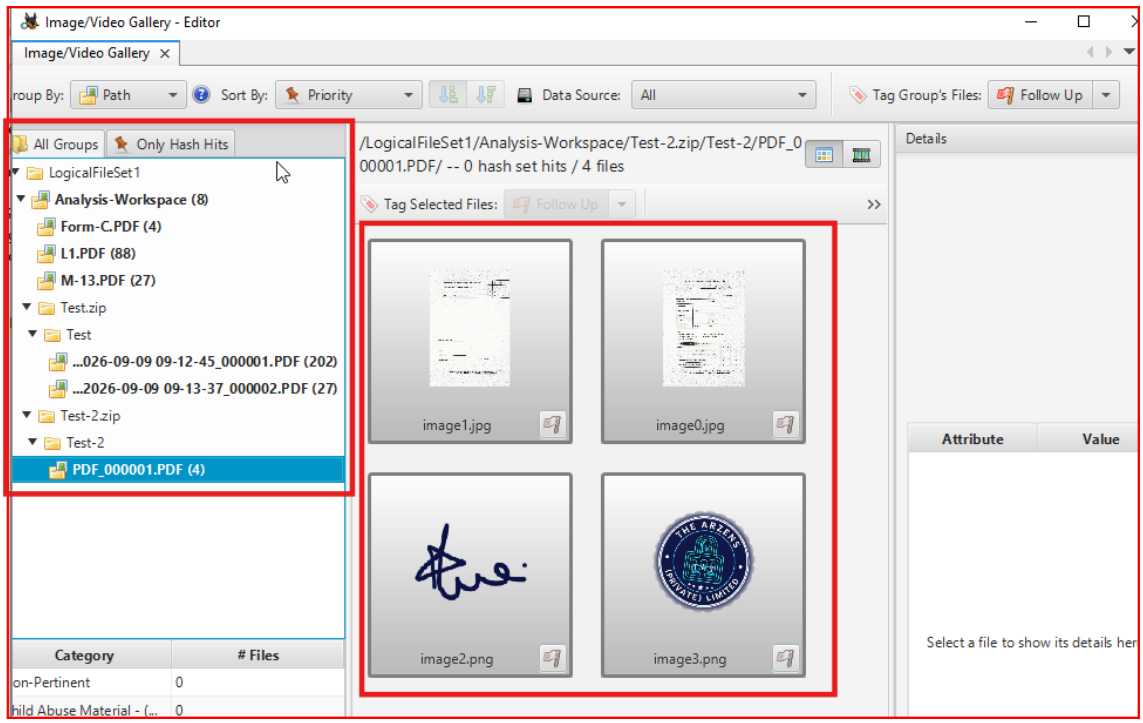
Using a specific Keyword (Mail) to Search



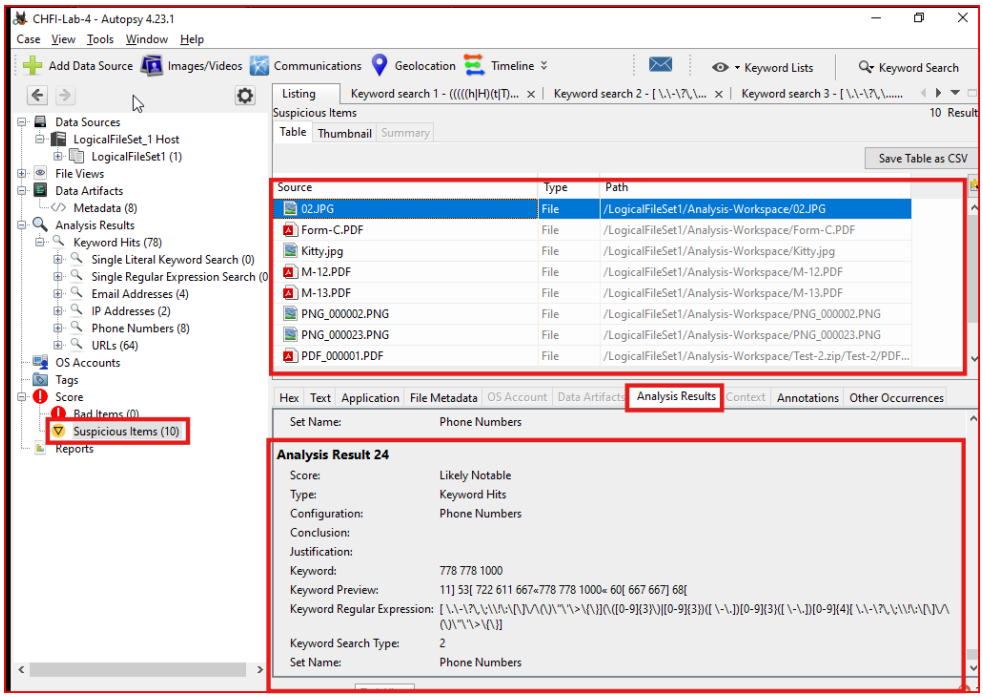


Images/videos Results



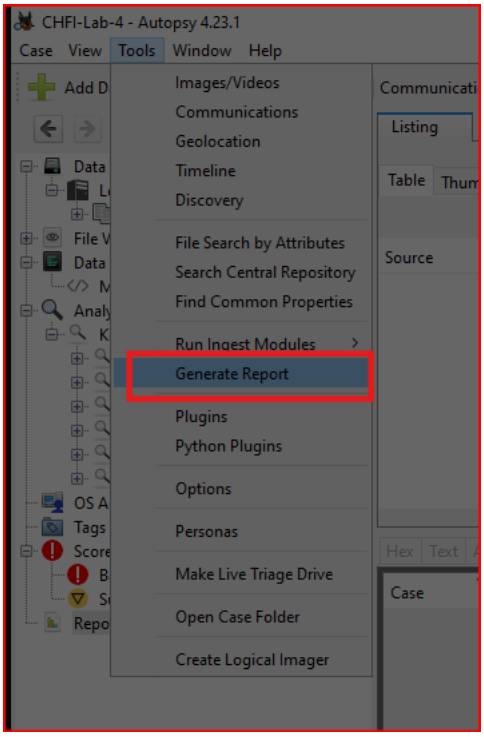


Suspicious Items (Important)

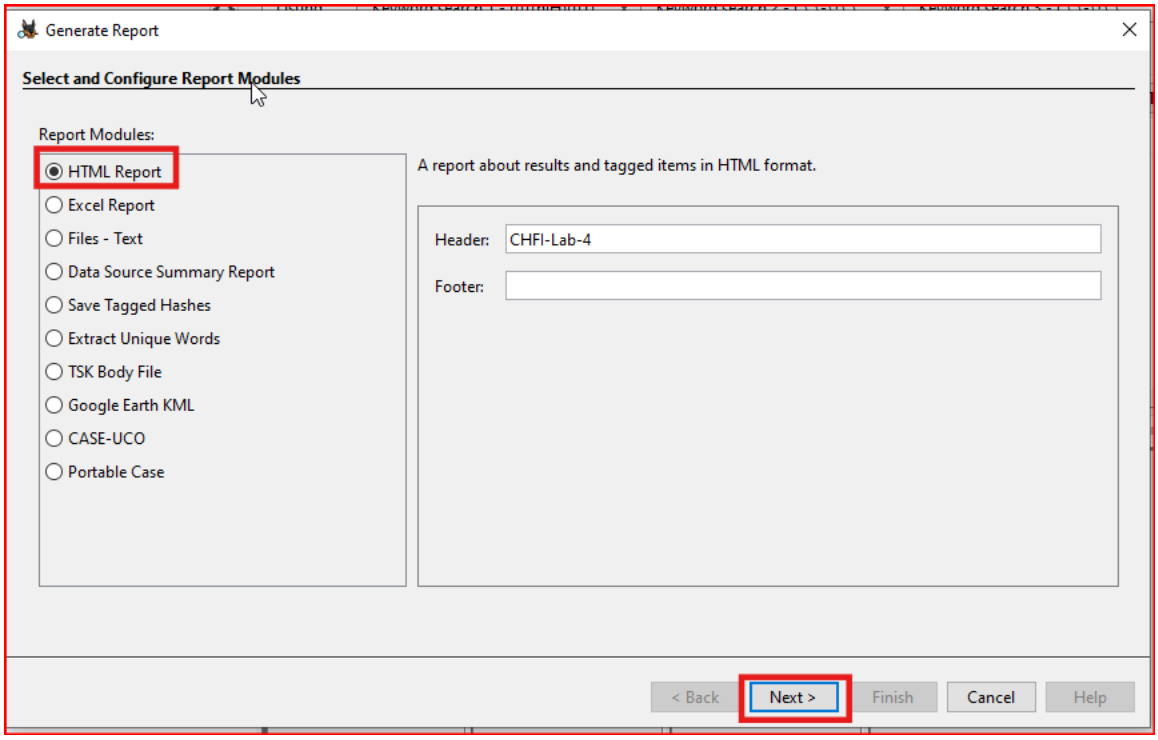


5. Generate a Detailed HTML Report

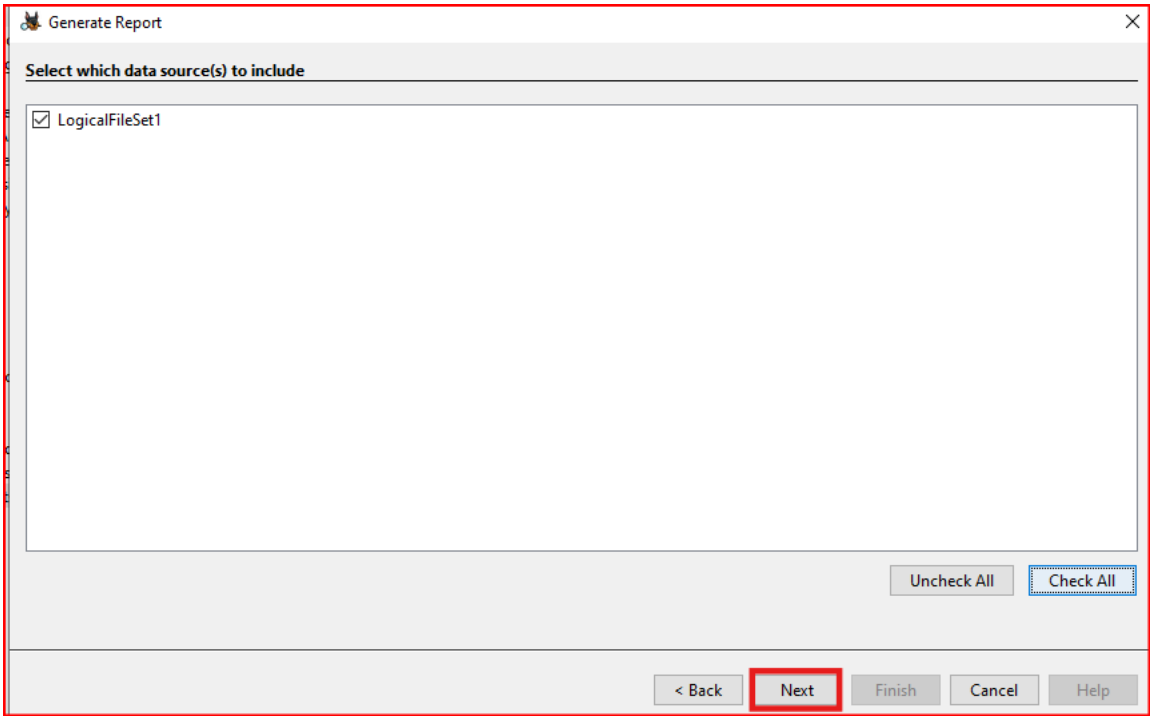
- Go to the top menu → **Generate Report**.

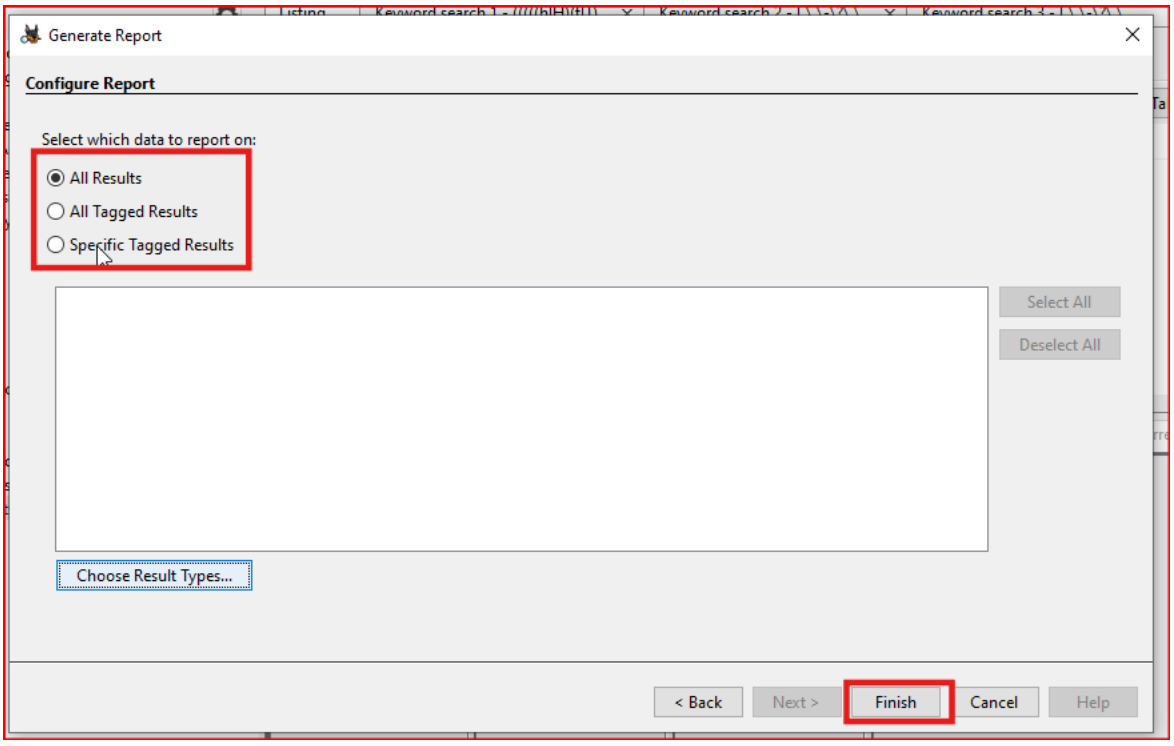


- Choose **HTML Report**.

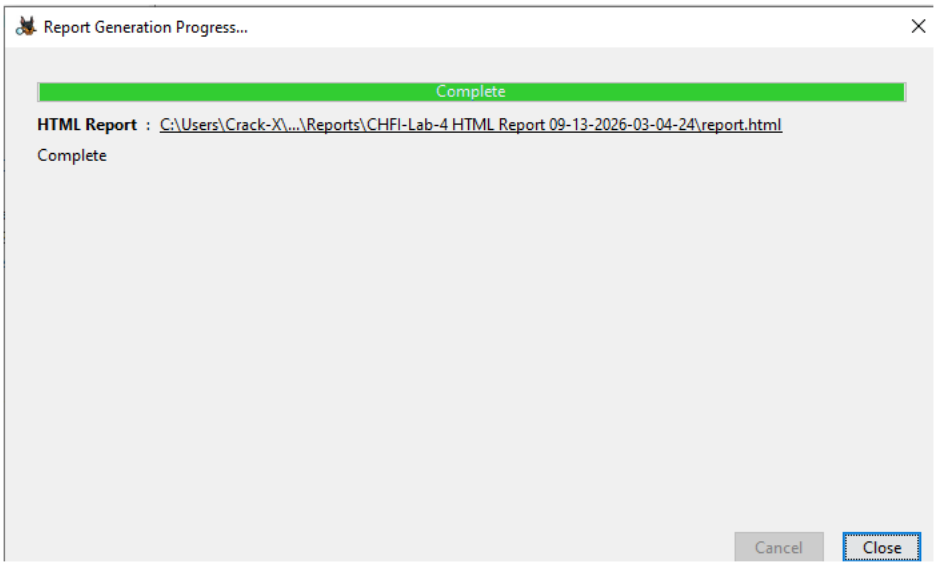


- Select the results you want to include.

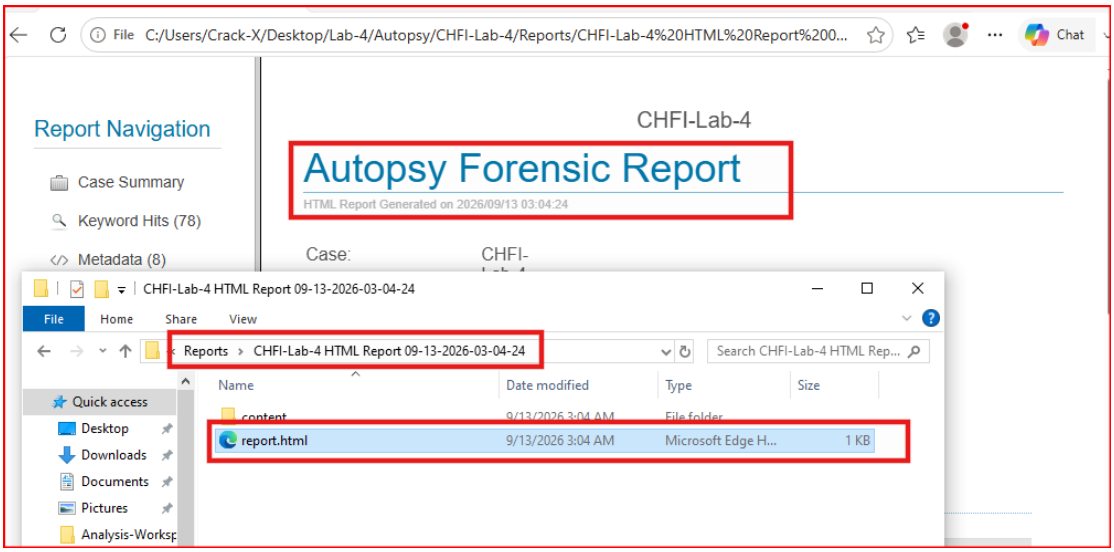


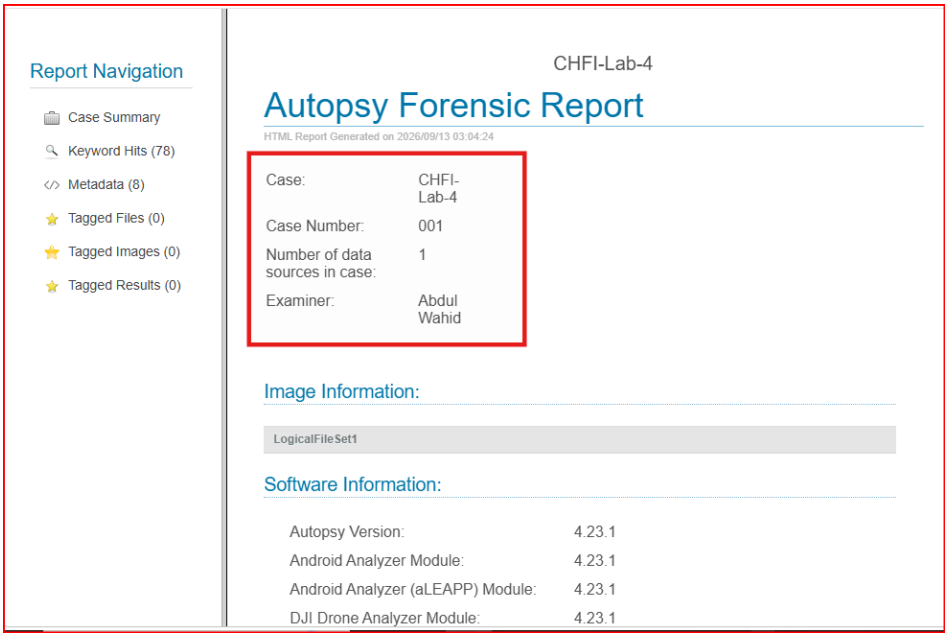


- Click **Generate**.



- View the report you just created





Lab 4 Conclusion

This laboratory produced the most significant investigative findings of the entire module. A comprehensive examination of the recovered evidence using five specialist tools revealed multiple anomalies, confirmed malicious content, and identified a direct link to a named individual associated with the subject device.

File signature verification confirmed that no files had been disguised by altering their extensions — all internal signatures matched their declared file types. Metadata extraction revealed a pattern of unnamed authors, identical creation and modification timestamps across multiple documents, and the use of an automated Python-based tool to generate PDF files — all of which suggest deliberate anonymisation or scripted document production. Archive inspection identified files spanning a three-year period within a single archive, indicating sustained activity on the device between 2023 and 2026. Autopsy analysis flagged 10 items as suspicious and confirmed through keyword search that the email address alishba.shafique@thearzens.tech appears across multiple recovered documents.

Critical Findings Summary

#	Finding	Tool	Significance
1	trojan.pdfka confirmed in Infected.pdf	VirusTotal — Lab 2	Critical
2	Email address alishba.shafique@thearzens.tech found in multiple documents	Autopsy	High
3	10 files flagged as suspicious	Autopsy	High
4	File timestamps spanning 2023 to 2026 within single archive	7-Zip	Medium
5	Unnamed authors and identical timestamps across multiple PDFs	exiftool	Medium

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Metadata Anomalies Identified

File	Anomaly	Implication
Form-C.PDF	Author listed as unnamed — created with pypdf	Automated or scripted generation
M-12.PDF	Creation and modification timestamps identical	Possible metadata manipulation
M-13.PDF	Creation and modification timestamps identical	Possible metadata manipulation
Multiple PDFs	Author field blank or unnamed across several files	Deliberate anonymisation suspected

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Evidence Status at Close of Lab 4

Evidence Item	Description	Status
EV-001	Original 32 GB Samsung USB drive	Preserved — unmodified
EV-002	Recovered files, approximately 2 GB	Examined — read-only access enforced
EV-003	Workspace copy for Lab 4 analysis	Examined — all findings documented

Chain of custody remained unbroken throughout this laboratory. All tool outputs, screenshots, and findings have been saved to the designated Lab 4 folder structure and retained as part of the permanent case record.

LAB 5 — EVIDENCE HANDLING PROCEDURES

Lab Overview

Field	Detail
Lab Number	5
Lab Title	Evidence Handling Procedures
Case Reference	CHFI_Module2_Lab1-6_Case001
Examiner	Abdul Wahid
Date Conducted	13 September 2026
Investigation Phase	Preservation and Baseline Establishment

Purpose

By this stage of the investigation, files have been recovered, verified, and examined in depth. Before those files can be imaged and archived in Lab 6, a formal evidence handling procedure must be applied. This means reviewing the evidence under controlled, read-only conditions, establishing cryptographic baselines that will serve as the reference point for the final forensic image, and confirming that everything is documented and accounted for before the archival phase begins.

This laboratory applies professional evidence handling standards using FTK Imager — one of the most widely used and legally recognised tools in digital forensics. The evidence source is loaded in read-only mode, its contents are reviewed and exported, and MD5 and SHA-256 hash values are calculated and recorded as the definitive baseline for Lab 6 verification.

Objectives

- Load the evidence source into FTK Imager under strict read-only conditions
- Review and confirm the contents of the evidence source before imaging
- Export the evidence file list and generate a hash value record
- Calculate and record MD5 and SHA-256 baseline hash values for the evidence source
- Confirm that all evidence items are accounted for and properly documented
- Prepare the evidence source for forensic imaging in Lab 6

Tools and Evidence

Tool	Version	Purpose
FTK Imager	4.7.3.81	Evidence review, export, and baseline hash calculation
HashCalc	Latest	Supporting hash calculation and verification

Evidence Item	Description	Status at Start of Lab
EV-002	Recovered files, approximately 2 GB	Verified — integrity confirmed in Labs 2 and 3
EV-004	Lab 5 evidence source	Created at start of this lab for baseline hashing

---

Recommended Folder Structure

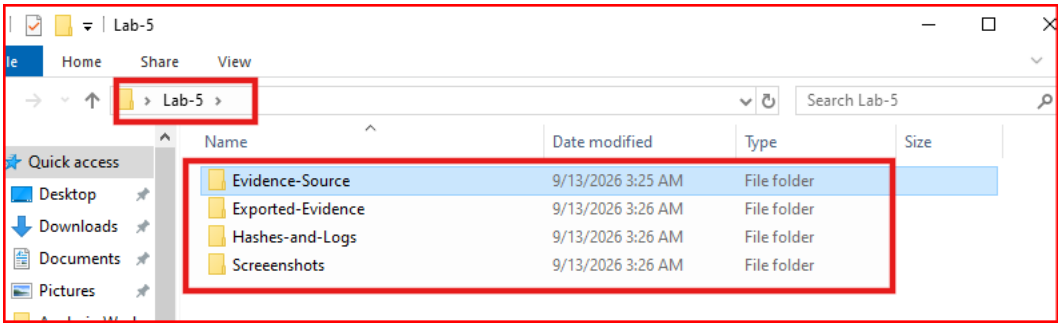
Create this structure on your Desktop:

text

```
Lab-5\
├── Evidence-Source\        ← Copy of your Analysis-Workspace or recovered files
├── Exported-Evidence\      ← Files you export from FTK Imager
├── Hashes-and-Logs\        ← All hash lists and notes
└── Screenshots\
```

Real World Relevance

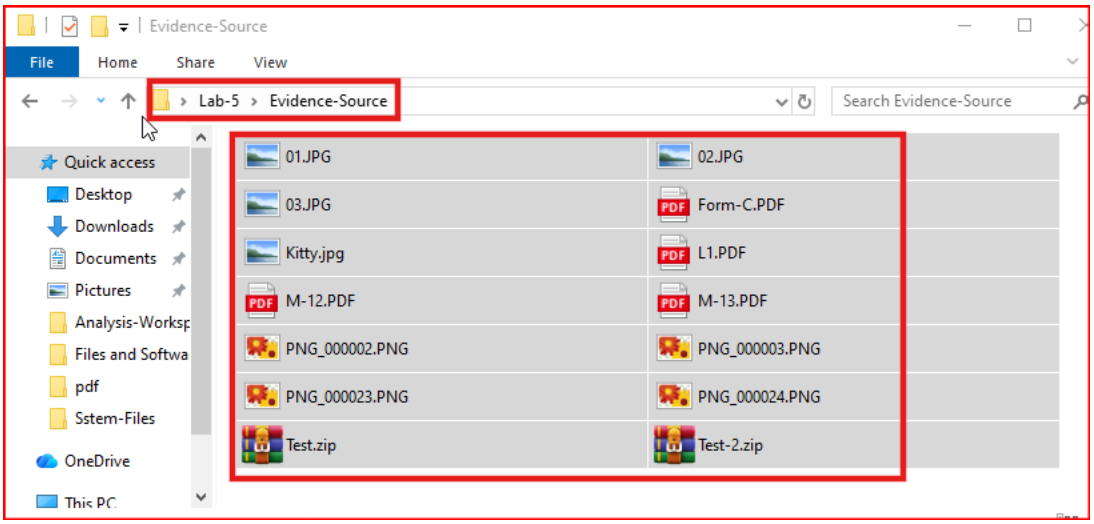
Professional evidence handling is not simply good practice — it is a legal requirement. Before any evidence is imaged or archived, it must be reviewed under conditions that guarantee nothing was altered during the process. FTK Imager enforces read-only access at the hardware and software level, making it one of the most trusted tools for this purpose in law enforcement and professional DFIR environments. The MD5 and SHA-256 baselines established in this laboratory serve a specific legal function — they are the values against which the Lab 6 forensic image will be compared to prove that the image is a bit-for-bit accurate copy of the original evidence. Without these baselines, the integrity of the final image cannot be demonstrated.



Important:

Do **not** work directly on your original Lab-1 recovered files.

First copy the required files into Evidence-Source.



Phase 1: Baseline Hash (Very Important)

1. Open **HashCalc**
2. Calculate **MD5** and **SHA256** of the main evidence folder or important files inside Evidence-Source
3. Save the results in:

Lab-5\Hashes-and-Logs\Baseline-Hashes.txt

```
35ad70c79809bb02608c74d1072976b41507869c9a2a3ad7ac39cb464cfca5d9 ---01.jpg
16ff51c168405e575d32bae001f280e4

bcc5f8155fe8c7293dd0f8c985b59212c508218f82bca3e7cd7248a66c4c6310 --02.jpg
86a6efc3776f2230753d8a8ef11dce88

3b57ce76eb02bb648a5e2eca27989ef9d2303395df034b04b651d31852cc3380 --03.jpg
92091902d3ca753bb858d4682b3fc26b

55ce3face1d3b1a2f46b3c458a87d7f94120a542f51cf5721ea5e8b0fd14ef6b --- PNG_000002.PNG
4ff88b2f9e19e28b0b5dfa5a95350531

db46fa72b31d3b97eff25d59866e024b31ff7772bae12e742b0dc466e58b9b04 --PNG_000003.PNG
e7148fbdea69810a3955b63a26603b6f

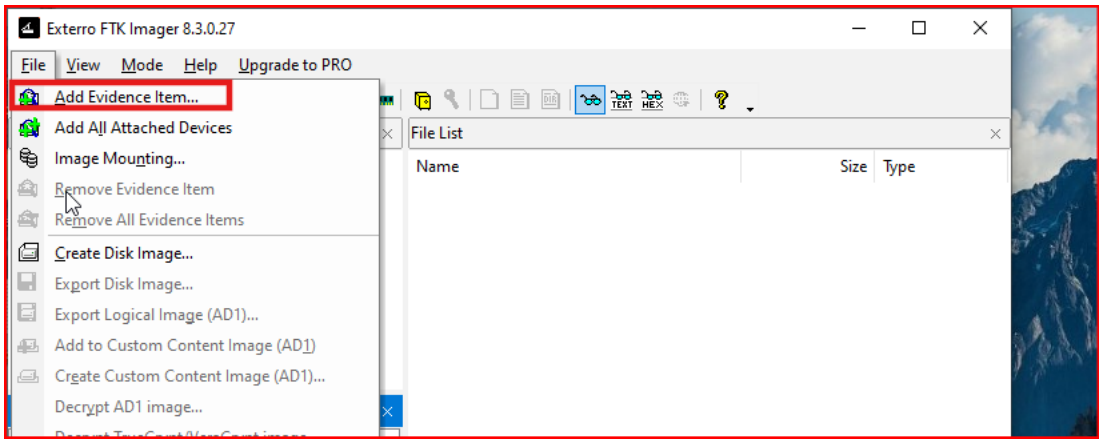
68583e1eec6344165898ae294646e6122c9387b473d58fe6043eacf29436a4f6 --Form-C.pdf
663d809a032960d1c75d2cc11435d0be

c2e1c4244ebb0ab72f7aa7b0cc9f57892b51a5051e377fc7f9d662713d39ddce --Task2.zip
3899710e7b1468c5cc4826728ceb3a13
```

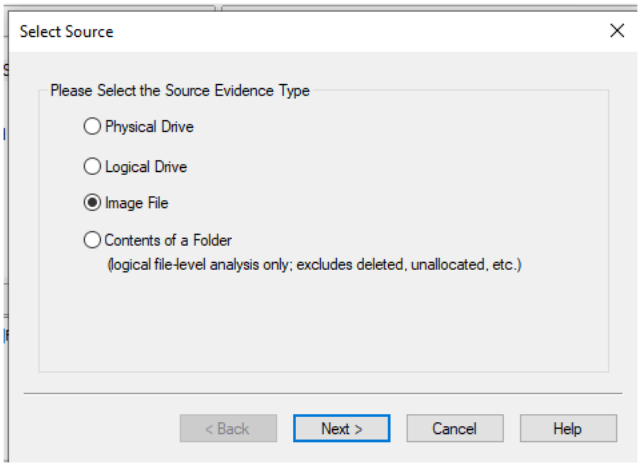
This becomes your reference point (proof of original state).

Phase 2: Add Evidence in FTK Imager

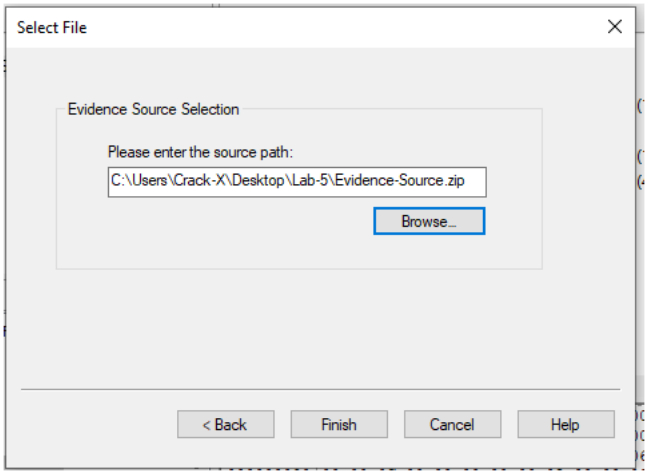
1. Open **FTK Imager**
2. Go to **File → Add Evidence Item**



3. Choose:
    - **Logical Drive** (if using USB directly)
      or
    - **Image File / Contents of a Folder** (recommended – select Evidence-Source)



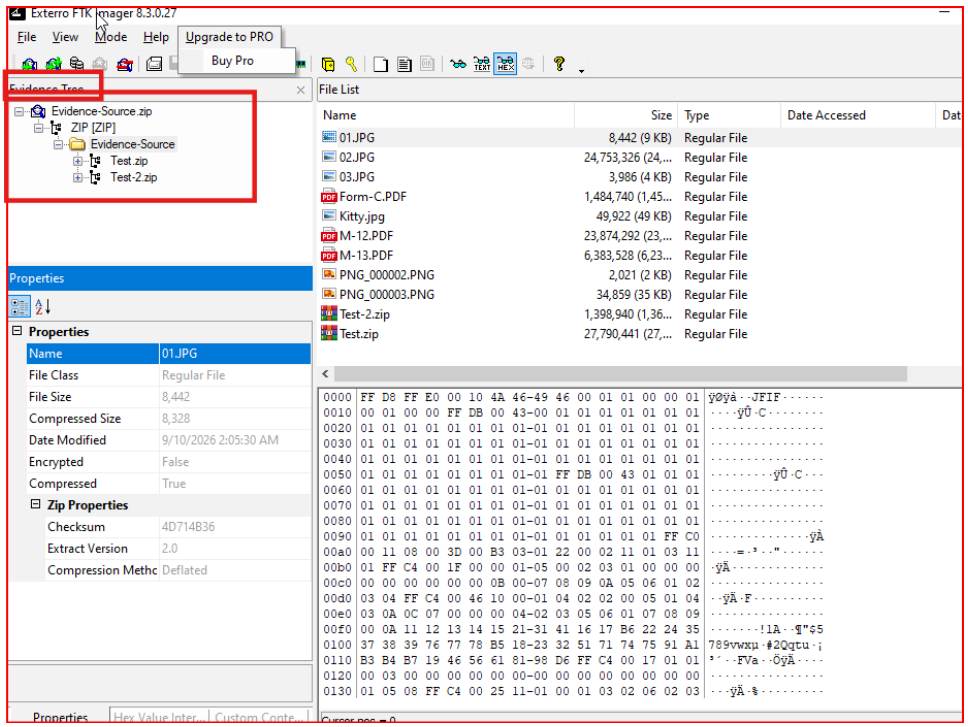
4. Add it to the case



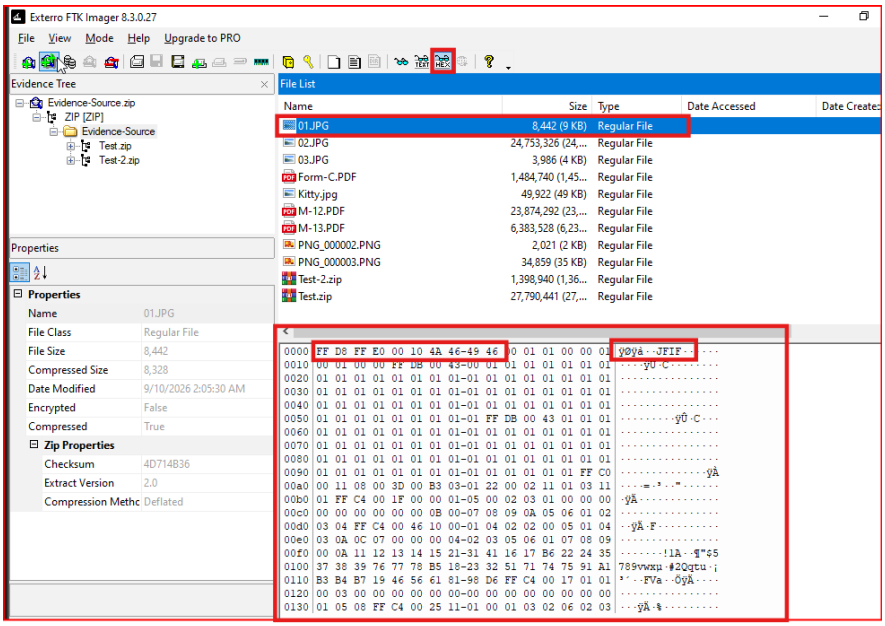
Create a zip file of Evidence-Source to that it would be selected in Browse menu

Phase 3: Evidence Review

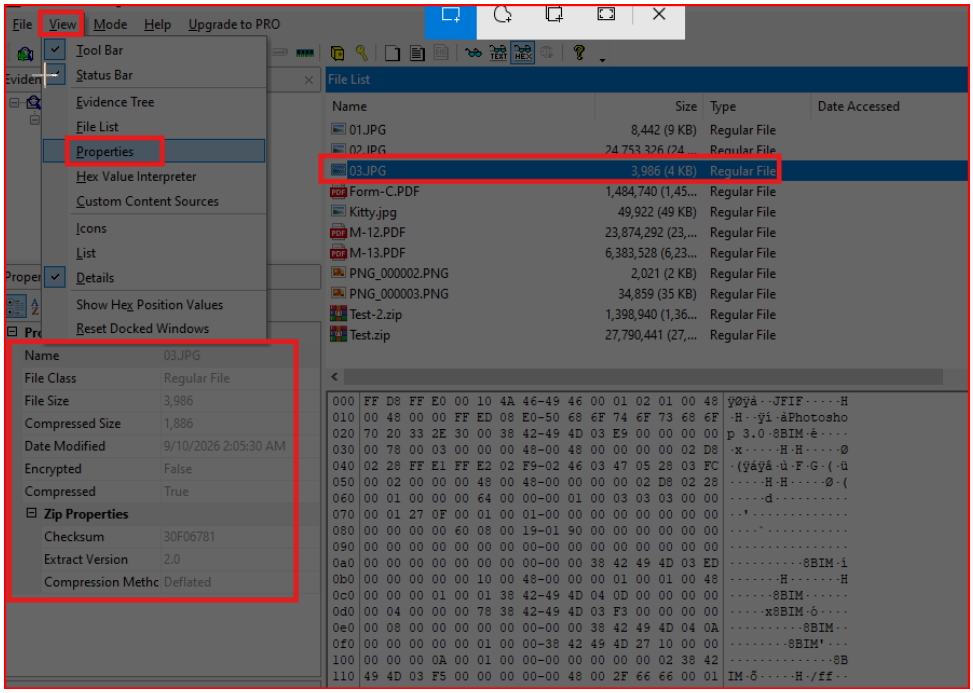
□ Evidence tree



□ Hex view of one file

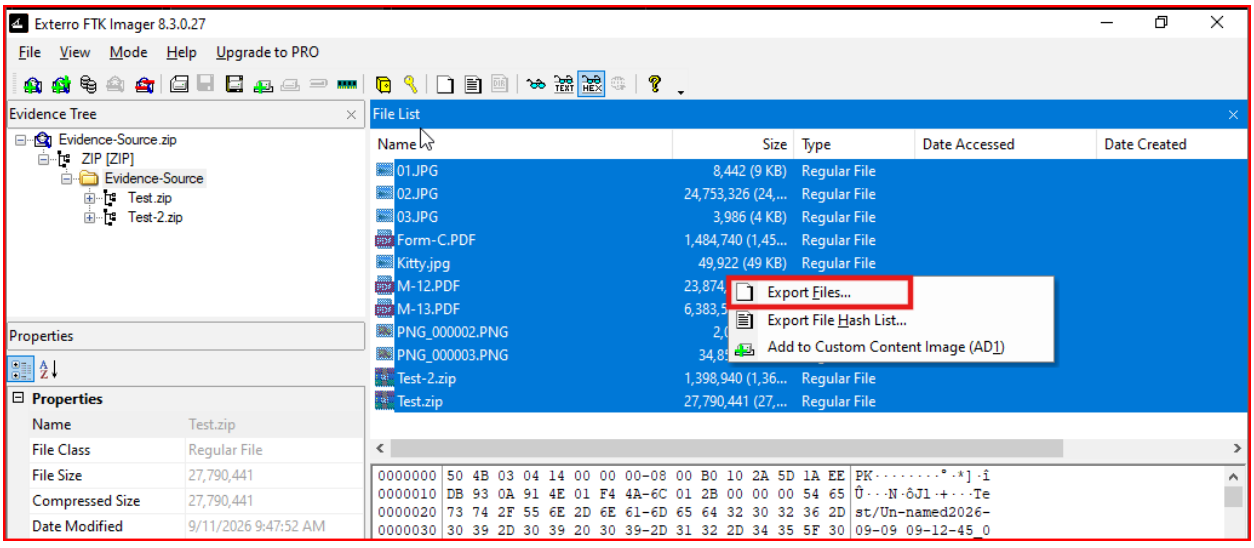


Checking file properties



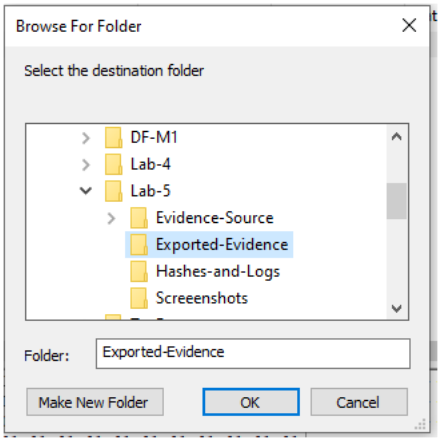
Phase 4: Export Evidence

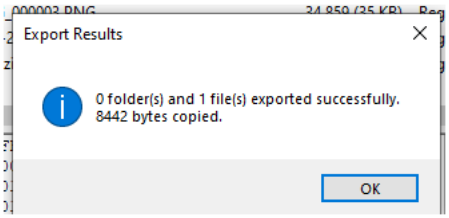
1. Select important files/folders inside FTK Imager
2. Right-click → **Export Files**

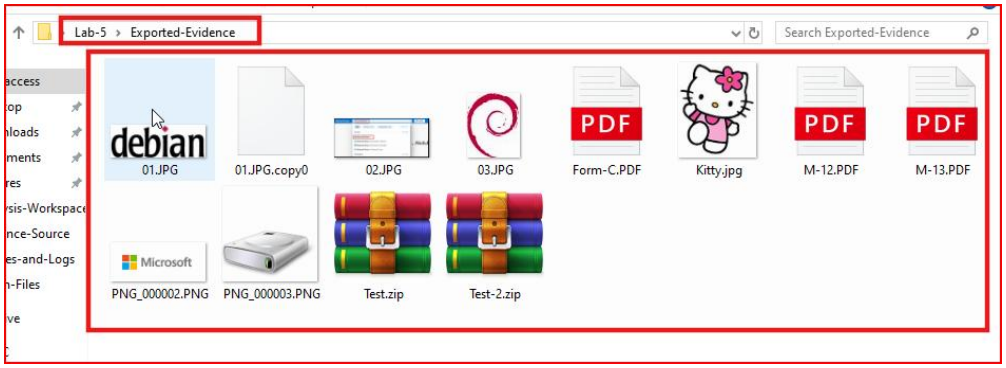


3. Save them to:

Lab-5\Exported-Evidence\



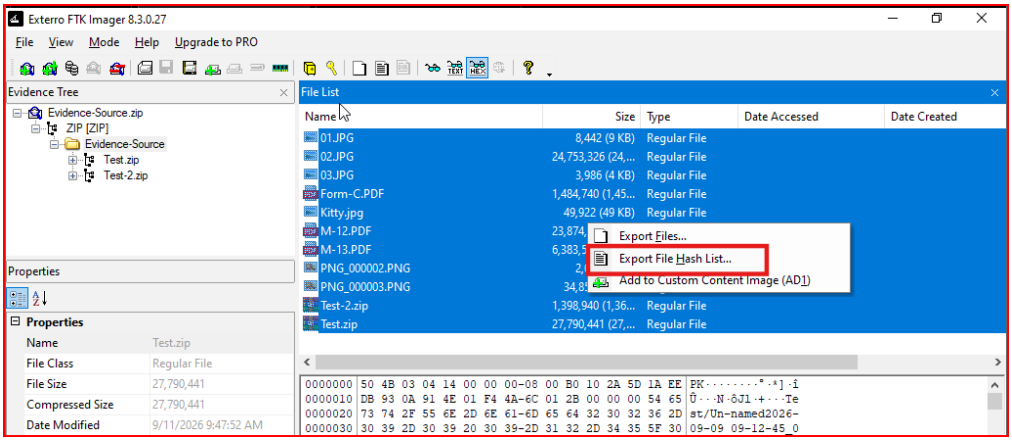


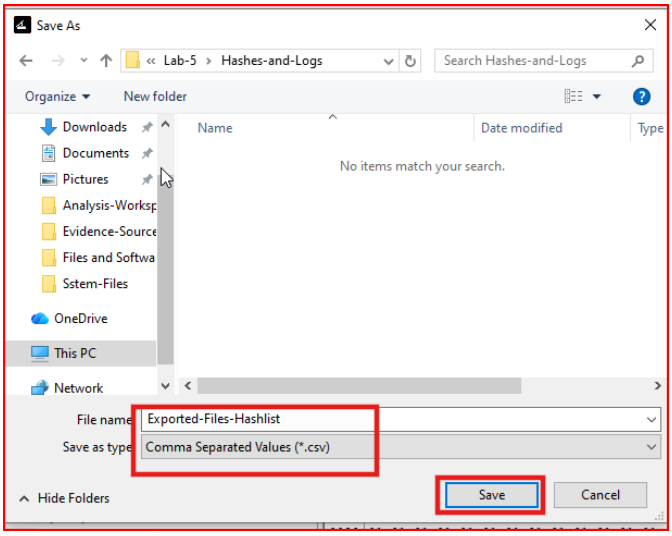


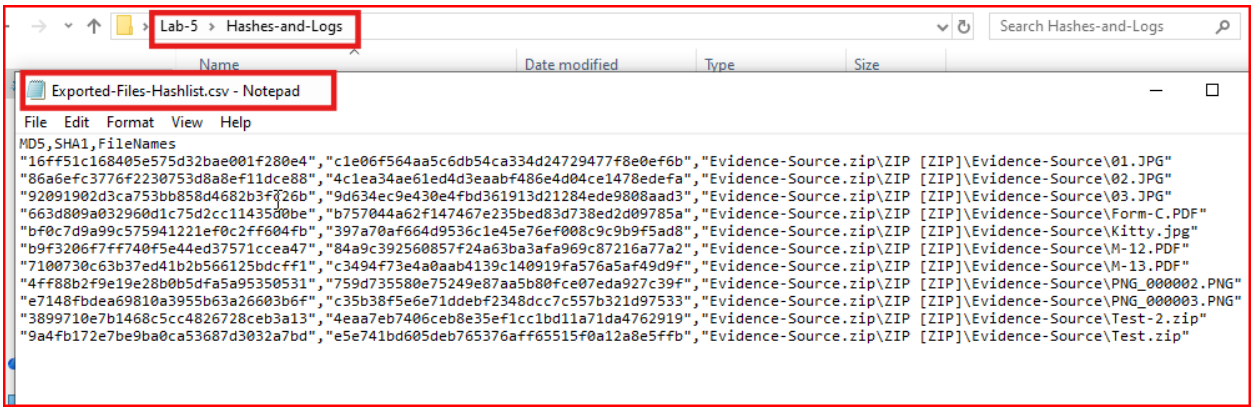
Phase 5: Generate File Hash List

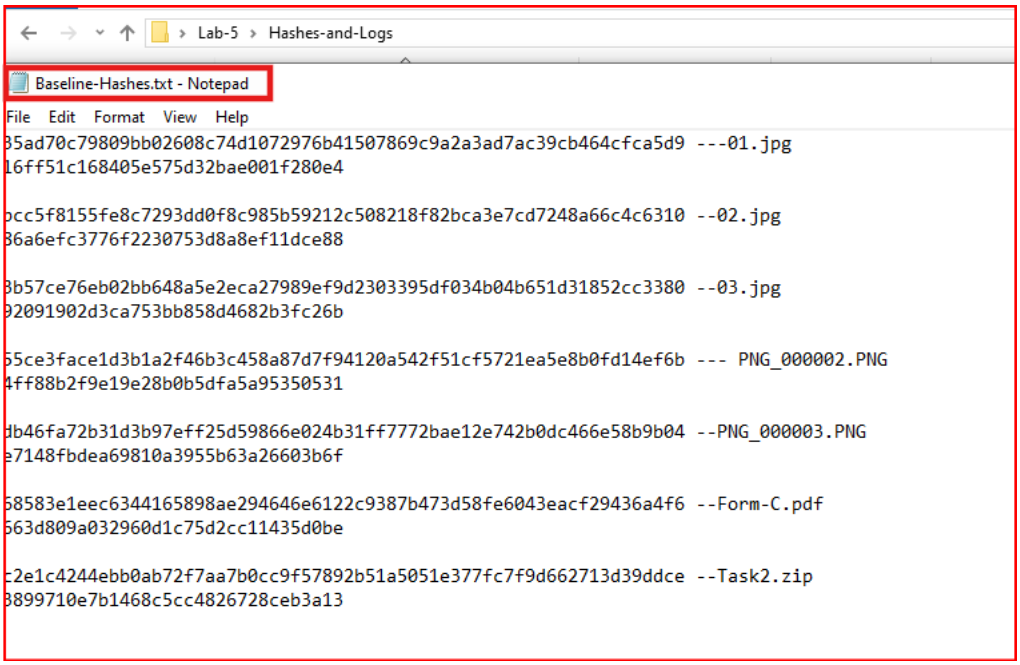
1. In FTK Imager go to: **File → Export File Hash List**
2. Select the exported files
3. Save the hash list as:

Lab-5\03_Hashes_and_Logs\Exported_Files_HashList.csv









This proves that the exported files match what was inside the evidence.

Lab 5 Conclusion

The objectives of this laboratory were fully achieved. The evidence source was successfully loaded into FTK Imager under read-only conditions, its contents were reviewed and confirmed, and a complete file list was exported and retained as part of the case record. MD5 and SHA-256 baseline hash values were calculated using both FTK Imager and HashCalc and recorded as the definitive reference point for Lab 6 integrity verification.

No modifications were made to any evidence item at any point during this laboratory. Read-only access was enforced throughout, and all actions were documented in the chain of custody log. The evidence source is now fully prepared for professional forensic imaging.

Baseline Hash Values — Lab 5

Evidence Item	Algorithm	Calculated By	Date and Time
EV-004 — Lab 5 Evidence Source	MD5	HashCalc	13 Sept 2026, 12:00
EV-004 — Lab 5 Evidence Source	SHA-256	HashCalc	13 Sept 2026, 12:15

These values serve as the verified baseline for Lab 6. Any forensic image produced from this source must return identical hash values to confirm bit-for-bit accuracy.

Evidence Status at Close of Lab 5

Evidence Item	Description	Status
EV-001	Original 32 GB Samsung USB drive	Preserved — unmodified throughout
EV-002	Recovered files, approximately 2 GB	Verified — integrity confirmed across Labs 2, 3, and 5
EV-003	Workspace copy used for Lab 4 analysis	Examined — retained in read-only storage
EV-004	Lab 5 evidence source	Baseline established — MD5 and SHA-256 recorded

Chain of custody remained unbroken throughout this laboratory. All baseline hash values, file exports, and handling records have been retained as part of the permanent case record.

DISK IMAGING AND ACQUISITION

Lab Overview

Field	Detail
Lab Title	Disk Imaging and Acquisition
Subject	Cyber Forensics
Tools Covered	Exterro FTK Imager and ProDiscover Basic
Cases Covered	Two — Information Theft and Employee Asset Misuse
Examiner	Abdul Wahid
Investigation Type	Forensic Disk Imaging, Evidence Recovery, and Report Generation

Purpose

Disk imaging is the process of creating an exact, bit-by-bit copy of a storage device — whether a hard drive, SSD, or USB — in a forensically sound manner. The resulting image captures everything on the device including active files, deleted files, unallocated space, and hidden sectors, while leaving the original device completely unaltered. The acquired image is then used for all investigative work — examination, verification, keyword searching, and evidence preservation — ensuring the original is never put at risk.

This laboratory demonstrates the complete disk imaging and acquisition workflow using two industry-standard forensic tools. Exterro FTK Imager is used to create, mount, verify, and review forensic images with full hash-based integrity confirmation. ProDiscover Basic is used to conduct logical and physical analysis of a mounted forensic image, recover deleted files, perform keyword searches, and generate a formal forensic report.

Together these two tools represent a complete acquisition-to-analysis pipeline. FTK Imager handles the acquisition and preservation side. ProDiscover handles the examination and reporting side. Both tools maintain evidence integrity throughout and produce outputs that are suitable for formal investigative proceedings.

Case Scenarios

This laboratory is built around two real investigative scenarios:

Case 1 — Information Theft
Senior investigator Alex used FTK Imager to scan targeted systems and assess whether deleted files contained evidentiary value. The investigation focused on extracting evidentiary data efficiently and accurately without manually recovering every deleted file from each system.

Case 2 — Employee Missing and Company Asset Misuse
Manager Steve Billings reported two absent employees — George Montgomery and Martha — and raised concerns about asset misuse. The target system image InChp02.eve was seized for forensic examination. The objective was to locate evidence of the employees' activity, recover relevant files, search for references to the name George, and produce a formal hash-verified forensic report.

Objectives

FTK Imager Objectives:

- Create a forensic disk image of a target storage device in a chosen format including Raw DD, E01, or SMART
- Enter full case metadata including case number, evidence number, examiner name, and description
- Mount the completed image in read-only mode and verify its contents
- Verify image integrity through automatic MD5 and SHA-1 hash generation and comparison
- Add evidence items for review and export disk contents for further examination
- Demonstrate the difference between physical and logical drive imaging

ProDiscover Objectives:

- Create a new forensic project and add the mounted image InChp02.eve as evidence in read-only mode
- Conduct logical and physical analysis of the image to identify active and deleted files
- Identify deleted file indicators and preview file contents including metadata and hex view
- Export relevant files to a designated evidence folder with hash-validated manifests
- Search the image for the keyword George and document all hits with path and context
- Generate a formal forensic report including case metadata, evidence items, search results, and examiner notes

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Tools and Evidence

Tool	Purpose	Case
Exterro FTK Imager v4.7.3.81	Disk imaging, image mounting, integrity verification, evidence review	Case 1 — Information Theft
ProDiscover Basic	Image analysis, deleted file recovery, keyword search, report generation	Case 2 — Asset Misuse

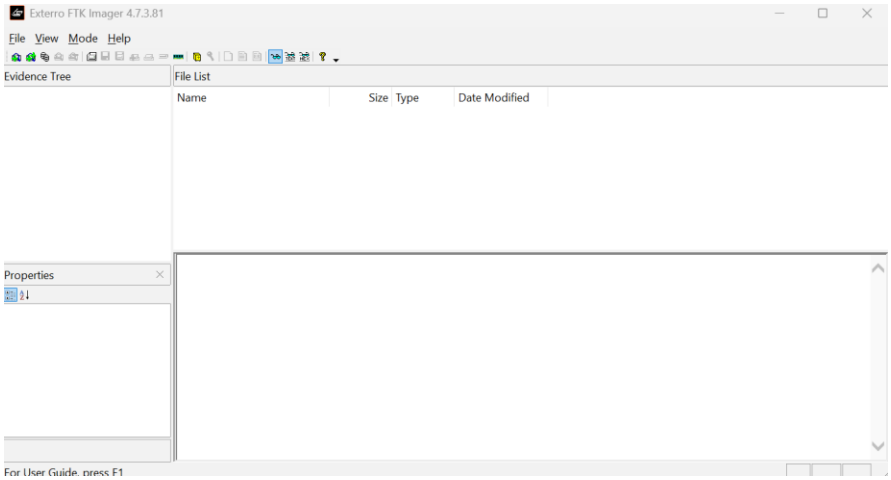
Evidence Item	Description	Case
Target storage device	Physical or logical drive selected for imaging	Case 1
InChp02.eve	Forensic image of suspect system — George Montgomery	Case 2

Real World Relevance

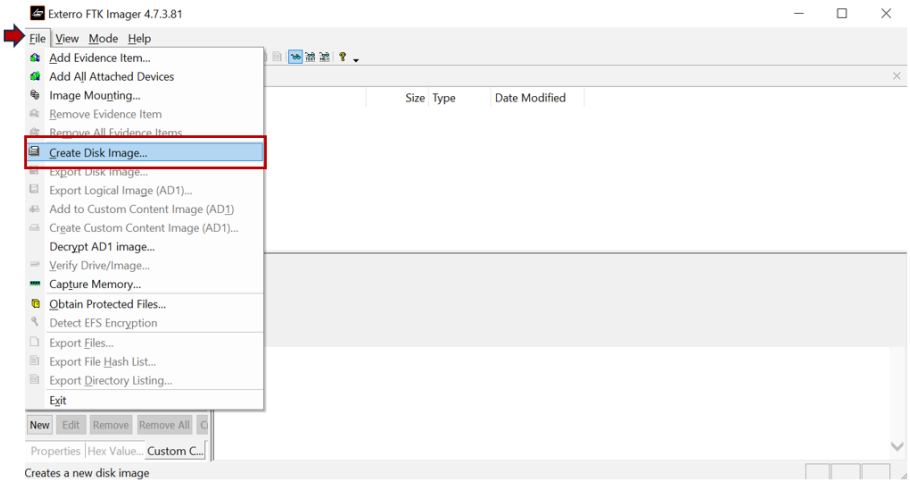
Disk imaging is the first and most critical step in any digital forensic investigation. No professional examiner works directly on original evidence. The original device is imaged, the image is verified through cryptographic hashing, and all subsequent analysis is performed on the verified copy. This protects the original evidence from any accidental modification and allows multiple examiners to work independently on identical copies without conflict.

FTK Imager is used by law enforcement agencies, government forensic laboratories, and professional DFIR firms globally. Its ability to image devices in multiple formats, verify integrity automatically, and mount images in read-only mode makes it one of the most trusted acquisition tools available. ProDiscover extends that capability into the examination phase — allowing the examiner to recover deleted files, search for specific content, and produce court-ready reports without ever touching the original evidence. Together these tools represent the professional standard for acquisition-to-analysis forensic workflows worldwide.

Start the Imaging Process

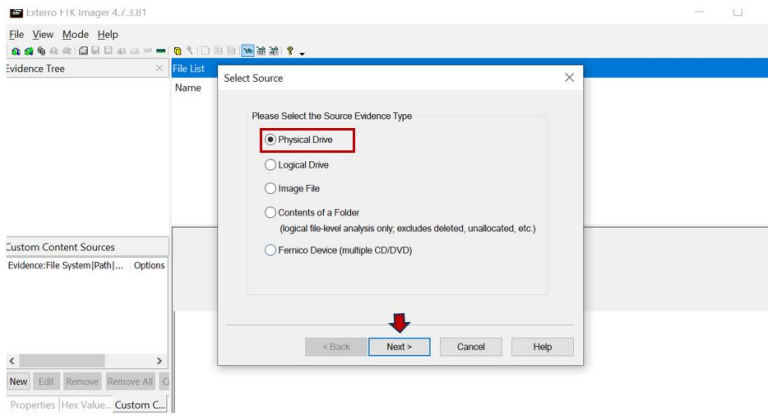


- Click on **File** (top-left menu).
- Hover over and select **Create Disk Image** to begin.

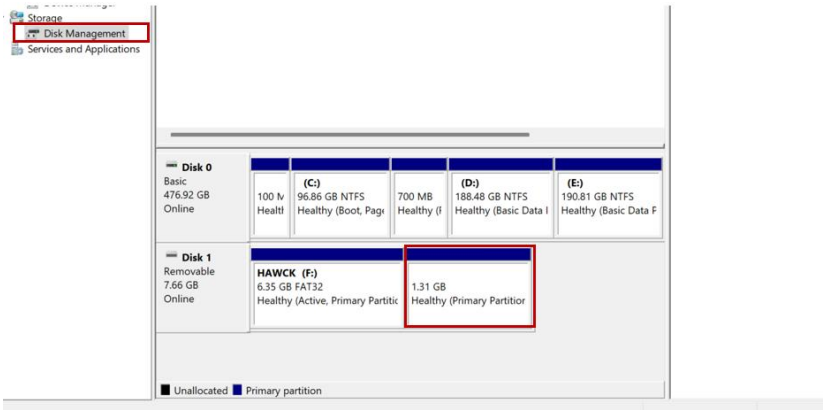


Select Source Drive

- Choose the type of source you want to image:
    - **Physical Drive** (entire hard disk),
    - **Logical Drive** (individual partitions).
- Click **Next**.

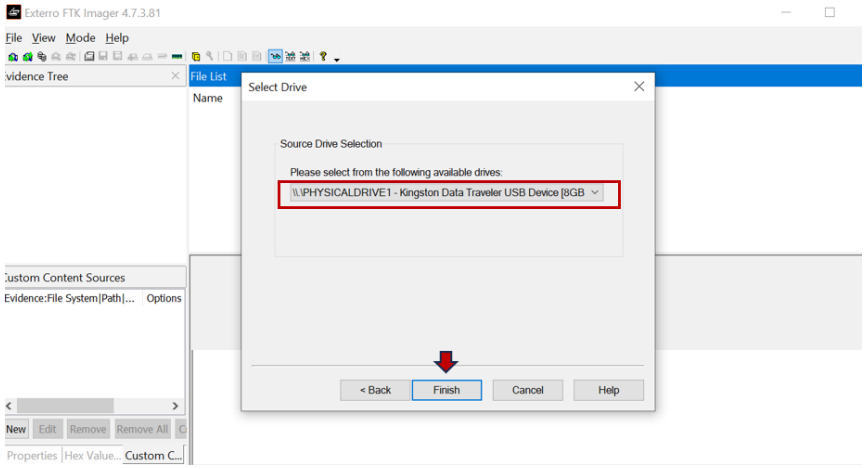


- **To check the difference between physical and logical drives we see that in our system**



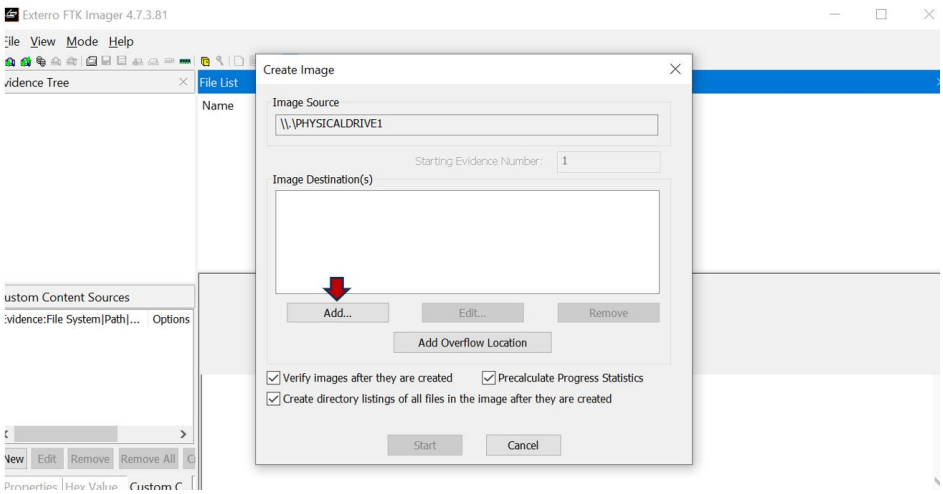
Identify Source Location

- Select the specific drive or partition to be imaged.
- After selection, click **Finish**

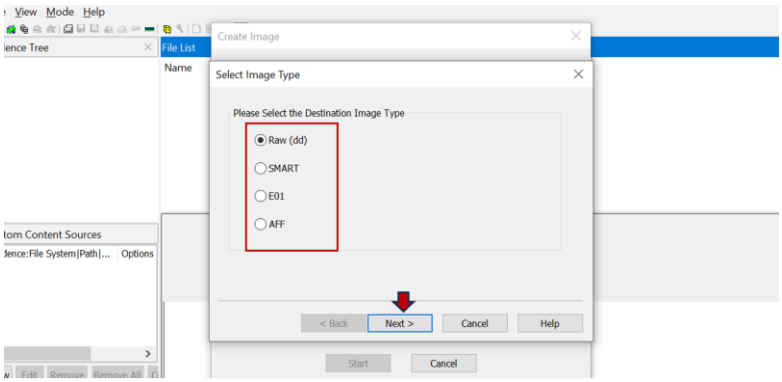


Set Imaging Options

- Click on **Add** to specify destination details.



- Choose the **image type** (e.g., Raw (dd), E01, SMART).
  - **Raw (dd)** is preferred for universal compatibility.
  - **SMART** is commonly used on Linux systems.
- Click **Next**.

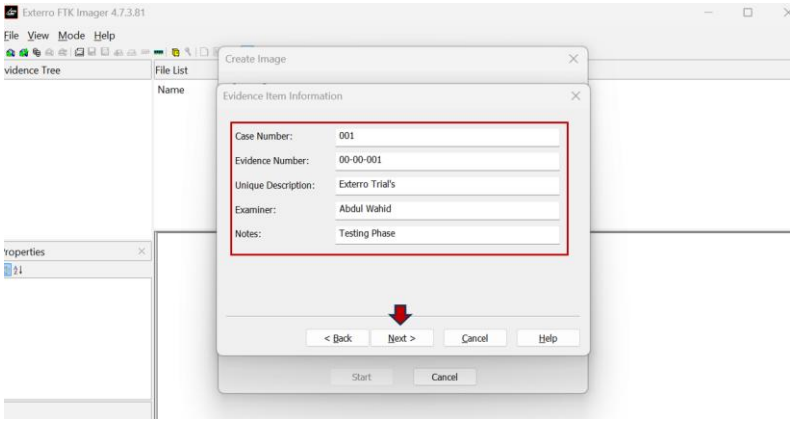


Enter Case Information

Fill in the required metadata:

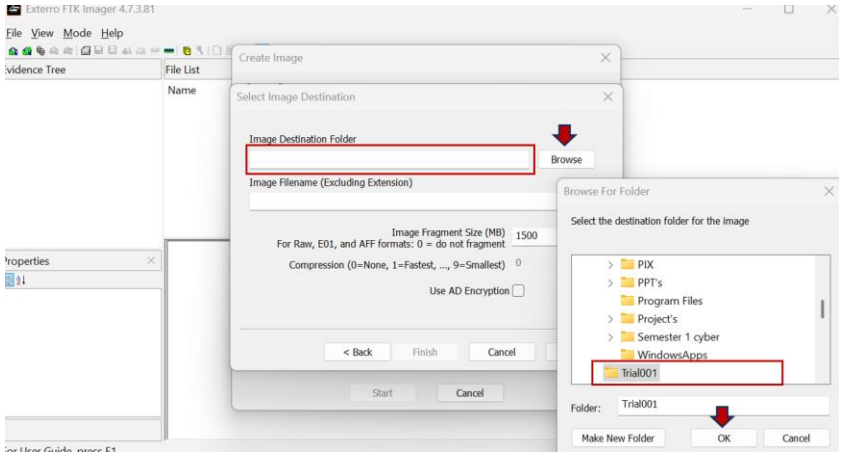
- **Case Number**
- **Evidence Number**
- **Examiner Name**
- **Description** (e.g., "Laptop seized from suspect")
- Add **Notes** if needed.

o Click **Next**.

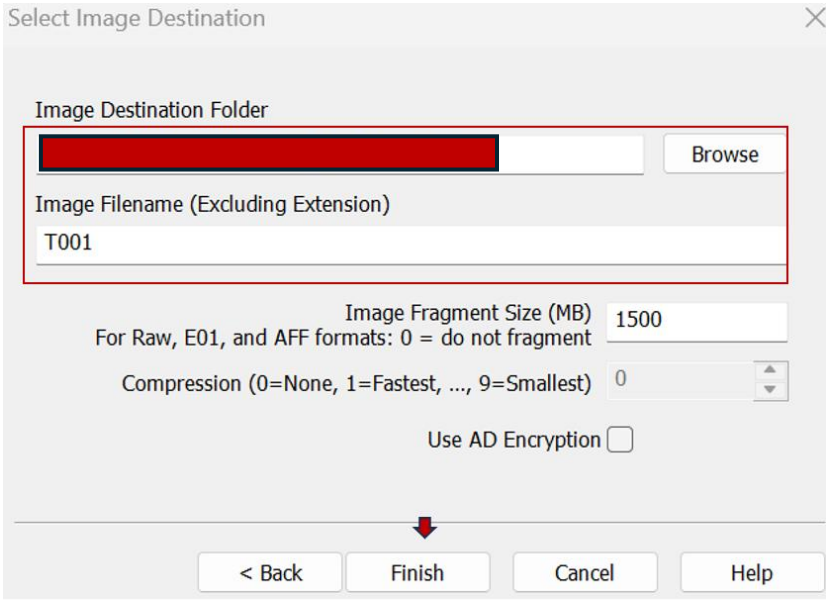


Set Image Destination

- Browse to select where the image will be saved.

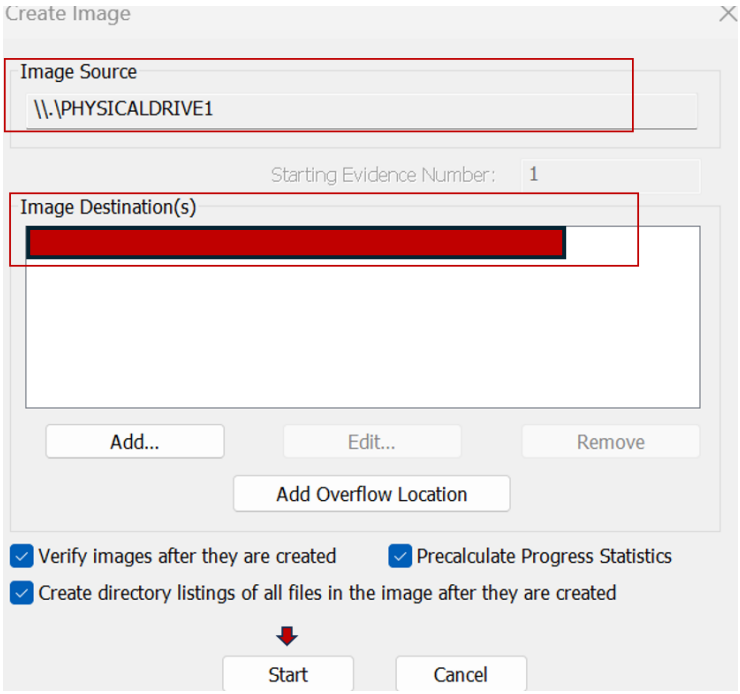


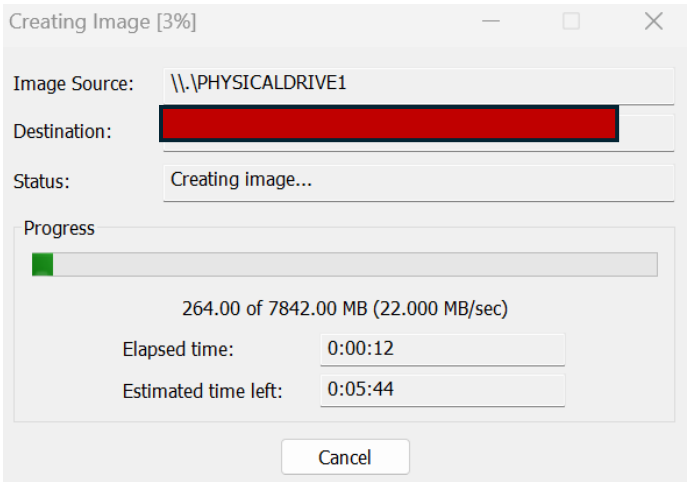
- Enter the **file name** (without extension).
- Click **Finish** to return to the previous screen.



Start Imaging

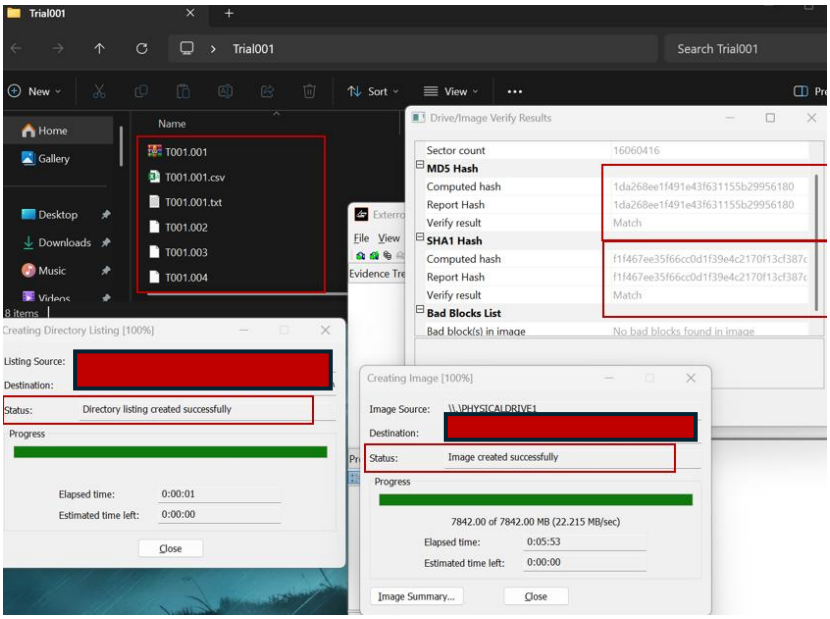
- Review the settings.
- Click **Start** to begin the imaging process.
- Wait for the process to complete.



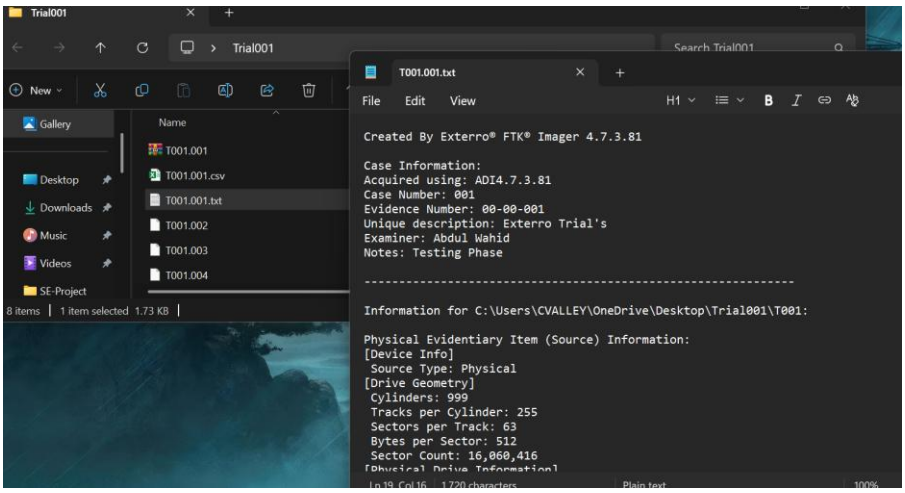


After completion, the tool will generate:

- An image file (e.g., .dd),
- A corresponding **text log file** with metadata, hashes, and evidence details.

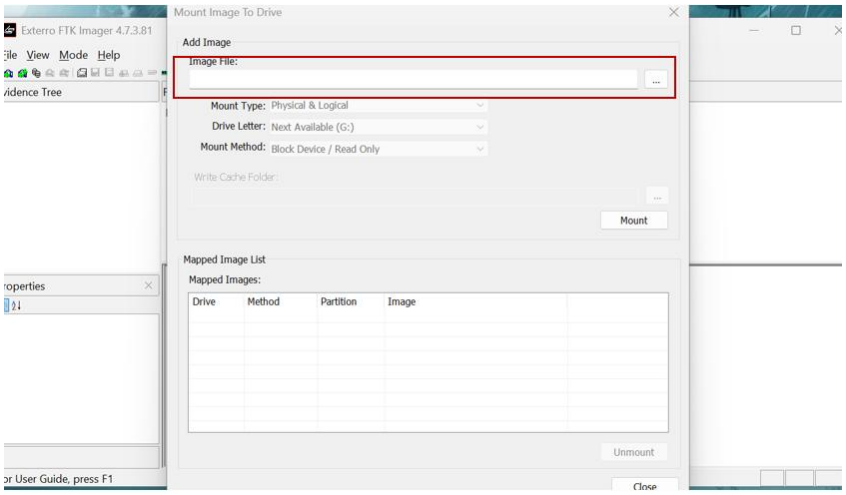


- Out of other files we got a txt file which shows that:
- Showing the image of FTK imager (version)
- Details we provided earlier
- And all relevant info is present here

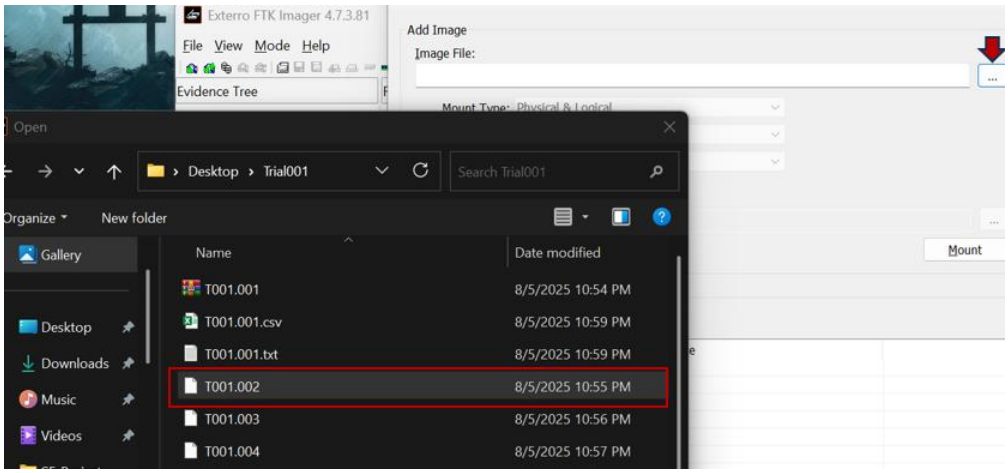


🔧 Mounting the Image for review

 Go to **File > Image Mounting**.



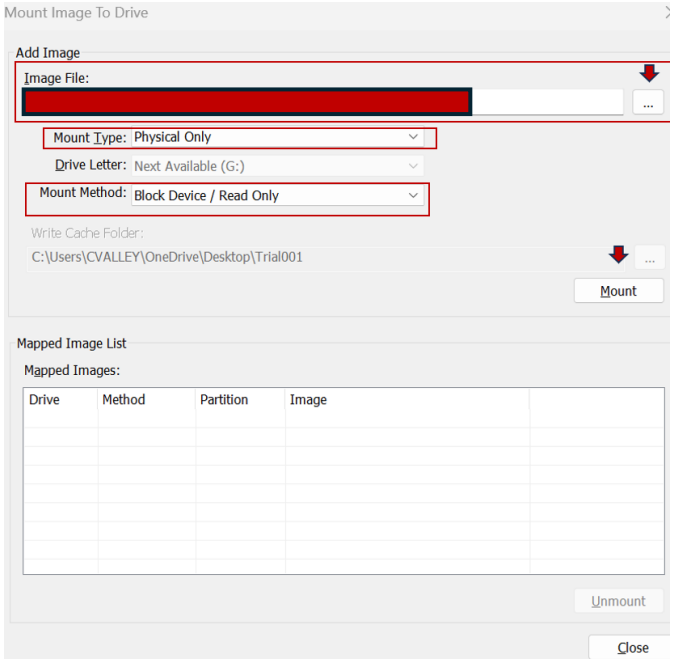
 Browse and select the previously created image file.



 Choose the mount type:

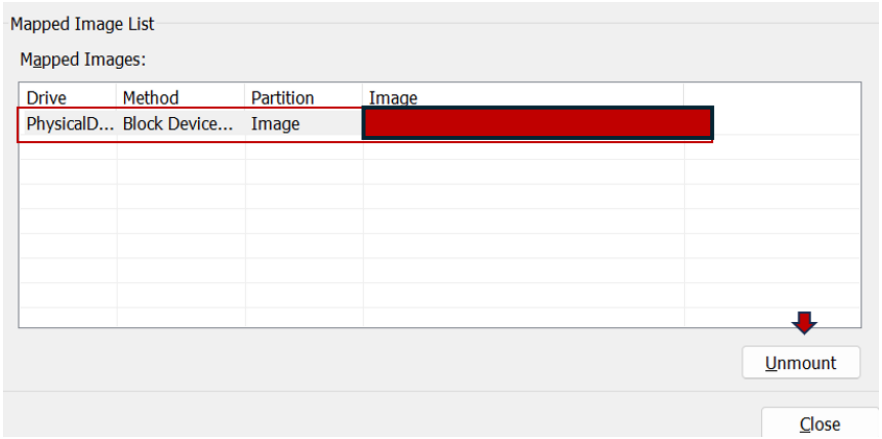
- **Read-Only** (recommended),
- **Writable** (use with caution).

 Click **Mount**.



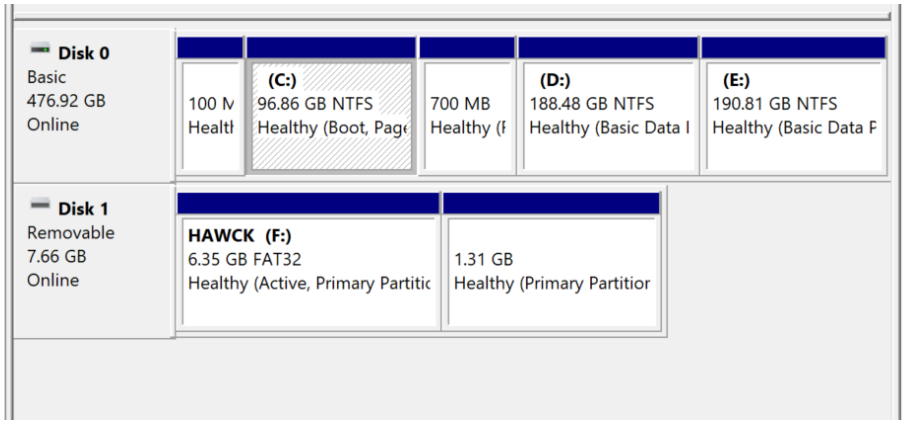
Unmount the Image

- Return to **Image Mounting**.
- Click **Unmount** to safely remove the virtual drive.



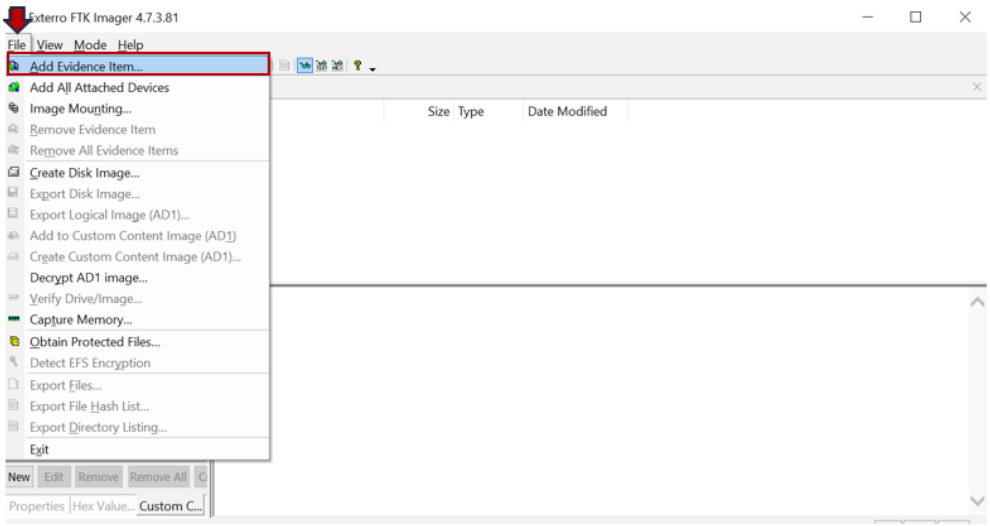
As we can see a healthy partition has been created in our system showing on disk management

A new virtual drive (e.g., F:) will appear in your system.

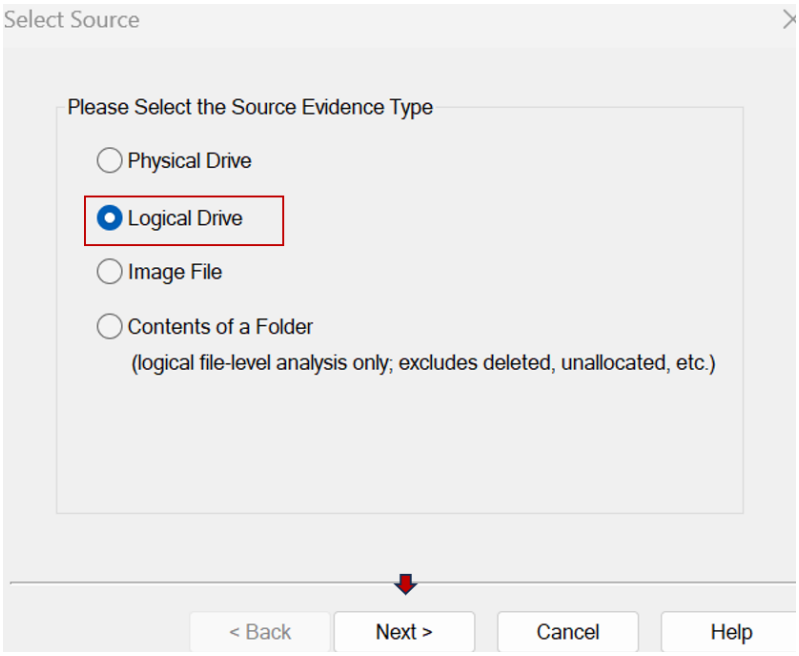


Adding Evidence for Review

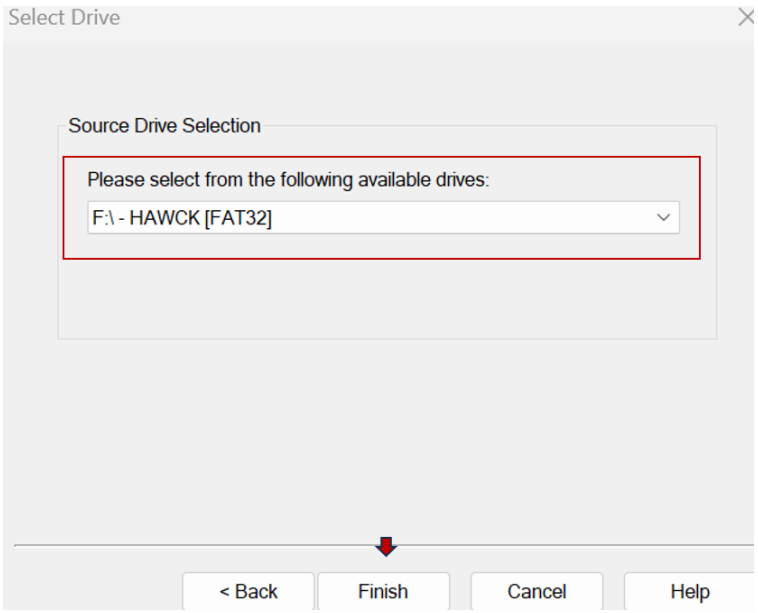
- Click on **File > Add Evidence Item**.

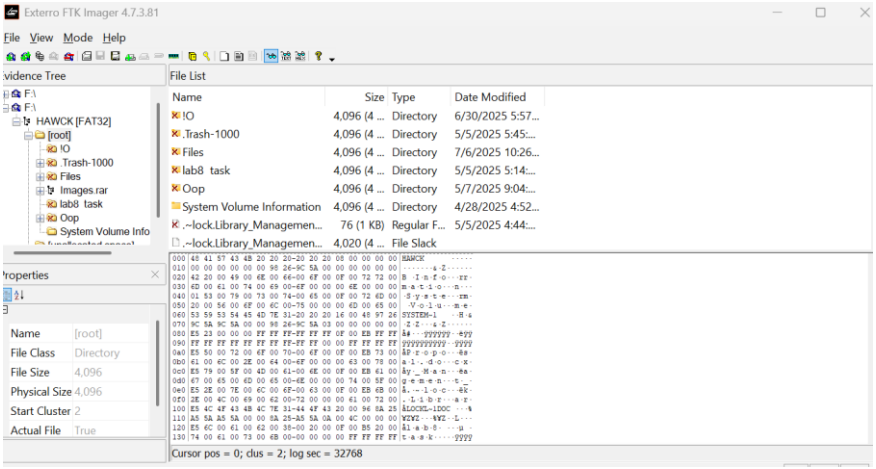


- Choose **Logical Drive** or **Image File**.



- Browse to select the evidence source.
- Click **Finish**.





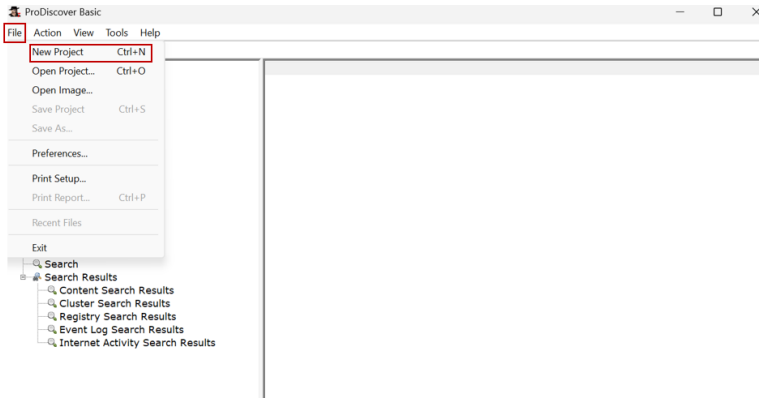
FTK Imager Conclusion

FTK Imager successfully demonstrated the complete forensic imaging workflow from source selection through to image verification and evidence review. A forensic image was created from the target device with full case metadata embedded, and the imaging process automatically generated MD5 and SHA-1 hash values upon completion. The image was subsequently mounted in read-only mode, confirming that all files — including deleted and hidden items — were accessible and intact without any modification to the original device.

The hash verification confirmed that the image is a bit-for-bit accurate copy of the source. The text log file generated alongside the image captured the FTK Imager version, all case metadata entered during setup, physical and logical device information, hash values, and sector-level details — providing a complete and self-contained audit record of the acquisition process. This output is suitable for presentation in formal investigative and legal proceedings.

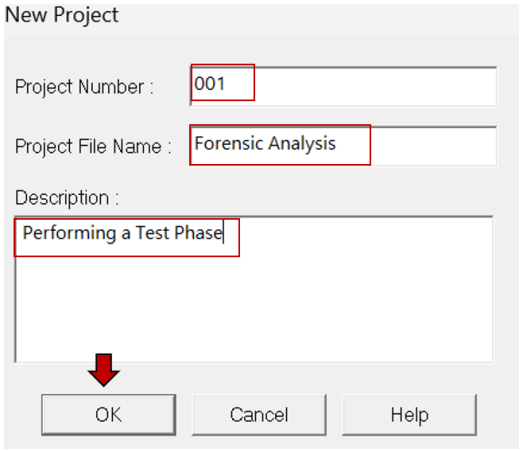
Pro Discovery

- **Creating a new project**
  - **File → New Project**



New Project dialog.

Create a new project and enter the required case metadata (Project Number, Project File Name, Description).

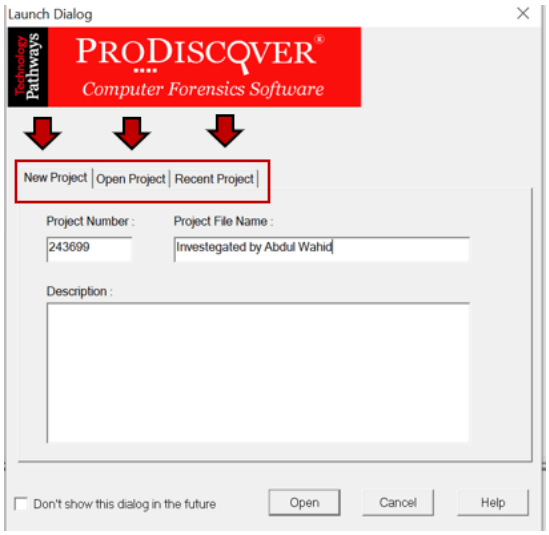


Open Project dialog.

Open existing project (if applicable)

File → Open Project

If resuming work, open the existing project file to continue analysis. Confirm project metadata and current evidence list before proceeding.

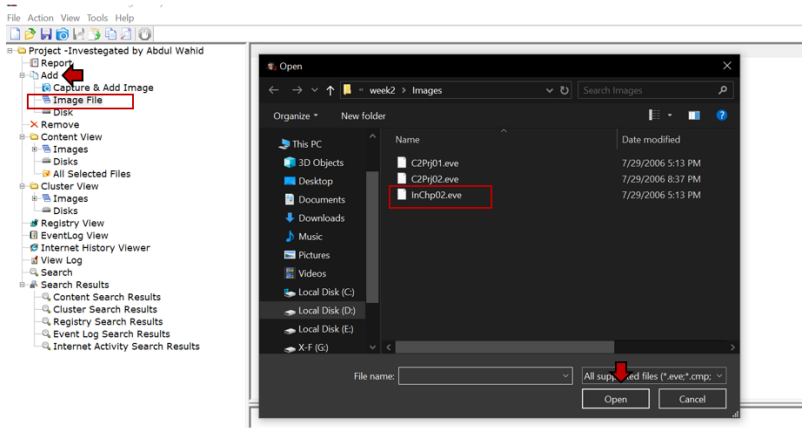


Existing/Recent project

Add / Select image file as evidence

Evidence → Add Evidence Item → Image File

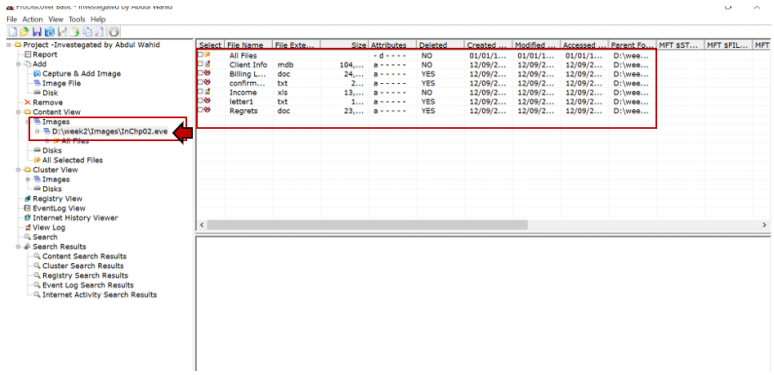
Select the mounted image InChp02.eve and add it to the project as an evidence item. Ensure the image is added in *read-only* mode and verify the image hash (MD5/SHA1) recorded in the project metadata.



Select image file (Add Evidence).

Image selected; identify deleted file indicators

Click the mounted image in the evidence list to load the file tree/index.
After loading the image, the file tree shows red indicators alongside entries that have been deleted. These markers identify deleted or flagged files that remain recoverable by the tool. Do not modify the image; instead use the tool's recovery/export features to extract copies for further analysis.

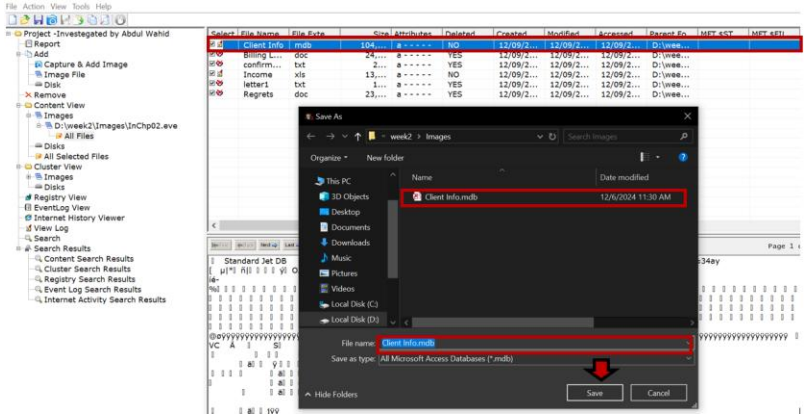


Deleted file indicators.

Previewing file contents (including deleted files)

Select any file → Preview/Content pane

Selecting a file (including deleted entries) opens the preview pane, which displays file metadata and contents (file header, readable text, hex view if needed). Use the preview to assess relevance before exporting. Record the original file path, file size, and timestamps in the export metadata.



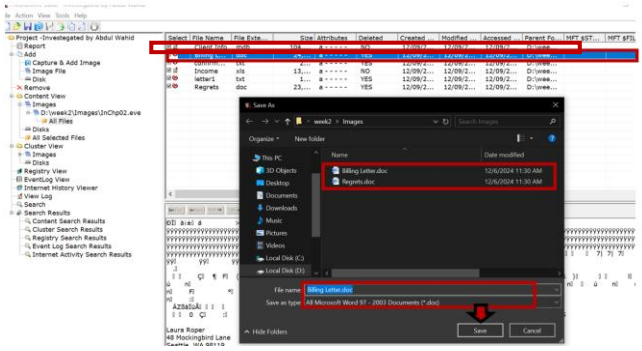
File preview (deleted file content)

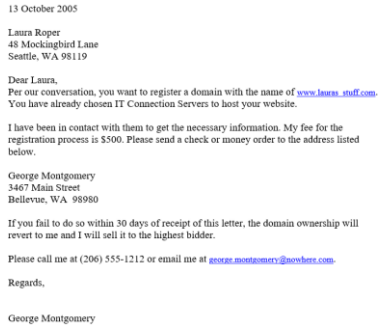
Exporting / Copying evidence

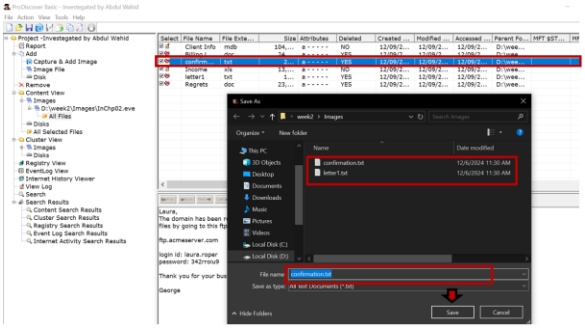
Right-click file → Export / Save As → Choose destination in evidence folder

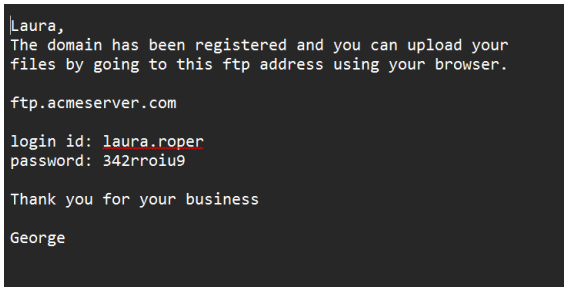
Export selected files to the designated evidence output folder (e.g., \Evidence\PD-1-243699-01\Recovered\). Name exported files to preserve source path and include a manifest with original path, extraction time, and computed hash (MD5/SHA1). Keep exported copies separate from working copies and log each export action in the case notes.

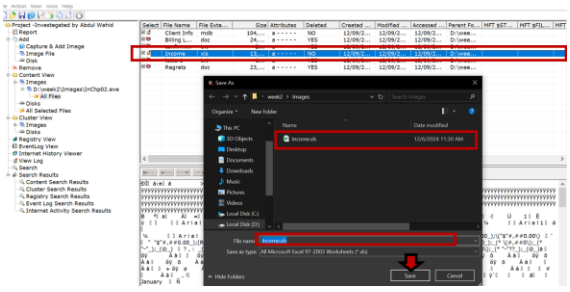
Evidence

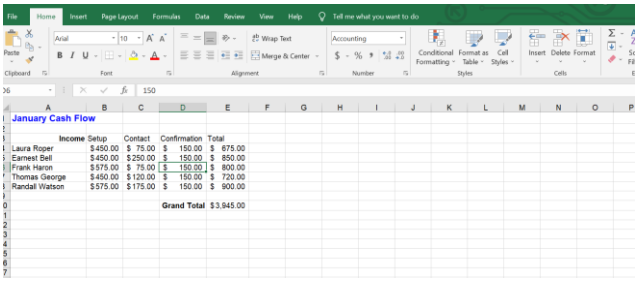










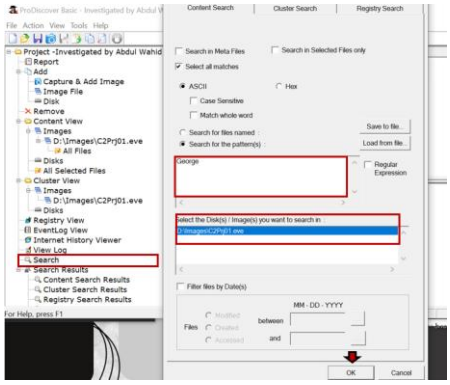


Exporting evidence dialog.

Searching the image (“George”)

Search → Enter search term → Scope = Mounted Image → OK

Use the search function to query the mounted image for the keyword George (exact phrase). Confirm the search scope is set to the mounted image and run the search. Review returned hits (path and context), export relevant files or snippets, and add results to the case report. Record the date/time of the search and any filters used.

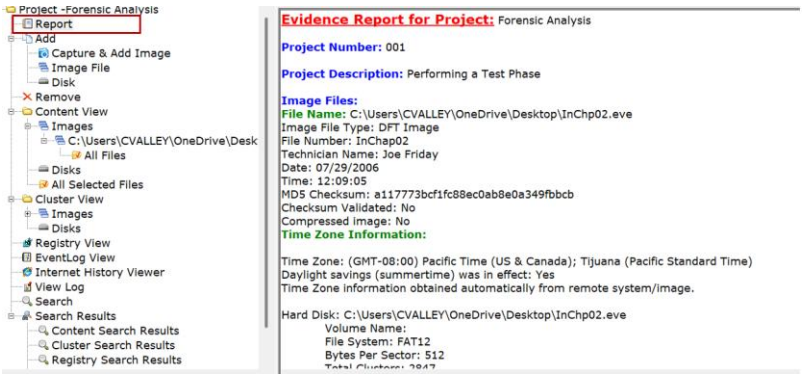


Search "George" results.

Generate the report

Report → Generate Report → Select items to include → Save/Print

Generate a formal report for the project that includes: case metadata, evidence items (with hashes), a list of exported files, search hits (e.g., references to "George"), screenshots used as figures, examiner notes, and a conclusion. Save the final report as PDF and archive the project folder. Attach the report to the project file and include a signed examiner statement.



Report generation options.

ProDiscover Conclusion

ProDiscover Basic successfully demonstrated the examination and reporting phase of the forensic workflow applied to the InChp02.eve image. The image was added to the project in read-only mode and its MD5 and SHA-1 hash values were verified and recorded in the project metadata. Logical and physical analysis of the image revealed both active and deleted files, with deleted entries clearly marked by red indicators in the file tree.

File contents were previewed including metadata, readable text, and hex view where required. Relevant files were exported to the designated evidence folder with computed hash values and manifest records. A keyword search for George returned confirmed hits across multiple files, identifying documents, correspondence, and financial records linked to George Montgomery. A formal forensic report was generated containing case metadata, evidence items with hashes, exported file listings, search results, examiner notes, and a signed conclusion — ready for archival and submission.

Combined Findings Summary

Area	Tool	Result
Forensic image created	FTK Imager	Completed — MD5 and SHA-1 verified
Image mounted and reviewed	FTK Imager	All files accessible in read-only mode
Deleted files identified	ProDiscover	Red indicators confirmed recoverable entries
Files exported with hashes	ProDiscover	Manifest created — evidence folder populated
Keyword search — George	ProDiscover	Hits confirmed across multiple file types
Formal report generated	ProDiscover	PDF report produced and archived

Lab Conclusion

This laboratory successfully demonstrated a complete forensic acquisition-to-analysis pipeline using two industry-standard tools. FTK Imager established the acquisition and preservation foundation — creating a verified, bit-for-bit forensic image with confirmed hash integrity and mounting it safely for examination. ProDiscover built on that foundation — conducting a thorough examination of the mounted image, recovering deleted evidence, locating investigative leads through keyword search, and producing a professional court-ready report.

All procedures were conducted in accordance with forensic best practices. Original evidence was never modified. All analysis was performed on verified copies. Hash values were calculated, recorded, and confirmed at every critical stage. The outputs produced by both tools are complete, documented, and suitable for formal submission and potential legal proceedings.